INSTRUCTIONS

C

CONTENTS

BX-JBXDTP/ACM

JBX FILER

This manual applies to the following JEOL products:

JBX-6300 series, JBX-9000MV,
JBX-9000MV/II, JBX-9300FS,
JBX-3030MV, JBX-3040MV

IEBBX-JBXDTP/ACM-2F
NOV2006-06110459
Printed in Japan

JEOL
Serving Advanced Technology

C

BX-JBXDTP/ACM

JBX FILER

Please be sure to read this instruction manual carefully, and fully understand its contents prior to the operation or maintenance for the proper use of the instrument.

NOTICE

- The following actions must be avoided without prior written permission from JEOL Ltd. or its subsidiary company responsible for the subject (hereinafter referred to as "JEOL"): modifying the instrument; attaching products other than those supplied by JEOL; repairing the instrument, components and parts that have failed, such as replacing pipes in the cooling water system, without consulting your JEOL service office; and adjusting the specified parts that only field service technicians employed or authorized by JEOL are allowed to adjust, such as bolts or regulators which need to be tightened with appropriate torque. Doing any of the above might result in instrument failure and/or a serious accident. If any such modification, attachment, replacement or adjustment is made, all the stipulated warranties and preventative maintenances and/or services contracted by JEOL or its affiliated company or authorized representative will be void.
- Replacement parts for maintenance of the instrument functionality and performance are retained and available for five years from the date of installation. Thereafter, some of those parts may be available for a certain period of time, and in this case, an extra service charge may be applied for servicing with those parts. Please contact your JEOL service office for details before the period of retention has passed.
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- After installation or delivery of the instrument, if the instrument is required for the relocation whether it is within the facility, transportation, resale whether it is involved with the relocation, or disposition, please be sure to contact your JEOL service office. If the instrument is disassembled, moved or transported without the supervision of the personnel authorized by JEOL, JEOL will not be held responsible for any loss, damage, accident or problem with the instrument. Operating the improperly installed instrument might cause accidents such as water leakage, fire, and electric shock.
- The information described in this manual, and the specifications and contents of the software described in this manual are subject to change without prior notice due to the ongoing improvements made in the instrument.
- Every effort has been made to ensure that the contents of this instruction manual provide all necessary information on the basic operation of the instrument and are correct. However, if you find any missing information or errors on the information described in this manual, please advise it to your JEOL service office.
- In no event shall JEOL be liable for any direct, indirect, special, incidental or consequential damages, or any other damages of any kind, including but not limited to loss of use, loss of profits, or loss of data arising out of or in any way connected with the use of the information contained in this manual or the software described in this manual. Some countries do not allow the exclusion or limitation of incidental or consequential damages, so the above may not apply to you.

- In some cases, this instrument, the software, and the instruction manual are controlled under the “Foreign Exchange and Foreign Trade Control Law” of Japan in compliance with international security export control. If you intend to export any of these items, please consult JEOL. Procedures are required to obtain the export license from Japan’s government.

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MANUFACTURER

JEOL Ltd. 1-2, Musashino 3-chome, Akishima, Tokyo 196-8558 Japan
Telephone: 81-42-543-1111 Facsimile: 81-42-546-3353 URL: http://www.jeol.co.jp

Note: For servicing and inquiries, please contact your JEOL service office.

NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

– CAUTION – :	Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
✍ :	Additional points to remember regarding the operation.
☞ :	A reference to another section, chapter or manual.
1, 2, 3 :	Numbers indicate a series of operations that achieve a task.
◆ :	A diamond indicates a single operation that achieves a task.
File:	The names of menus, commands, or parameters displayed on the screen are denoted with **bold** letters.
File–Exit :	Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–). For example, **File–Exit** means selecting **Exit** from the **File** menu.
Ctrl :	Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

Mouse pointer:	A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
Click:	To press and release the left mouse button.
Right-click:	To press and release the right mouse button.
Double-click:	To press and release the left mouse button twice quickly.
Drag:	To hold down the left mouse button while moving the mouse.

CONTENTS

SAFETY PRECAUTIONS

Before using the instrument, be sure to read this instruction manual, and upon fully understanding its contents, use the instrument properly. This section, “Safety Precautions”, contains important information about safety.

- **The safety indications and their meanings are as follows:**

⚠ **DANGER:**	An imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ **WARNING:**	A potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠ **CAUTION:**	A potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or a situation that could result in serious damage to facilities or acquired data.

- **Labels bearing the following symbols are attached to dangerous locations on the instrument. Do not touch any of these locations with your hands or anything else.**


Beware of electric shock


Beware of getting caught


Beware of fire


Beware of laser


Do not disassemble

Examples of symbols

- **Use the instrument properly within the scope of the purpose and usage described in its brochures and manuals.**
- **Never remove or disable protective parts, safety devices, or other safety features.**
- **Never make modifications that include disassembling the instrument or installing substitute parts.**
- **Never disconnect the grounding wire or move it from the prescribed position. Failure to follow this instruction could result in electric shock.**
- **When you dispose of the instrument or liquid or other waste, follow all applicable laws and regulations, and dispose of it in a proper manner without polluting the environment.**
- **Be sure to read the “Safety Precautions” section of the manuals for the accessories attached to or built into the instrument.**
- **If anything is unclear, please contact your local JEOL service office.**

1 OVERVIEW OF JBXFILER

Here is an overview of the Unix/OpenVMS version of the pattern-data preparation program (hereinafter referred to as JBXFILER).

1.1 Introduction to JBXFILER

JBXFILER is software for preparing data in various electron-beam lithography systems in the JEOL JBX series. In other words, it is the user interface for the data preparation system of the electron-beam lithography system. You can perform each function of the data preparation system through the operation screens of JBXFILER.

1.2 Features of JBXFILER

The features of JBXFILER are as follows.

■ Format compatibility

- JBXFILER can handle the following data formats:
  JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, MAPJ52, JEOL52V2.1M, JEOL52V3.0, JEOL52V3.1, JEOL01, STREAM, JEOL02, PG3000, PG3600

  ✍ Only the Unix version of JBXFILER can handle the dimmed formats ( ).

  ✍ Only Version 3.804 and earlier versions of JBXFILER can handle the enclosed formats ( ).

■ Executing the programs of the data preparation system

You can execute each program of the data preparation system from JBXFILER. These are the executable programs:

eblist	ebdisplay
ebupdate	pread
ebinformation	ebpsp
ebbackup	ebfc5152v30
ebrestore	ebblock
ebsrcmrg	ebj02to52v30
ebconv	patrdy
ebrphis	ebj02to51
ebsave	

■ Graphical user interface

JBXFILER employs Motif for the graphical user interface (GUI), providing users with an operating environment integrated into the Motif style.

■ Independence of the execution module

JBXFILER (GUI) and the JBX commands (the execution module) are independent of each other. By executing the JBX commands directly from the shell prompt, you can execute the functions of the data preparation system without using JBXFILER.

☞ The JBX commands are processed in the background for executing each function of JBXFILER. For information on the JBX commands, refer to Section 8.1, “Help Window for the JBX Commands.”

■ Network compatibility

For files in the JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1 format, JBXFILER can process data created by a network computer with a different byte order.

2 BASIC OPERATION OF JBXFILER

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

This chapter explains the basic techniques for using JBXFILER.

2.1 Basic Techniques for Entering Parameters

● Entering text

Fig. 1 Text-input field

Clicking in a text-input field enables you to enter text using the keyboard. In some cases, you can enter only certain allowed characters.

● Toggle button

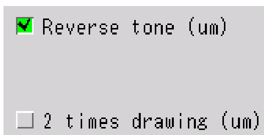


Fig. 2 Toggle buttons

Clicking on a toggle button changes its setting. When the button is green, it is selected.

● Radio buttons

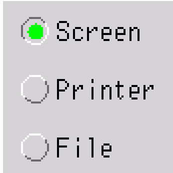


Fig. 3 Radio buttons

With radio buttons, you can select one from a group of options. You cannot select two or more options at a time. The button of the selected option is green.

● Scale bar

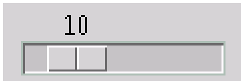


Fig. 4 Scale bar

You can specify the desired value by dragging the slider of a scale bar. The present value appears above the slider.

● Option menu

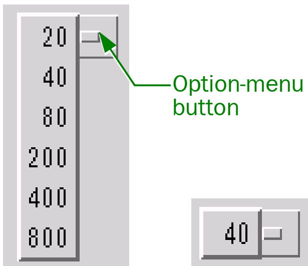


Fig. 5 Option menus

Clicking on an option-menu button opens its pull-down menu, listing the selectable items. You can select the desired item from the pull-down menu. As shown on the right in the figure, no pull-down menu appears if there is only one selectable item.

● Selecting from a list

You can select an item from the list in a selection area. Sometimes you can select multiple items.

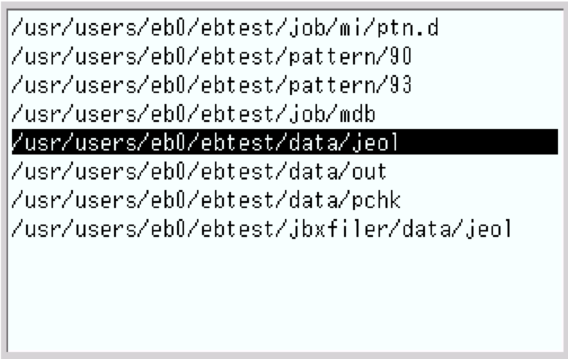


Fig. 6 Selection area

● Specifying a device or directory

The area where you can specify a device or directory consists of a text-input field and a button. Clicking on the **Select** button displays a list of devices for your selection.

Fig. 7 Output device specification area

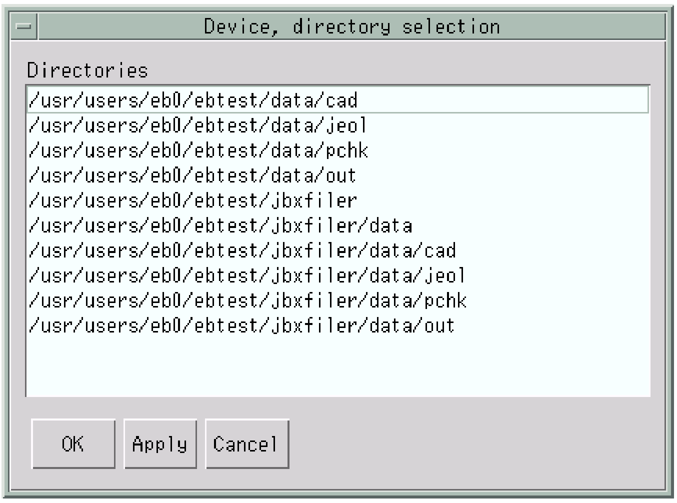


Fig. 8 Device, directory selection window

3 BASIC CONFIGURATION OF JBXFILER

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/・・・
Example: usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/・・・
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN・・・]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

This chapter explains the basic operation of JBXFILER.

3.1 Starting JBXFILER

◆ To start the filer, enter the following command after the prompt ($):

$ jbxfiler Enter

Upon starting, the following title window appears. Although this window will disappear after 3 seconds, you can also delete it by clicking on the **Close** button. After the title window disappears, the main window of JBXFILER appears.

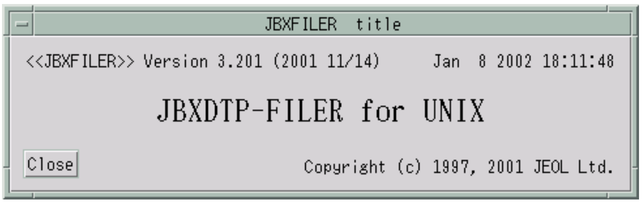


Fig. 9 JBXFILER title window

3.2 Configuration of the Main Window of JBXFILER

This section explains the components of the main window of JBXFILER.

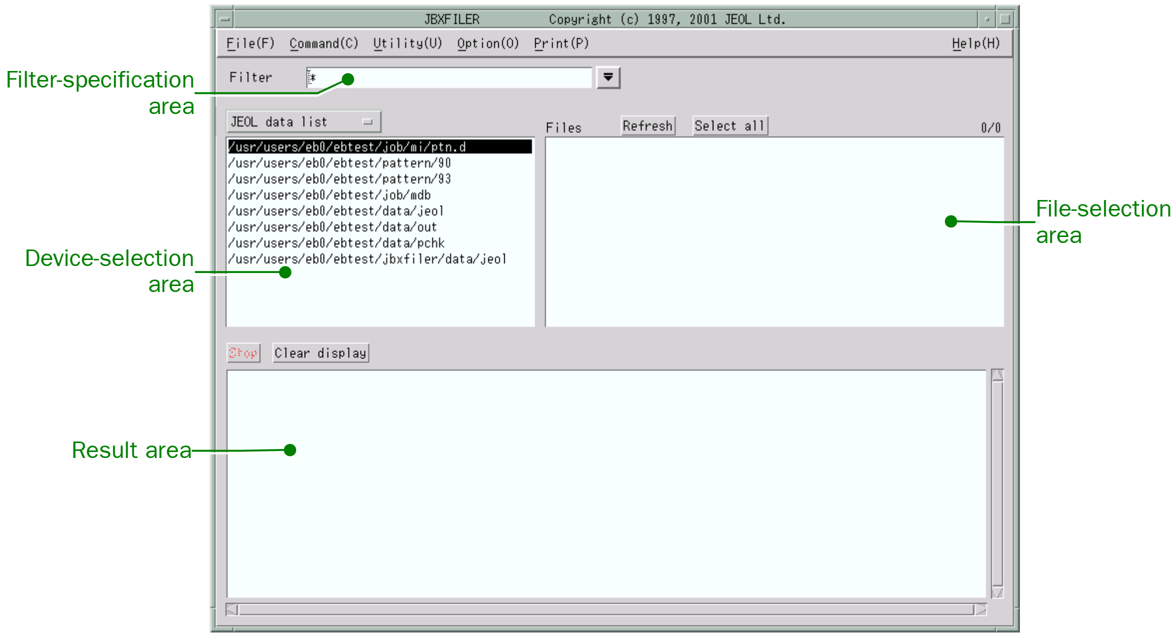


Fig. 10 Main window of JBXFILER

3.2.1 Menu bar

The menu bar of the main window includes the following six menus.

- File (F)
- Command (C)
- Utility (U)
- Option (O)
- Print (P)
- Help (H)

To display a menu, select the menu with the mouse or press the underlined shortcut key while holding down the Alt key.

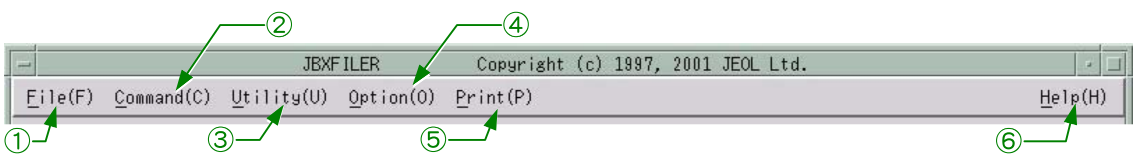


Fig. 11 Menu bar

① File (F) menu

This menu is used to manage files or to terminate JBXFILER.

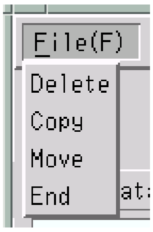


Fig. 12 File (F) menu

Delete: Deletes the files selected in the file-selection area. If no file is displayed, this item does not appear. Also, with no file selected, this item is not executable.

Copy: Copies the files selected in the file-selection area to the specified destination. If no file is displayed, this item does not appear. Also, with no file selected, this item is not executable.

Move: Moves the files selected in the file-selection area to the specified destination. If no file is displayed, this item does not appear. Also, with no file selected, this item is not executable.

End: Terminates JBXFILER.

② Command (C) menu

This menu is used to execute the JBX commands. Depending on the circumstances, either one of the following two pull-down menus will appear.

• When only a list of devices is being displayed

✍ Only the OpenVMS version of JBXFILER displays this menu.

Fig. 13 Command (C) menu (1)

List (PREAD): Lists the data processed by PREAD.

Rename (PREAD): Renames the data processed by PREAD.

Delete (PREAD): Deletes the data processed by PREAD.

• When a list of files is being displayed

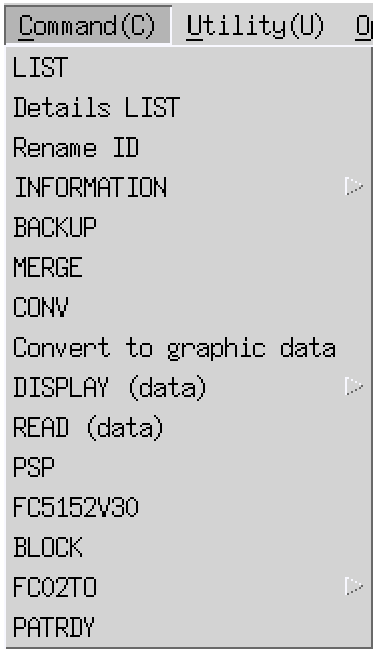


Fig. 14 Command (C) menu (2)

LIST: Displays information on the selected files.

Details LIST: Displays detailed information on the selected files.

Rename ID: Changes the ID names of the selected files.

INFORMATION: Displays detailed information, such as the dump list, library list, and data count, on the contents of each selected file.

BACKUP: Copies the selected files to a tape device.

MERGE: Merges CAD data.

CONV: Converts CAD data to JEOL formats.

Convert to graphic data:
Converts JEOL-format data to graphic data.

DISPLAY (data): Displays graphic data.

READ (data): Reads JEOL-format data and converts them to the scanner format data.

PSP: Separates patterns in a JEOL format data.

FC5152V30: Converts a JEOL51 data file to a JEOL52V3.0 data file.

BLOCK: Combines 2 to 9 JEOL data files.

FC02TO: Converts JEOL02 data files to JEOL52V3.0 or JEOL51 data files.

PATRDY Converts JEOL format data that contains map libraries to JEOL52 (V1.1/V2.1/V3.0) format data that does not contain map libraries.

✍ In the menu that appears when a list of files is being displayed, the following items are optional, so they might not appear:
CONV, FC5152V30, Convert to graphic data, BLOCK, DISPLAY (data), FC02TO, READ (data), PATRDY, PSP.

- When a list of magnetic-tape devices is being displayed

Fig. 15 Command (C) menu (3)

RESTORE: Copies the files from a tape to the specified directory.

③ Utility (U) menu

This menu is used to execute special functions.

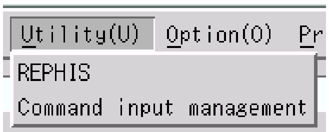


Fig. 16 Utility (U) menu

REPHIS: Reports the result of data conversion (CONV) (optional). Without a license for data conversion (CONV), this item does not appear.

Command input management:
Executes commands from JBXFILER.

④ Option (O) menu

This menu is used to execute the additional functions of JBXFILER.

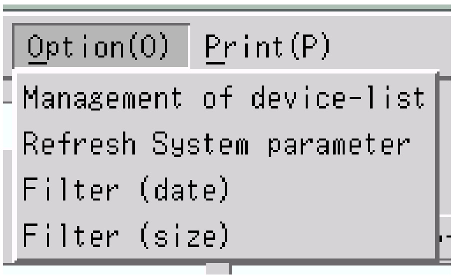


Fig. 17 Option (O) menu

Management of device-list: Lets you edit the device lists.

Device lists are divided into these three types:

- JEOL data list
  Specify the directory in which to save JEOL data.
- CAD data list
  Specify the directory in which to save CAD data.
- Magnetic-tape list
  Specify the magnetic-tape directory.

Refresh System parameter: Rereads the system-parameter file (ebsys.prm).

☞ For information on the contents of ebsys.prm, see “Appendix B: File Specifications” in “JBX Command Users Guide.”

Filter (date): Sets the file-selection condition (date).

Filter (size): Sets the file-selection condition (size).

⑤ Print (P) menu

This menu is used to print the contents of the result area. You can select from the following three methods of printing.

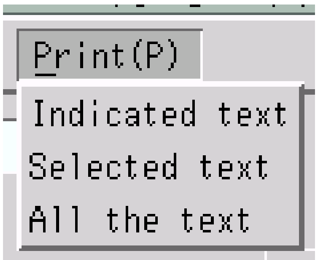


Fig. 18 Print (P) menu

Indicated text:	Prints the text on the result area.
Selected text:	Prints the text you specified in the result area.
All the text:	Prints all the text in the result area.

⑥ Help (H) menu

This menu is used to display information on the JBX commands.

Fig. 19 Help (H) menu

Version:	Displays the version information on JBXFILER and JBX commands.
JBX command:	Displays help information on the JBX commands using Netscape.

3.2.2 Filter-specification area

The filter-specification area consists of the filter-input field and pull-down menu for filter selection. Here, you can set the filter to use as the file-selection condition. You can use UNIX wildcards (?, *) and OpenVMS wildcards (%, *) in the filter, but you cannot use a format that specifies multiple conditions (that is, you cannot divide the filter by using commas). You can also select a filter from the pull-down menu. You can define the filters to display on the pull-down menu in the resource file.

☞ For details, refer to Appendix 2, “JBXFILER Resource File.”

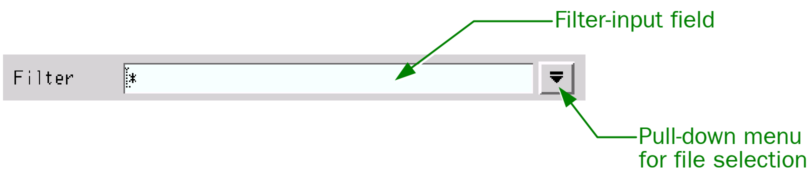


Fig. 20 Filter-specification area

3.2.3 Device-selection area

The device selection area consists of the option menu for selecting a device list, and a list of devices. In this example, the **JEOL data list** is selected as the device list.

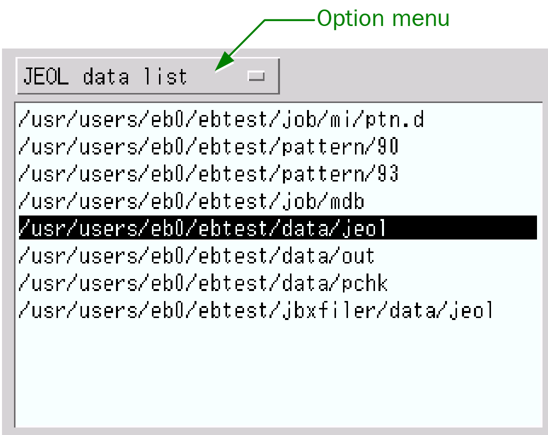


Fig. 21 Device-selection area

The option menu displays the following three device lists.

Device list	Information to be displayed
JEOL data list	Contents of disklist.
CAD data list	Contents of cdisklist.
Magnetic-tape list	Contents of mtlist.

The above files (dsiklist, cdisklist, mlist) store the contents in this area.

3.2.4 File-selection area

The file-selection area consists of the **Refresh** button, **Select all** button, file count, and a list of files.

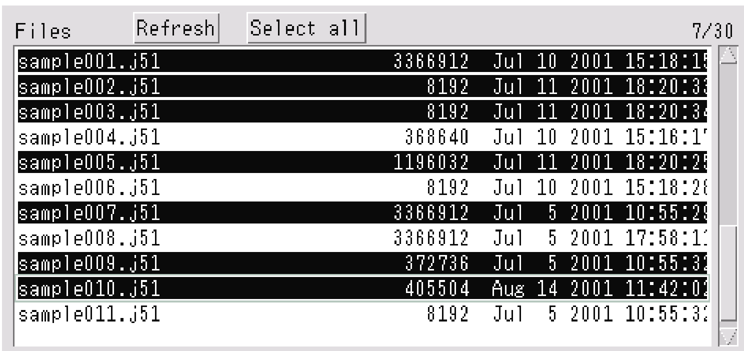


Fig. 22 File-selection area

Refresh: Displays the latest list of files.
Select all: Selects all the files in the list.

File count: Appears near the top right of the list. In the example in the figure, the file count is 7/30; the denominator 30 indicates the total number of files, and the numerator 7 indicates the total number of selected files.

3.2.5 Result area

The result area consists of the **Stop** button, **Clear display** button, and display field.

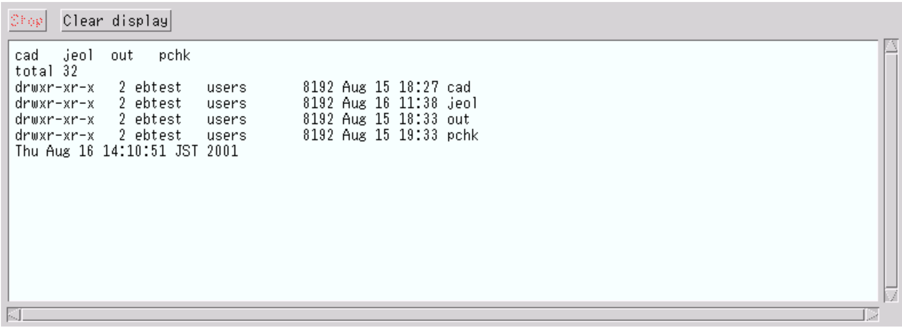


Fig. 23 Result area

Stop: Freezes the display field. As a result, the process being executed stops. This button is in effect during command execution. Normally, this button is dimmed.

Clear display: Clears the display field.

Display field: Displays the results of the execution of commands or JBXFILER. You can review anything that overflows out of the displayable area by using the vertical and horizontal scroll bars.

3.3 Terminating JBXFILER

To terminate JBXFILER, follow this procedure:

◆ Select **File(F)–End**.

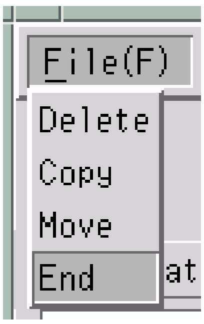


Fig. 24 Selecting End from the File (F) menu

Selecting **End** displays the End program confirmation dialog box.

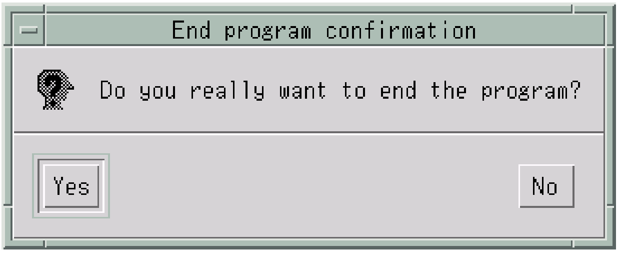


Fig. 25 End program confirmation dialog box

Yes: Terminates JBXFILER.
No: Returns to the main window of JBXFILER.

3.4 Selecting a Device

This section explains how to select a device. You can select the desired device by clicking on it in the device-selection area.

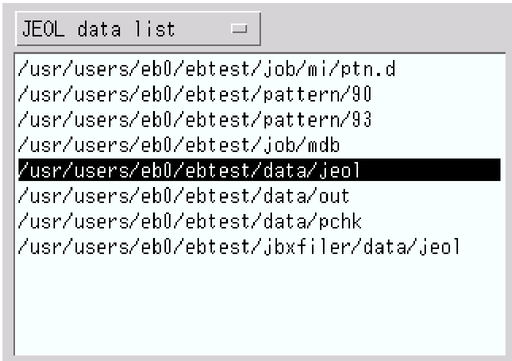


Fig. 26 Device-selection area

3.4.1 Selecting a device list

As shown in the following figure, three device lists exist on the option menu. Select the desired list from the option menu to display the contents of the list in the device-selection area.

Fig. 27 Device lists

JEOL data list: Lists the contents of disklist, which stores JEOL data.
CAD data list: Lists the contents of cdisklist, which stores CAD data.
Magnetic tape list: Lists the contents of mtlist, which is the magnetic-tape directory.

3.4.2 Selecting a device

You can use either one of the following two methods to select a device.

- Single clicking on the desired device
  This method selects the device and initializes the file-selection area. To list the files located on the selected device, click on the Refresh button of the file-selection area, or enter a selection condition in the filter-input field and then press the Return key.
- Double-clicking on the desired device
  This lists the files located on the selected device. You can use this only when you have selected JEOL data list or CAD data list. This method is unavailable if you have selected Magnetic tape list.

3.5 Selecting Files

This section explains how to select files from the list in the file-selection area. After you select a device by double clicking on it, the list of the files on the selected device appears. Here, you can select or deselect files, or redisplay the list.

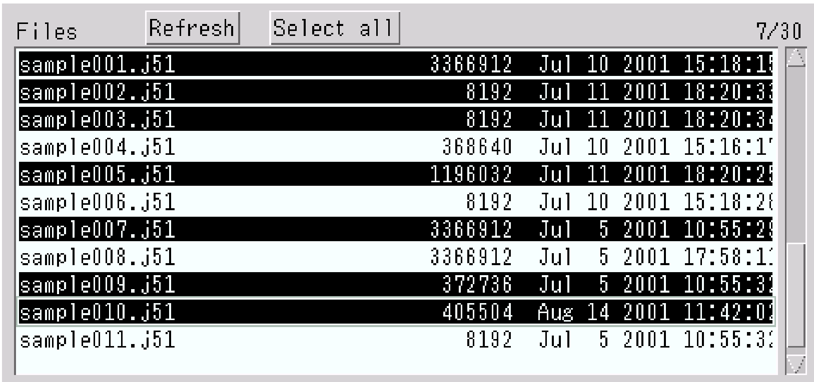


Fig. 28 File-selection area

3.5.1 Selecting and deselecting files

You can select or deselect files by clicking on them. If you have already selected some files and want to select one more, click on it while holding down the Ctrl key. Similarly, if you want to deselect one of the selected files, click on it while holding down the Ctrl key. After you click on a file, clicking on another file while holding down the Shift key selects all the files in between the two files.

3.5.2 Selecting all the files

To select all the files in the list, click on **Select all** button.

3.5.3 Redisplaying the files

To update the list, click on **Refresh** button. Use this button after you delete or move files.

3.6 Password

✍ Only the OpenVMS version of JBXFILER displays this window.
If the selected device has node specification and no password is written, the following password-entry window appears.

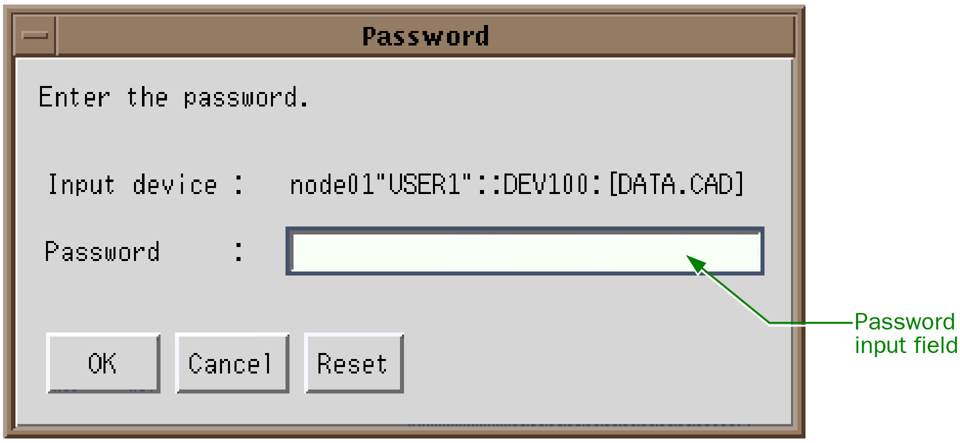


Fig. 29 Password entry

OK: Click this button after entering a password or no password exists. Clicking it is the same as pressing the Return key in the password-input field.

Cancel: Clicking this button cancels password entry and returns you to the previous window.

Reset: Clicking this button clears the password-input field. Use this button if you enter a wrong password.

In the password-input field, asterisks (*) appear instead of what you type. The system checks to see if the password is correct; if you enter a wrong password three times in a row, the system assumes that your access is unauthorized even if you enter a correct password afterward. This means that you cannot gain access for a given period of time determined by a system parameter at the network destination.

4 ADVANCED OPERATIONS OF JBXFILER

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

This chapter explains the advanced operations of JBXFILER。

4.1 Managing the Device Lists

You can change the contents of the device lists for the UNIX version (disklist, cdisklist, mtlist) and for the OpenVMS version (DISKLIST.PRM, CDISKLIST.PRM, MTLIST.PRM). In the main window of JBXFILER, selecting **Option(O)–Management of device-list** displays the Device-list-file management window. Selecting the list that you want to change from the option menu displays the selected list.

✍ The writing program uses the data processed by the data-read function (scanner-format data, PREAD-processed data), so these data are not managed by the device list displayed as a "JEOL data list" but by another method. Accordingly, when you use this function to register or modify a directory that contains the data processed by the data-read function, it is also necessary to register or modify the directory by using the setdtab command explained in Appendix 9.

✍ The setdtab command is available only in UNIX.

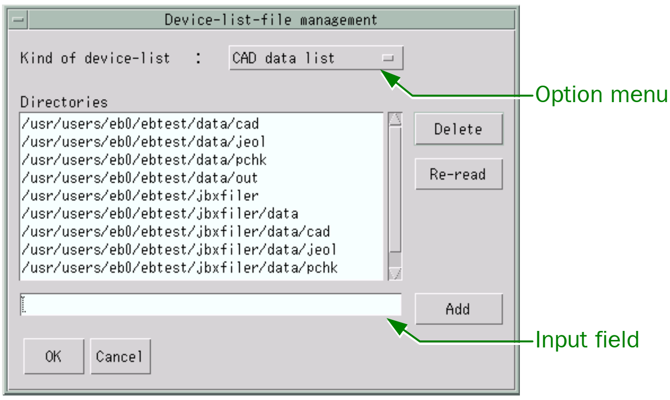


Fig. 30 Device-list-file management window

Add:	Appends the contents of the input field to the list.
Delete:	Deletes the lines that you have selected in the list.
Re-read:	Replaces the list being displayed with the one in the device-list file.
OK:	Replaces both the device list in the device-list file and that of JBXFILER with the one being displayed.
Cancel:	Cancels the changes you have made and returns to the main window of JBXFILER.

4.2 Rereading the System Parameters

JBXFILER can reread the system parameter file (ebsys.prm) after it starts. To do this, select **Option(O)–Refresh System parameter** in the main window of JBXFILER. This enables you to change the system parameters without restarting JBXFILER. The file is updated in the same way as when you restart JBXFILER. This operation does not display any messages.

4.3 Selecting a Filter

Entering a filter in the filter-input field in the main window of JBXFILER makes the file-selection area display only the files whose names pass through that filter. Also, in addition to directly entering a filter in the filter-input field by using the keyboard, you can also select a filter from the pull-down menu.

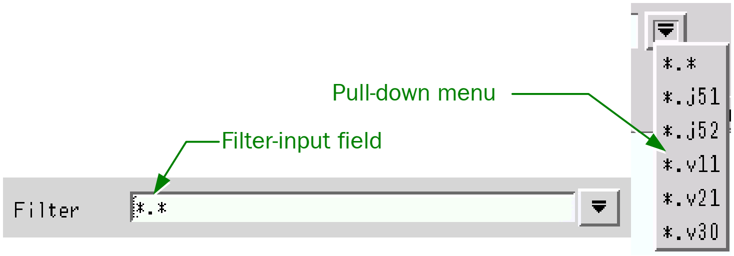


Fig. 31 Filter-specification area and its pull-down menu

☞ You can specify the items from the pull-down menu in the resource file. For details, refer to Appendix 2, “JBXFILER Resource File.”

4.4 Setting the File-Selection Condition

Besides using the filter-specification area, you can specify file-selection conditions regarding creation date or file size.

4.4.1 File-selection condition regarding creation date

You can specify a file-selection condition regarding creation date.

◆ In the main window of JBXFILER, select **Option(O)–Filter (date)**.
This displays the Filter (date) window.

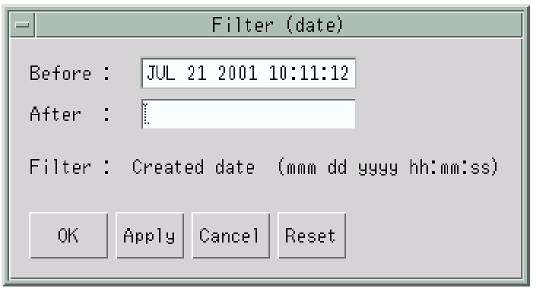


Fig. 32 Filter (date) window

Before: Only the data created before the date you enter here will appear in the file-selection area. The format for the UNIX version is “mmm dd yyyy hh:mm:ss” and the format for the OpenVMS version is “dd-mmm-yyyy hh:mm:ss.ss.”
Example: NOV 04 2005 16:30:10 (UNIX version)
Example: 04-NOV-2005 16:30:10.00 (OpenVMS version)
If you also enter a value in the After box, the file-selection area displays the data created after the date in the After box and before the date in the Before box.

After: Only the data created after the date you enter here will appear in the file-selection area. The format is the same as with the Before box. If you also enter a value in the Before box, the file-selection area displays the data created after the date in the After box and before the date in the Before box.

OK: Applies the condition that you have set and closes this window.

Apply: Applies the condition that you have set without closing this window.

Cancel: Cancels the condition that you have entered and closes this window.

Reset: Cancels the condition that you have entered and displays the previous condition.

4.4.2 File-selection condition regarding file size

You can specify a file-selection condition regarding file size.

◆ In the main window of JBXFILER, select **Option(O)–Filter (size)**.

This displays the Filter (size) window.

✍ In the OpenVMS version, you can select the allocation size.

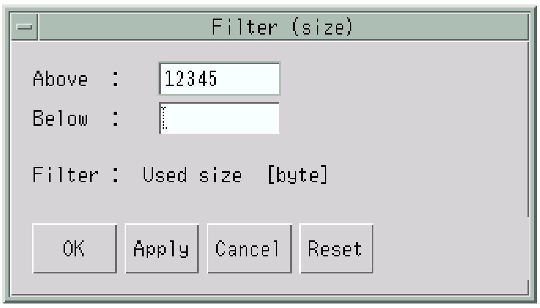


Fig. 33 Filter (size) window

Above: Only the data whose sizes are larger than or equal to the size you enter here will appear in the file-selection area. If you also enter a value in the Below box, the file-selection area displays the data whose sizes are larger than or equal to the size in the Above box and less than or equal to the size in the Below box.

Below: Only the data whose sizes are less than or equal to the size you enter here will appear in the file-selection area. If you also enter a value in the Above box, the file-selection area displays the data whose sizes are larger than or equal to the size in the Above box and less than or equal to the size in the Below box.

OK: Applies the condition that you have set and closes this window.

Apply: Applies the condition that you have set without closing this window.
Cancel: Cancels the condition that you have entered and closes this window.
Reset: Cancels the condition that you have entered and displays the previous condition.

4.5 Terminating Command Execution

During command execution, the **Stop** button is active as shown in the following figure (normally, the button is dimmed). Clicking on this button while it is active terminates command execution and freezes the display field in the result area.

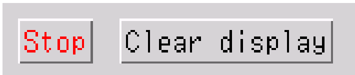


Fig. 34 Buttons in the result area

4.6 Operating in the Result Area

The result area displays the results of the execution of the functions of JBXFILER. This area always displays the latest line of the results. By using the scroll bar, you can review the previously displayed results. JBXFILER saves only the specified number of lines of the results. The resource file enables you to set the number of lines to save.

For information about how to modify the resource file, see Appendix 2, "JBXFILER Resource File."

The default number of lines is 5,000. If the number of lines of the results exceeds the specified number, the program discards the excess lines one by one, from the oldest line.

You can print the contents in the result area. For information on printing, refer to Chapter 7, "Printing."

4.7 Network

This function is available only in the OpenVMS version of JBXFILER.

If the device list contains a device that requires the NFS mount, the filer automatically performs the NFS mount of that device. To perform this function, the program writes the information required for the mount in the network-related file (see Appendix 6, "Network Setup File"). The program periodically checks the node name contained in that file to see if it is connectable; if it is not connectable, a message appears. The filer resource file enables you to change the time interval between connection checks (see Appendix 2, "JBXFILER Resource File).

NFS is short for Network File System.

5 OPERATING ON FILES

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

JBXFILER enables you to delete, copy, or move the files selected in the file-selection area. Here is the procedure:

1. Select the files to operate on.
Select files in the file-selection area.

2. Select an operation.
Select an operation from the File menu.

3. Setting parameters.
Set the parameters required for the operation. You do not have to set any parameter to delete files.

4. Start the operation.
Clicking on the **OK** button displays the following Confirmation window. In this window, clicking on the **Yes** button starts the operation, whereas clicking on the **No** button cancels the operation.

Fig. 35 Confirmation window

5.1 Deleting Files

You can delete the files selected in the file-selection area as follows.

1. Select the files to delete, and select **File(F)–Delete**.
The Confirmation window appears.

2. Click on **Yes**.
This deletes the selected files. The result of this operation appears in the result area.

5.2 Copying Files

You can copy the files selected in the file-selection area to the specified location as follows:

1. Select the files to copy, and select **File(F)–Copy**.

   The following COPY parameter window appears. The selection area lists the devices selected in JBXFILER.

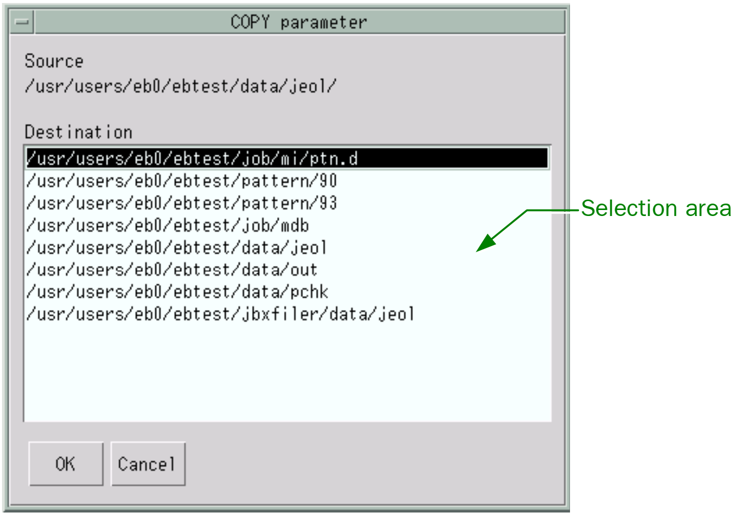


Fig. 36 COPY parameter window

2. In the selection area, select the destination device.
3. Click on the **OK** button to complete the parameter setting; the Confirmation window appears. Alternatively, click on the **Cancel** button to stop the parameter setting and return to the main window of JBXFILER.
4. Click on **Yes** in the Confirmation window.

   This copies the selected files. The result of this operation appears in the result area.

5.3 Moving Files

You can move the files selected in the file-selection area to the specified location as follows.

1. Select the files to move, and select **File(F)–Move**.

The following MOVE parameter window appears. The selection area lists the devices selected in JBXFILER.

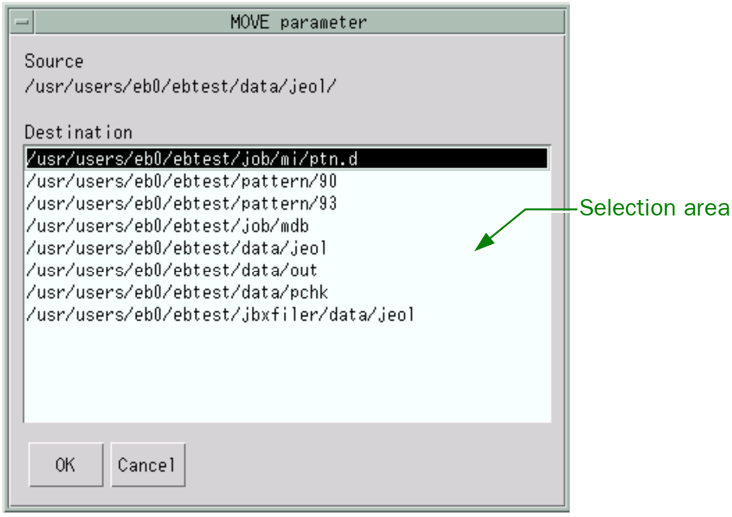


Fig. 37 MOVE parameter window

2. In the selection area, select the destination device.

3. Click on the **OK** button to complete the parameter setting; the Confirmation window appears. Alternatively, click on the **Cancel** button to stop the parameter setting and return to the main window of JBXFILER.

4. Click on **Yes** in the Confirmation window.

This moves the selected files. The result of this operation appears in the result area.

6 OPERATING THE UTILITY FUNCTIONS

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

This chapter explains the utility functions.

6.1 Statistical Processing of the Data Conversion History File Function (Optional)

Here is the procedure:

1. Select **Utility(U)–REPHIS**.

✍ This function appears on the Utility menu only if you have a license for data conversion (CONV).

The following parameter-setting window appears. The selection area lists the history files.

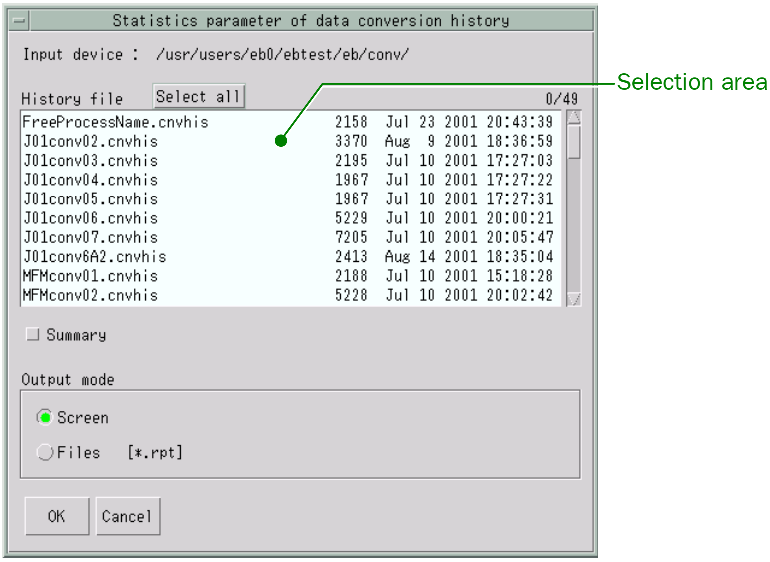


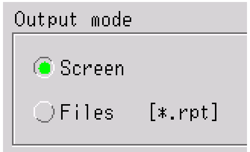
Fig. 38 Statistics parameter of data conversion history window

2. Select files from the selection area.

You can select files, deselect files, or select all the files just as you can do in the file-selection area in the main window of JBXFILER.

3. Select the destination to which to output the result of the execution.

Click on either the **Screen** or the **Files** radio button. Selecting **Screen** outputs the result to the result area, whereas selecting **Files** outputs the result to a file.



4. If you need the summary information, select the **Summary** button; the button turns green.

5. Click on the **OK** button to start execution.

Alternatively, click on the **Cancel** button to cancel the process and return to the main window of JBXFILER.

6.2 Command Input

1. In the main window of JBXFILER, select **Utility(U)–Command input management**.

The Command window appears. You can execute commands from this window. The program keeps a log of the commands executed from this window. From the log, you can recall a command and use it. Also, the JBX command that JBXFILER executed the last time is recorded here. You can modify the command that you have recalled.

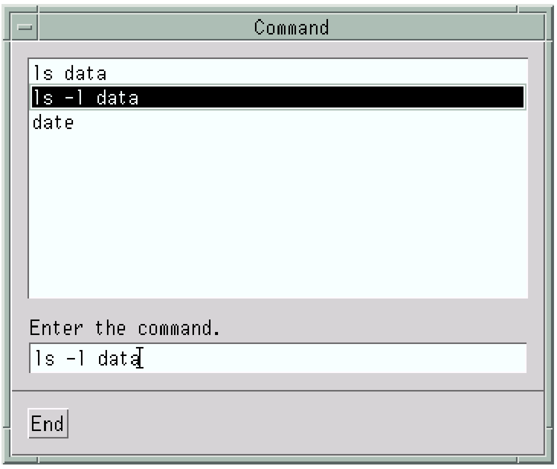


Fig. 39 Command window

As shown in this figure, the window lists the logged commands. If there are too many commands to display in the window, a scroll bar appears.

2. Select a command by using the arrow keys (↑/↓), or directly select it with the left mouse button.

3. Clicking on the **End** button completes the command input and returns you to the main window of JBXFILER.

7 PRINTING

You can print the contents of the result area. The program uses the printer specified in the resource file. If no printer is specified in the resource file, the program uses the default printer.

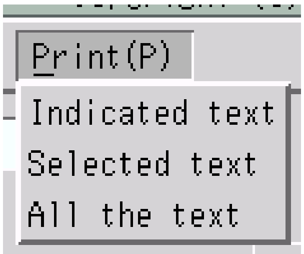


Fig. 40 Print (P) menu

7.1 Printing the Displayed Text

◆ Select **Print(P)–Indicated text**.
This outputs only the information displayed in the result area to the printer.

7.2 Printing the Specified Text

◆ Select **Print(P)–Selected text**.
This outputs the part you have specified in the result area to the printer.

7.3 Printing All the Text

◆ Select **Print(P)–All the text**.
This outputs all the information in the result area to the printer.

8 HELP

You can display information on the JBX commands.

8.1 Help Window for the JBX Commands

● UNIX version of JBXFILER

◆ Select **Help(H)–JBX command.**

This starts Netscape and opens the following help window for the JBX commands.

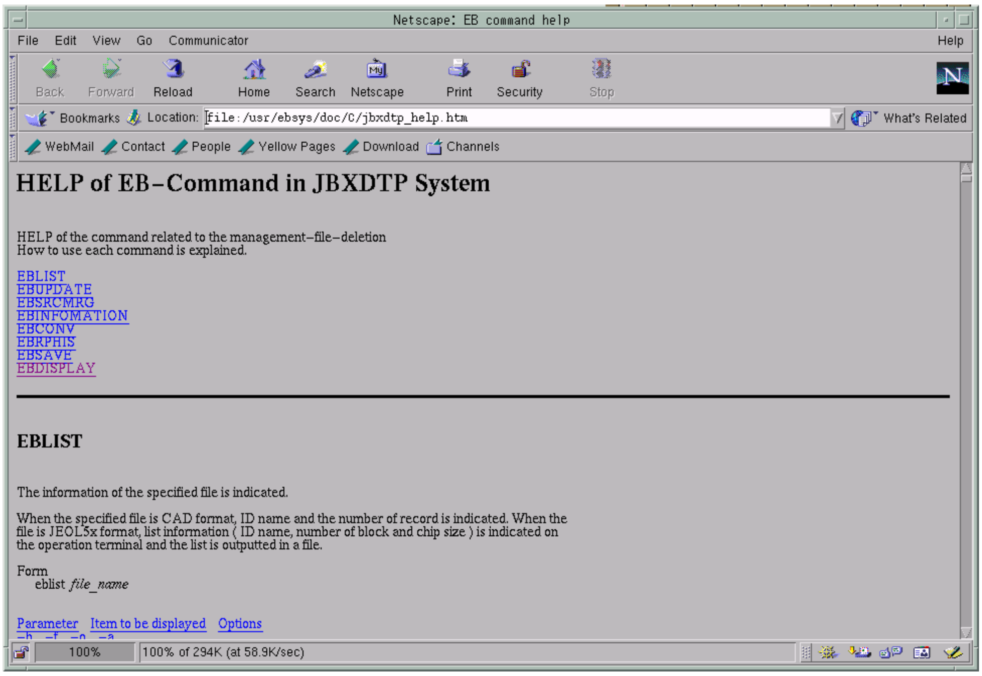


Fig. 41 Help window (UNIX version)

You can operate the help window as you browse a web site. By clicking on an underlined keyword, you can jump to the detailed information on the selected keyword.

Each detailed information window displays **back** and **goto Top** on the right. Clicking on **goto Top** returns you to the top of the help window (Figure 41 Help window (UNIX version)). Clicking on **back** returns you to the top of the JBX command to which the keyword belongs.

To close the help window, select **File– Exit** in the help window.

● OpenVMS version of JBXFILER

◆ Select **Help(H)–JBX command.**

This starts Help in the DEC Windows, displaying the following Help window for the JBX commands.

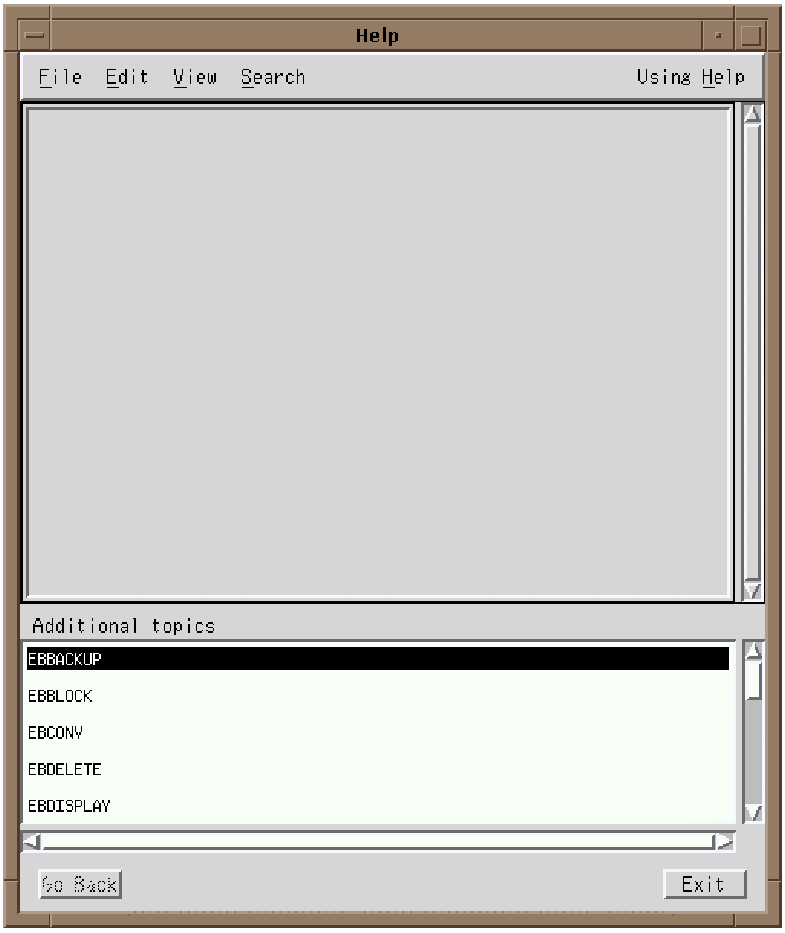


Fig. 42 Help window (OpenVMS version)

Go Back: Returns you to the previous screen.
Exit: Exits Help and returns you to the filer.

This window is exactly the same as that in the DEC Windows. Double-clicking on an item in the Additional topics box displays its details. By clicking on **Using Help**, you can display details on how to use Help.

8.2 Version Information on the JBX Commands

◆ Select **Help(H)–Version**.

This displays the version information on both JBXFILER and the JBX commands that JBXFILER uses.

Fig. 43 JBX command version list window

If a command is not installed, "NONE" is indicated in lieu of its version number. Also, if the acquisition of a command version number fails, dotted lines (------) appear instead.

9 LISTING

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern

Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern

OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

This chapter explains the method of operating the list function and the detailed list function.

9.1 LIST Function

◆ Select desired files in the file-selection area, and select **Command(C)–LIST**.

This displays the results of the execution of the list function in the result area. If the selected files are in the JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1 format, the program creates bspoolhhmmss.dat in the directory where the selected file is located. The letters hhmmss represent the execution time of the LIST function, namely hh: hours, mm: minutes, ss: seconds.

You can also execute the LIST function by double-clicking on a file; this creates bspool.dat.

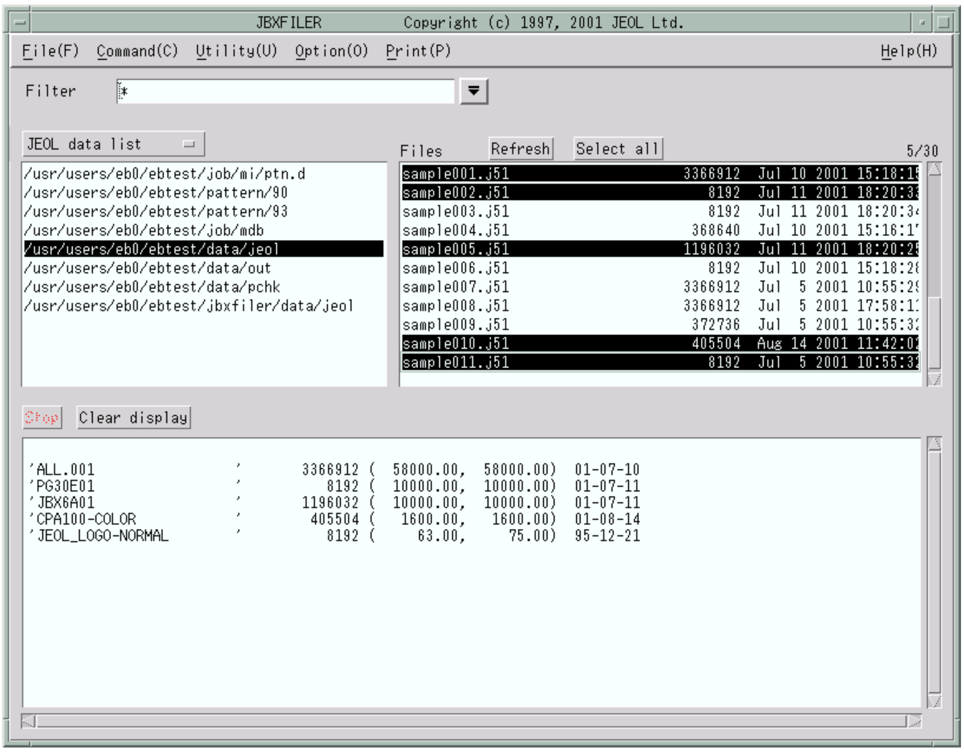


Fig. 44 JBXFILER window, after the execution of the LIST function

9.2 Detailed LIST Function

◆ Select the desired file in the file-selection area, and select **Command(C)–Details LIST**.

This displays the results of the execution of the Detailed LIST function in the result area. The program creates fspoolhhmmss.dat in the directory where the selected file is located. The letters hhmmss represent the execution time of the Detailed LIST function, namely hh: hours, mm: minutes, ss: seconds.

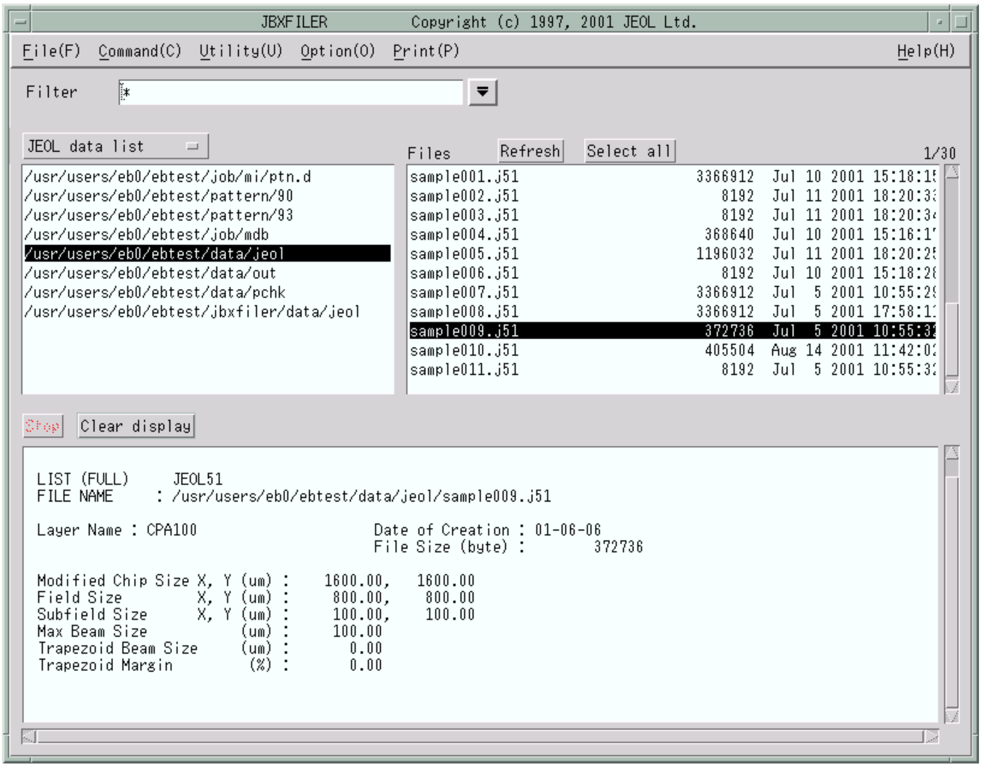


Fig. 45 JBXFILER window, after the execution of the Detailed LIST function

10 RENAMING

The renaming function lets you change the ID name of the specified CAD or JEOL data file. This chapter explains the method of operating the rename function.

10.1 Setting Parameters

1. Select desired files in the file-selection area, and select **Command(C)–Rename ID**.

This displays the following parameter-setting window where you can specify new ID names. The selection area lists the selected files and their present ID names.

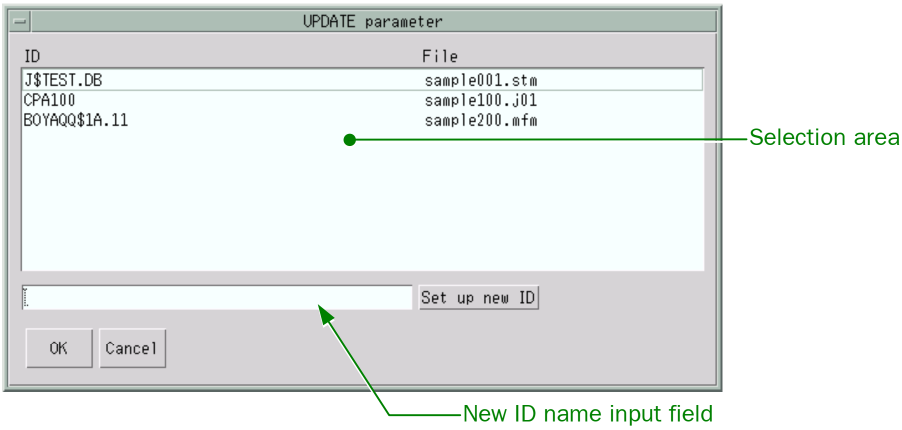


Fig. 46 UPDATE parameter window

New ID name input field:	Lets you enter a new ID name.
Set up new ID:	Sets the new ID name.
OK:	Starts renaming.
Cancel:	Cancels renaming and returns you to the JBXFILER main window.

2. In the list, select (click on) the file you want to rename.

This moves the focus to the new ID name input field. The name of the selected file appears in the new ID name input field.

3. Enter a new ID name, and click on the **Set up new ID** button.

✍ Instead of using the **Set up new ID** button, you can also set the new ID name by entering the new ID name and then pressing Enter.

10.2 Command Execution

After you set the parameters, clicking on the **OK** button changes the name.

☞ The results output to the result area are the same as with a JBX command. Refer to "JBX Command User's Guide."

Clicking on the **Cancel** button cancels the process and returns you to the main window of JBXFILER.

11 INFORMATION PROCESSING

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

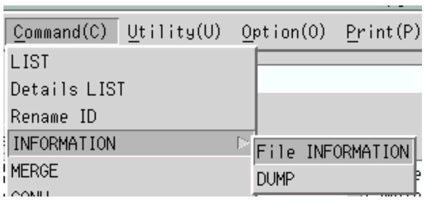
This chapter explains the method of operating the information-processing function.

11.1 File-Information Function

The file-information function outputs the information on the specified STREAM or JEOL data file. For STREAM format, it outputs the library and structure information; for JEOL data, it outputs the shot count or the contents of the data.

◆ Select the desired files in the file-selection area, and select **Command(C)–INFORMATION–File INFORMATION.**

✍ This function is available only for STREAM, JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, and JEOL52V3.1 files.



11.1.1 Setting parameters

The following parameter-setting window appears, so set the parameters.

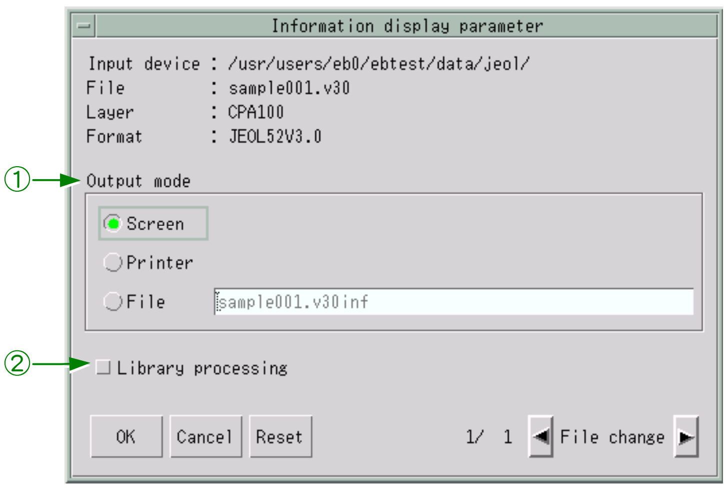


Fig. 47 Information display parameter window

① Output mode

Here you can specify the destination to which to output the result of the execution of the file-information function.

Screen: Outputs to the result area.
Printer: Outputs to the printer.
File: Outputs to a file.

② Library processing

This button lets you specify whether or not to perform library processing.

✍ This button appears only if the file format is JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1.

• Nesting information

You can specify whether or not to process the reference structures of SREF and AREF of STREAM data. This button appears only for STREAM format (in Figure 47, the window displays “Nesting information” in place of “Library processing”).

☞ For information on the relationships between the formats and files, see the following table.

• Format names and file names

Format	Place of creation	File name
STREAM	Location of the input file	input_filename.stminf
JEOL51		input_filename.J51inf
JEOL52		input_filename.j52inf
JEOL52V1.1		input_filename.v11inf
JEOL52V2.1		input_filename.v21inf
JEOL52V3.0		input_filename.v30inf
JEOL52V3.1		input_filename.v31inf

11.1.2 Command execution

OK: After you set the parameters, clicking on the **OK** button executes the file-information function.

☞ The results output to the result area are the same as with a JBX command. See “JBX Command User’s Guide.”

Cancel: Cancels the process and returns you to the main window of JBXFILER.

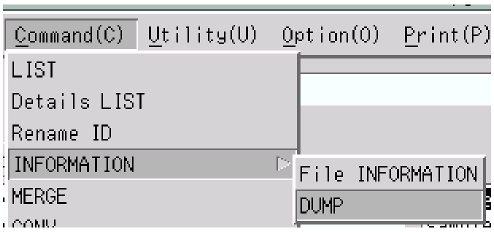
Reset: Resets the parameters to the settings that were current when you clicked on the **File change** button. If you have not clicked on the File change buttons, the default settings are restored.

11.2 DUMP Function

The dump function dumps the contents of the specified JEOL data file.

◆ Select desired files in the file-selection area, and select **Command(C)–INFORMATION–DUMP.**

✍ This function is available only for JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, and JEOL52V3.1 files.



11.2.1 Setting parameters

The following parameter-setting window appears, so set the parameters.

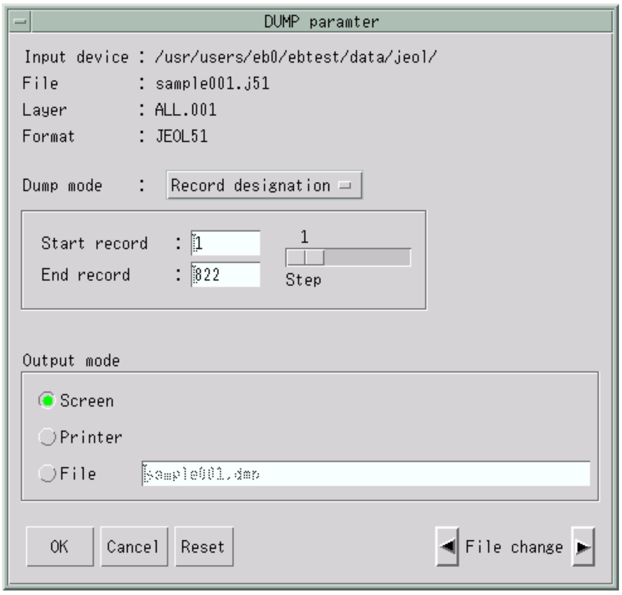


Fig. 48 DUMP parameter window

■ Dump mode

Here you can set the dump mode.

Fig. 49 Dump mode option menu

- **Record designation**

  You can set the range to dump with records. The default settings are as follows:

  Start record: 1

  End record: Final record in the file

  Step: 1

  ✍ You cannot specify a number larger than the final record number.

Fig. 50 Record designation area

- **Range designation**

  If you select **Range designation** as the dump mode:

  You can set the upper-left and lower-right coordinates of the range to dump. The unit is μm.

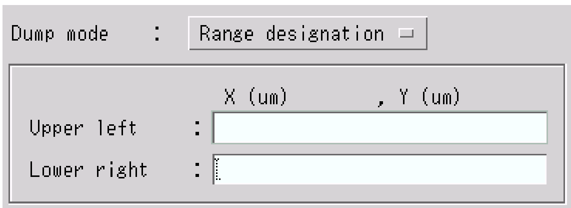


Fig. 51 Range designation area

- **Field designation**

  If you select **Field designation** as the dump mode:

  You can specify the range to dump as field coordinates. The unit is μm.

Fig. 52 Field designation area

■ Output mode

Here you can specify the destination to which to output the result of the execution of the dump function.

Screen: Outputs to the result area.
Printer: Outputs to the printer.
File: Outputs to a file. The default file name is the input file name with the file name extension .dmp.

11.2.2 Command execution

OK: After you set the parameters, clicking on the **OK** button executes the DUMP function.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked on the **File change** button. If you have not clicked on the File change buttons, the default settings are restored.

12 BACKING UP

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern

Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern

OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The backup function copies the specified CAD or JEOL data file to a magnetic tape. This chapter explains the method of operating the backup function.

◆ Select desired files in the file-selection area, and select **Command(C)–BACKUP**.

All the selected files must be in the same format. Also, the total number of the selected files must be 99 or less.

12.1 Setting Parameters

The following parameter-setting window appears, so set the format-dependent parameters.

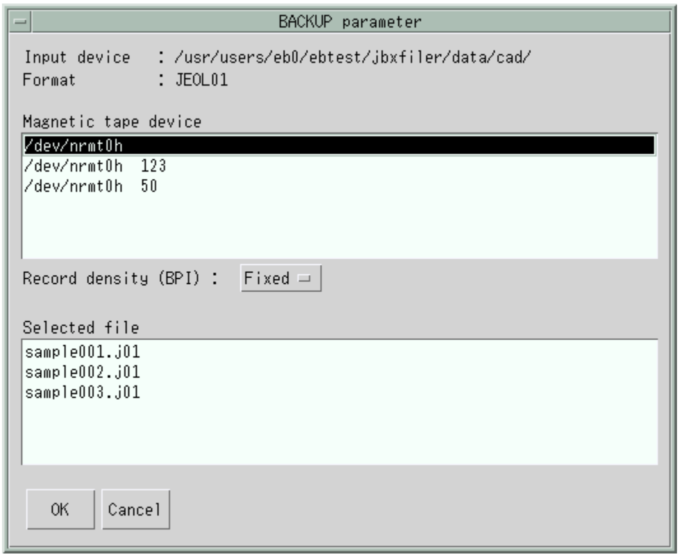


Fig. 53 BACKUP parameter window

● Common parameters

- Magnetic tape device

  You can select the magnetic tape device to use for backup.

- Record density (BPI)

  You can select **Fixed**, **800**, **1600**, or **6250** for the recording density. If a DAT or any other device having a fixed record density is selected, select **Fixed**.

● JEOL51 parameters

Here are the parameters that appear when backing up JEOL51 format data.

✍ These parameters do not appear when backing up data in a format other than JEOL51 format.

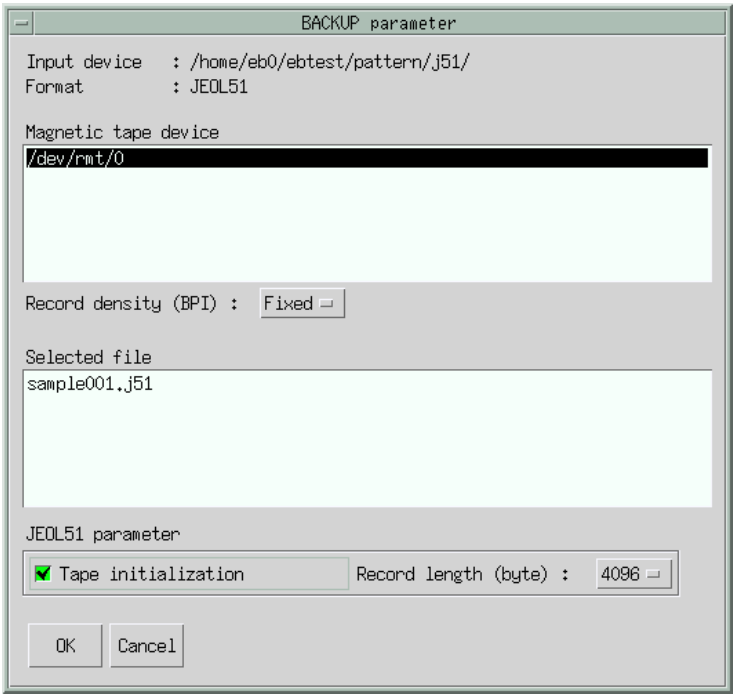


Fig. 54 BACKUP parameter window (with JEOL51 format data specified)

- Tape initialization

  You can specify whether or not to initialize the tape before backup. Selecting (clicking on) the Tape initialization button will initialize the tape (the button turns green).

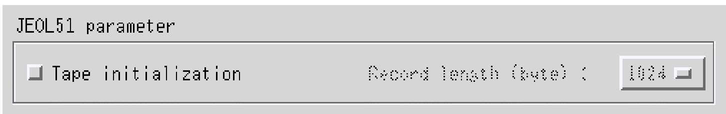


Fig. 55 JEOL51 parameter specification area (with Tape initialization not selected)

- Record length

  The option menu lets you select **1024**, **2048**, or **4096** bytes for the record length.

  ✍ The record length can be specified only if Tape initialization is selected.

Fig. 56 JEOL51 parameter specification area (with Tape initialization selected)

● JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, JEOL52V3.1 parameters

Here are the parameters that appear when backing up JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1 format data.

✍ These parameters do not appear when backing up data in a format other than JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1 format.

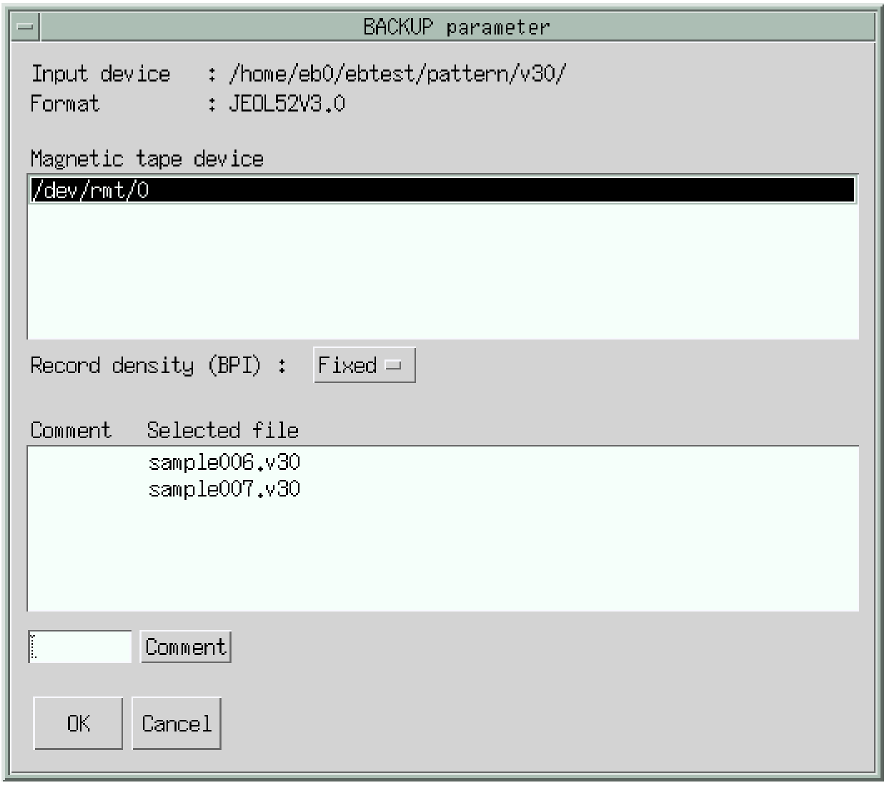


Fig. 57 BACKUP parameter window (with JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1 format data specified)

• Comment

You can specify the comment to be written in the directory block of the magnetic tape.

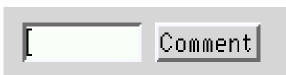


Fig. 58 Comment input field

Comment input field: Lets you specify a comment with up to eight characters. If the file selected in the Selected file area has a comment, it appears here.

Comment: Sets the comment in the Selected file area. With no file selected in the list, nothing takes place.

12.2 Command Execution

OK: After you set the parameters, clicking on the **OK** button displays the BACKUP confirmation window.

Fig. 59 BACKUP confirmation window

Yes: Executes the BACKUP function.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

No: Returns you to the BACKUP parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

13 RESTORING

The restore function copes the specified CAD or JEOL data file from the magnetic tape onto the specified disk. This chapter explains the method of operating the restore function.

1. Select **Magnetic tape list** from the option menu in the device-selection area, and select **Command(C)–RESTORE**.
2. Confirm that a magnetic tape is loaded, and click on **Yes** in the confirmation window.

   Clicking on **No** cancels the restoration and returns you to the main window of JBXFILER.

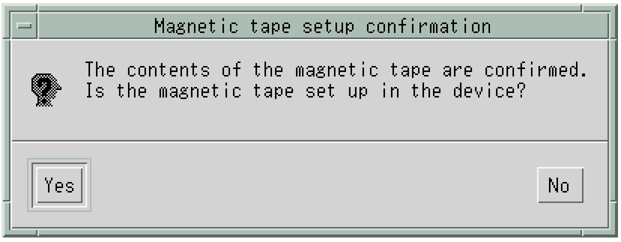


Fig. 60 Magnetic tape setup confirmation window

13.1 Setting Parameters

A parameter-setting window appears, so set the parameters.

● Common parameters

- Output device

  You can select the destination directory. This is a required parameter and you cannot omit it.
- Output file

  You can specify the output file name of the file to be restored.

  ☞ The file name input field lets you specify the file name. For details, refer to the corresponding part in "ID name specification" or "Appearance number specification."

  If you specify nothing here, the program creates a file name using the ID name of the file to be restored.

● Specifying ID names

For PG3000, PG3600, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, JEOL52V3.1 formats, the RESTORE parameter window shown in Figure 61 lets you set the parameters. This window lists the ID names of the data recorded on the tape.

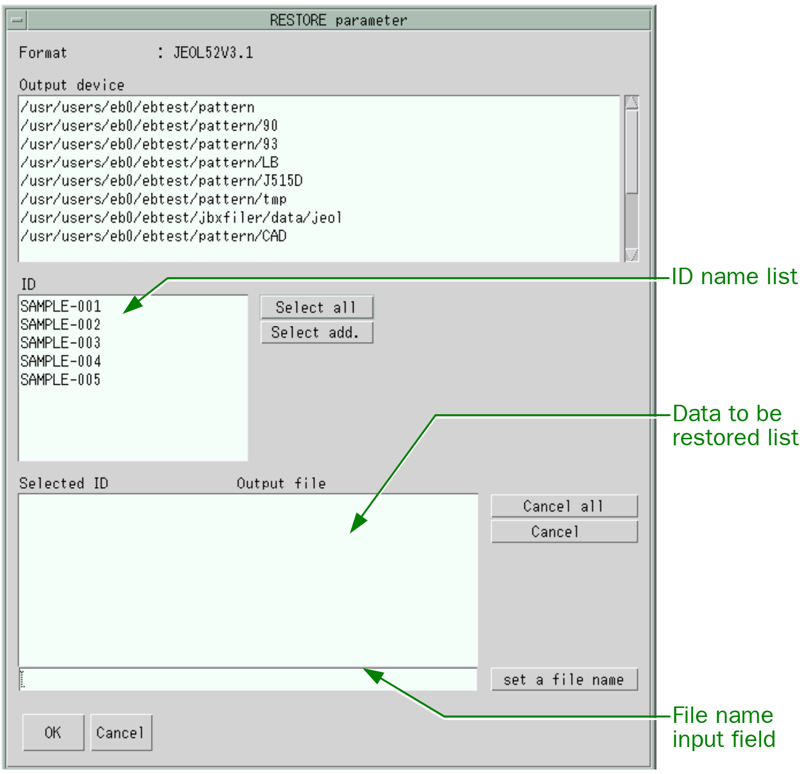


Fig. 61 RESTORE parameter window (for specifying ID names)

Select all:	Selects all the ID names displayed in the ID name list.
Select add.:	Moves the ID names selected in the ID name list to the data to be restored list.
Cancel all:	Clears the data to be restored list and initializes the ID name list.
Cancel:	Returns the ID names selected in the data to be restored list to the ID name list.
set a file name:	Sets the output file name for the ID name selected in the data to be restored list. This button is in effect when a file name is specified in the file name input field.
File name input field:	Lets you specify an output file name for the file to be restored. If a file name is already specified for the number selected in the data to be restored list, it appears here.

● Specifying sequential numbers of appearance

For JEOL01, STREAM, and JEOL51 formats, use the RESTORE parameter window shown in Figure 62 to set the parameters.

This window lists the sequential numbers of appearance. Because tapes in the formats handled here have no index information on their contents, numbers from 1 to 99 always appear. Accordingly, you can specify numbers that do not have the corresponding data in the tape; if you specify such numbers, the JBX command cancels them.

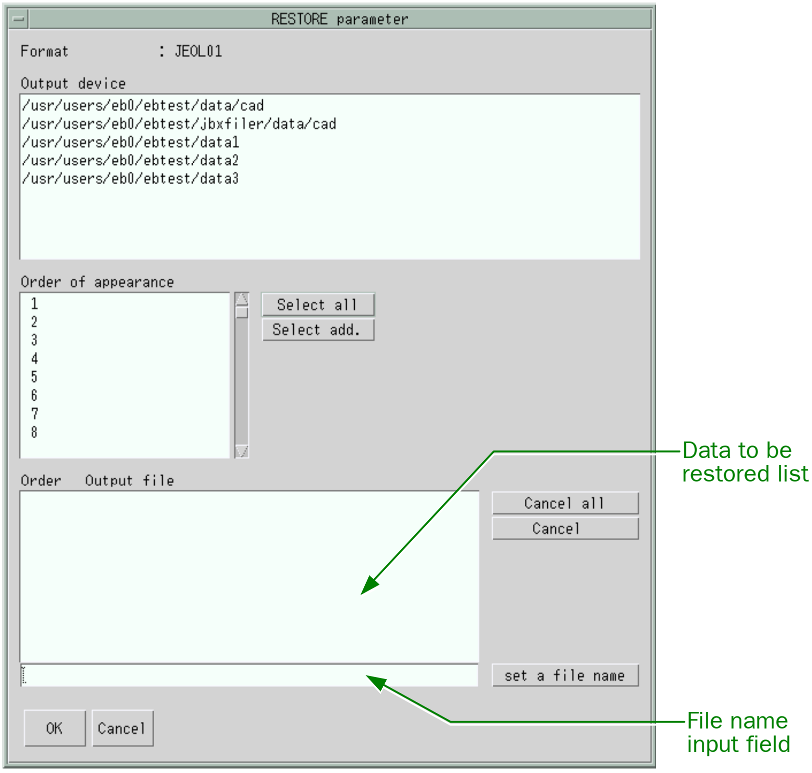


Fig. 62 RESTORE parameter window (for specifying sequential numbers of appearance)

Select all: Selects all the numbers in the Order of appearance area.

Select add.: Moves the numbers selected in the Order of appearance area to the data to be restored list.

Cancel all: Clears the data to be restored list and initializes the Order of appearance area.

Cancel: Returns the numbers selected in the data to be restored list to the Order of appearance area.

set a file name: Sets the output file name for the number selected in the data to be restored list. This button is in effect when a file name is specified in the file name input field.

File name input field: Lets you specify an output file name for the file to be restored. If an output file name is already specified for the number selected in the data to be restored list, it appears here.

13.2 Command Execution

OK: After you set the parameters, clicking on the **OK** button displays the RESTORE confirmation window.

Fig. 63 RESTORE confirmation window

Yes: Executes the RESTORE function.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

No: Returns you to the RESTORE parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

14 MERGING

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The merge function merges two to nine specified files in JEOL01, PG3000, or PG3600 format. This chapter explains the method of operating the merge function.

◆ Select desired files in the file-selection area, and select **Command(C)–MERGE**.

This function can handle up to nine files. If you specify 10 or more files, the function disregards the 10th and later files (in such a case, a message appears).

This function is available only for JEOL01, PG3000, and PG3600 files.

14.1 Setting Parameters

Selecting **MERGE** displays the following parameter-setting window, so set the parameters.

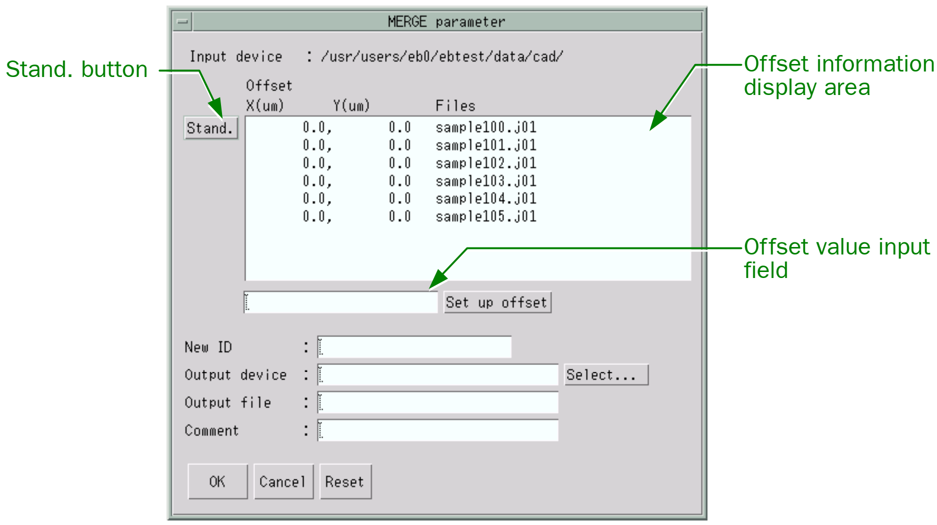


Fig. 64 MERGE parameter window

Changing the standard file

You can change the standard file by clicking on the **Stand.** button. With no file selected, clicking on this button makes the second file become the standard file. With a file selected, clicking on this button makes it become the standard file. Changing the standard file moves the previous standard file to the end of the list. You cannot set the offset of the standard file; when a file becomes the standard file, its offset values change to (0, 0).

Specifying the offset	You can set the X- and Y-direction offset values of the chip origin to be merged. However, you cannot specify offset values for the standard file (the file at the top).
Set up offset:	Sets the offset values for the file selected in the offset information display area.
Offset value input field:	You can enter the offset values (X, Y) in µm. This field displays the offset values of the file selected in the offset information display area.
New ID field	You can specify a new ID name for the merged file. If you leave this field blank, the program uses the same ID name as that of the standard file (the top file). This field appears only for JEOL01 data.
Output device field	You can specify the device and directory to which to store the output file after merging. This is a required parameter and you cannot omit it.
Output file field	You can specify the name of the file to be ouput after merging. If you leave this field blank, the program creates a file name using the ID name.
Comment field	You can specify the comment to write in the output file.
Data unit buttons	You can set the data unit for PG3000/PG3600 data. These buttons appear only for PG3000/PG3600 data.

14.2 Command Execution

OK: After you set the parameters, clicking on the **OK** button executes the merge function.

☞ The results output to the result area are the same as with a JBX command. See “JBX Command User’s Guide.”

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters you changed back to the initial settings.

15 DATA CONVERTING (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The data-conversion function converts a CAD, JEOL52, JEOL52V1.1, JEOL52V3.0 or JEOL52V3.1 data file to the specified JEOL data format file. This chapter explains the method of operating the data-conversion function.

◆ Select desired files in the file-selection area, and select **Command(C)–CONV**.

15.1 Setting Parameters (in the STREAM Parameter Window)

If the selected file is in STREAM format, the following STREAM parameter window appears before you set the parameters for data conversion.

✍ You can check the information required for setting the parameters in the STREAM parameter window by selecting **Command(C)–INFORMATION–File INFORMATION** in the main window of JBXFILER.

☞ For details on **File INFORMATION**, see Section 11.1, “File-Information Function.”

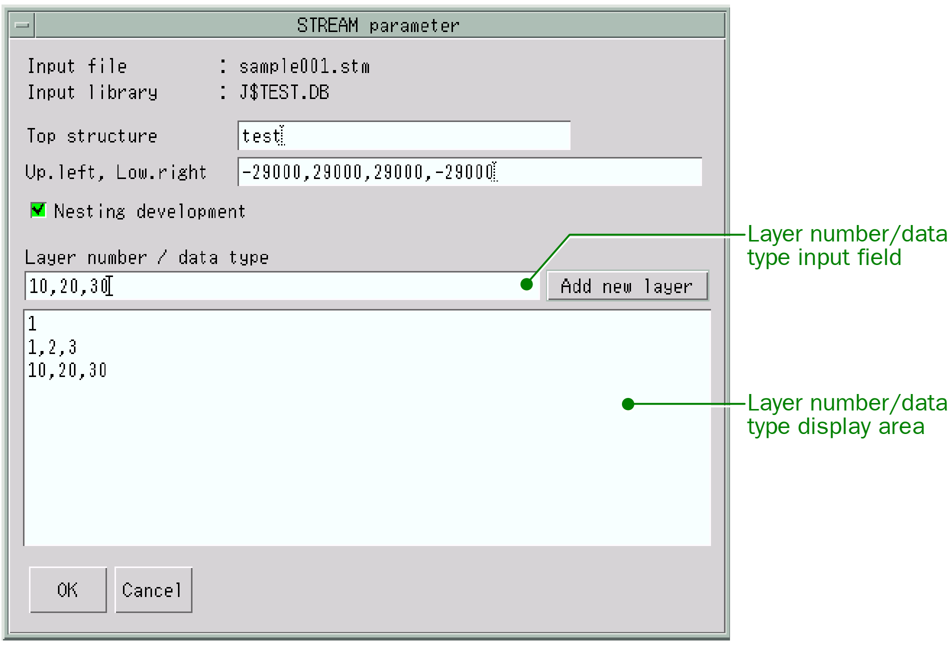


Fig. 65 STREAM parameter window

Top structure input field — Lets you set the name of the structure to develop.

Up. left, Low. right input field — Lets you set the area to develop by entering the upper-left and lower-right coordinates of the STREAM data in µm.

Nesting development: Lets you specify whether or not to develop other structures that are referenced by the structure specified in the Top structure input field. With this button selected (the button is in green), the program performs nesting development. If no other structure is referenced, deselecting this button reduces the processing time.

Layer number/data type input field

Lets you specify the layer numbers and data types to develop. After clicking on this field, you can enter them using the keyboard. For one STREAM file, you can specify up to 32 items (one item corresponds to one line in the layer number/data type display area; clicking on the Add new layer button appends one line to the list in the layer number/data type dispalyarea).

Layer number/data type input field:

You can enter layer numbers and data types according to the format shown below.

Add new layer: Lets you set the layer numbers and data types to develop. The layer number/data type display area displays the settings. The settings do not take effect unless you click on this button.

Format: Layer-【/Datatype, ...】

Layer-No: Layer number in the STREAM data

Datatype: Data type in the STREAM data

✍ You can omit the items in the brackets 【 】.

Specification method:

1. Single specification

   You can specify one layer number (/data type).

   Example: 1/1

2. Multiple specification

   You can specify multiple layer numbers (/data types) by separating them with commas.

   Example: 1,3/10,5

3. Consecutive specification

   You can specify consecutive layer numbers and data types by connecting the first and last values with hyphens.

   Example: 1-5/10-15

15.2 Command Execution

OK: Displays the CONV parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

15.3 Setting Parameters (in the CONV Parameter Window)

The following CONV parameter window appears, so set the parameters.

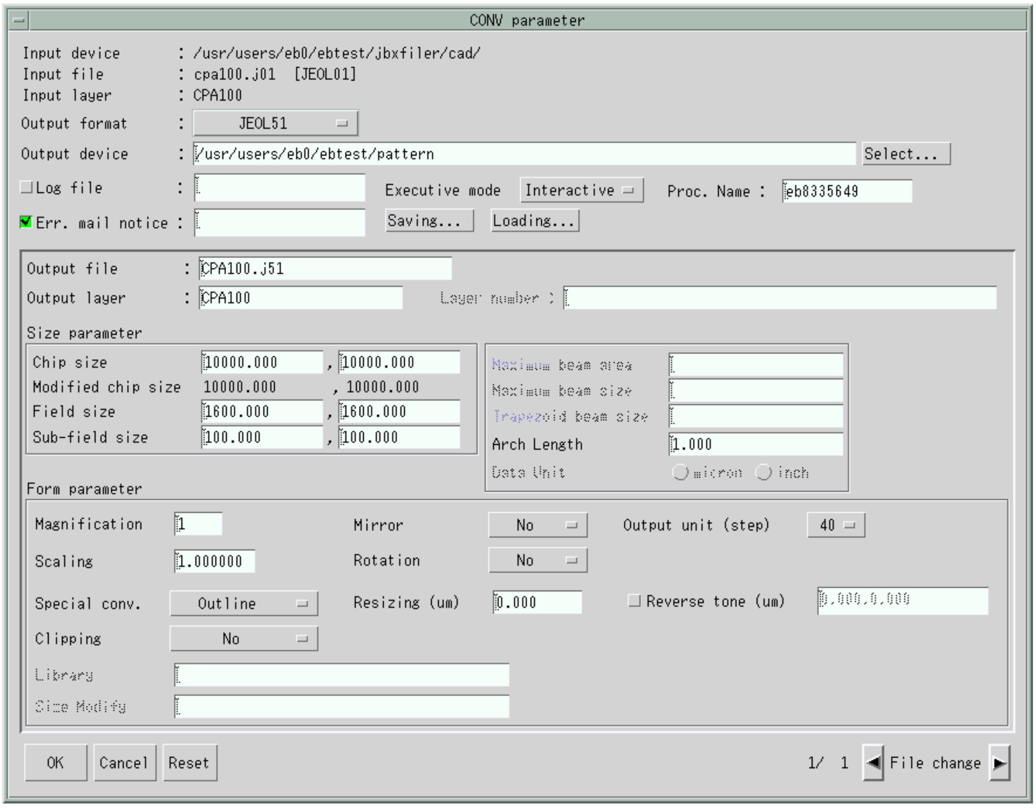


Fig. 66 CONV parameter window

Output format — Lets you specify the format of the output data.

Output device — Lets you specify the device and directory in which to save the output data file after conversion.
✍ This is a required parameter and you cannot omit setting it.

Log file — Lets you specify whether or not to output a log file at each startup to a destination other than the common directory in the system. With this button selected (the button is in green), the program outputs a log file at each startup. You can specify a log file name together with the device and directory. The default settings are as follows:

Directory:	JBXFILER starting directory
File name:	ebconv
File type:	.log

Err. mail notice — Lets you specify whether or not to send the error-detection notification mail. With this button selected (the button is in green), the program sends the mail. With no mail address specified, JBXFILER sends the mail to the user who has started it.

Saving: Saves the settings in the CONV parameter window in a file.

☞ For details, see Section 15.5, "Loading and Saving the Parameters."

Loading: Loads the saved data and applies them to the CONV parameter window.

☞ For details, see Section 15.5, "Loading and Saving the Parameters."

Output file: Lets you specify the output file name using up to 84 characters. You cannot enter characters unusable in a file name. If you leave this box blank, the program creates a file name using the output layer name.

Output layer: Lets you specify the layer name to be output to the converted file. You cannot enter characters unusable in a file name. If you leave this box blank, the layer name is the same as that of the input file. However, for STREAM format, the name is the structure name with a serial number appended.

Layer number: Displays the layer numbers and data types specified in the STREAM parameter window. You can revise the layer numbers and data types here.

✍ This box is unavailable for formats other than STREAM format (the indication is dimmed).

Size parameter

Chip size: Lets you specify the range of the input data in μm. The default value depends on the input format (☞ See the following table).

Default chip sizes

Input format	Default chip size
JEOL01	10000 × 10000 (μm)
STREAM	Calculated from the data region of the conversion structure.
PG3000	10000 × 10000 (μm)
PG3600	10000 × 10000 (μm)
JEOL52	Same chip size as with the input data file.
JEOL52V1.1	Same as above
JEOL52V3.0	Same as above
JEOL52V3.1	Same as above

Field size: Lets you specify the size of the field to be converted in μm.

Sub-field size: Lets you specify the size of the sub-field to be converted in μm.

Indication

Modified chip size: Displays the modified chip size calculated from the specified chip size and the form parameters.

Maximum beam area* — Lets you specify the maximum beam area of one shot in rectangle writing, in the range 0.0025 to 156.25 μm^2.

✍ This box is available if JEOL51 (6A) is selected as the output format.

Maximum beam size* — Lets you specify the maximum beam size of a pattern that can be written at one time, in the range 0.05 to 12.5 µm.

✍ This box is available if JEOL51 (6A) is selected as the output format.

Trapezoid beam size* — Lets you specify the length of the shorter side for trapezoid writing, in the range 0.0 to 1.0 µm.

✍ This box is available if JEOL51 (6A) is selected as the output format.

Arch Length — Lets you specify the arc length for converting rings to polygons, in the range 0.05 to 1000.0 µm.

✍ This box is available if the input format is JEOL01 and the output format is JEOL51.

Form parameter — Contains the following 12 parameters.

Magnification — Lets you specify the reticle magnification for converting reticle data.

Scaling — Lets you specify the scaling value for calculating the overall magnification.

Mirror — Lets you specify whether or not to perform mirroring. Select an item from the option menu.

Rotation — Lets you specify whether or not to perform rotation. Select an item from the option menu.

Output unit (step) — Lets you specify the number of steps per micrometer (µm). You can change the output unit only if the output format is JEOL51, JEOL52V3.0 (9000), JEOL52V3.0 (9300), JEOL52V3.0 (6300), or JEOL52V3.1. For JEOL51, select an output unit from the option menu. For JEOL52V30 or JEOL52V31, the area shown in Figure 67 appears; you can specify whether or not to specify the output unit. Clicking on the button (the button turns green) changes the output unit to the specified value. You can enter the output unit using the keyboard. With the button unselected (the button is dimmed), the program uses the default value.

Default output units

Output format	Default output unit
JEOL52V3.0 (9000)	See “Table of the parameter limits and default values for the JBX-9000MV” in Appendix 7, “Tables of the Limits on the Data-Conversion Parameters.”
JEOL52V3.0 (9300)	See “Table of the parameter limits and default values for the JBX-9300FS” in Appendix 7, “Tables of the Limits on the Data-Conversion Parameters.”

* At present, the data conversion of the JBX command does not support this parameter; even if you change the value on the screen, the program uses the default value instead.

Output format	Default output unit
JEOL52V3.0 (6300)	See "Table of the parameter limits and default values for the JBX-6300 series" in Appendix 7, "Tables of the Limits on the Data-Conversion Parameters."
JEOL52V3.1	Unit for the chip size = 1000 Unit for the position set value = 100 Unit for the pattern data = 40

✍ If you are a JBX-6300xx user (xx: Last characters following JBX-6300.), please read "JBX-6300 series" in this manual as "JBX-6300xx."

Fig. 67 Output unit specification area for JEOL52V3.0

Micro size — Lets you specify the micro size for the pattern data.

✍ If the output format is JEOL51, the Micro size box does not appear.

Special conv. — Lets you specify whether or not to perform outlining or shot-rank correction. Select an item from the option menu.

- **Outline**: Specifies the execution of outlining and sets the shot-time correction rank of the output data to 0.
- **No outline**: Specifies that outlining is not executed.
- **Shot modu. 1**: Specifies the execution of shot-time correction.
- **Shot modu. 2**: Specifies the shot-time correction rank and the execution of shot-time correction. This item appears on the option menu only if the output format is JEOL51 or JEOL51 (6A).
  JEOL51: You can specify two ranks for area patterns and for line patterns.
  JEOL51 (6A): You can specify only the rank for area patterns.

Resizing — Lets you specify the amount of resizing. If you specify 0.0, resizing is not performed.

Reverse tone — Lets you specify whether or not to perform reverse tone. With this button selected (the button is in green), the program performs reverse tone. You can enter X/Y frame values (the widths of the painted part on the chip periphery) using the keyboard.

Clipping — Lets you specify whether or not to perform clipping. Select an item from the option menu. Selecting **Positive** or **Negative** displays the window coordinates input field where you can enter the bottom-left and top-right coordinates of the window with the bottom left of the chip at the origin.

Library — Lets you specify the name of the library parameter file for conversion with library. The program searches ${HOME}/eb/prm or the directory indicated by the environmental variable CNVUSRPATH.

You cannot specify this file name if the output format is JEOL51 or JEOL51 (6A) or if library conversion is not supported.

For details on the library parameter file, see Appendix 10, “Library Parameter File.”

2 times drawing — The area shown in Figure 68 appears if the output data format is JEOL52V3.0 (9000) and two-time writing is supported. With this button selected (the button is in green), the program performs two-time writing. You can specify X/Y shift amounts, in this order, in the range 0.0 to 4.0 µm, using the keyboard.

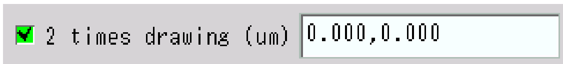


Fig. 68 2 times drawing parameter setting area

15.4 Command Execution

OK: After you set the parameters, clicking on the **OK** button executes the data-conversion function.

The results output to the result area are the same as with a JBX command. See “JBX Command User’s Guide.”

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked on the File change button. If you have not clicked on the File change buttons, the default settings are restored.

15.5 Loading and Saving the Parameters

In the CONV parameter window, clicking on the **Loading…** or **Saving…** button displays the Parameter file selection window. In the Parameter file selection window, selecting a file loads the contents of the file or saves the parameters in the file.

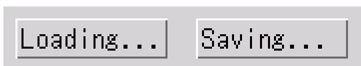


Fig. 69 Loading and Saving buttons

Fig. 70 Parameter file selection window

15.5.1 Settings

Filter: Lets you enter a filter.

Directories: Lets you specify the directory of the file to load or save. The selected directory appears in the Selection box.

Files: Lets you select the file to load or save. The selected file appears in the Selection box. When saving, the specified file is overwritten. If you want to crate a new file for saving, type a file name directly in the Selection box.

Selection: Lets you specify the name of the directory and file to load or save.

15.5.2 Command execution

OK: Executes loading or saving.

Filter: Updates the Files box by using the filter entered in the Filter box. Instead of using this button, you can also press Enter with a filter in the Filter box.

Cancel: Cancels the process and returns you to the CONV parameter window.

16 CONVERTING TO GRAPHIC DATA (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

This chapter explains the method of operating a feature that converts JEOL data files to data files for graphic display (PCHK).

◆ Select desired files in the file-selection area, and select **Command(C)–Convert to graphic data.**

16.1 Setting Parameters

The following parameter-setting window appears, so set the parameters.

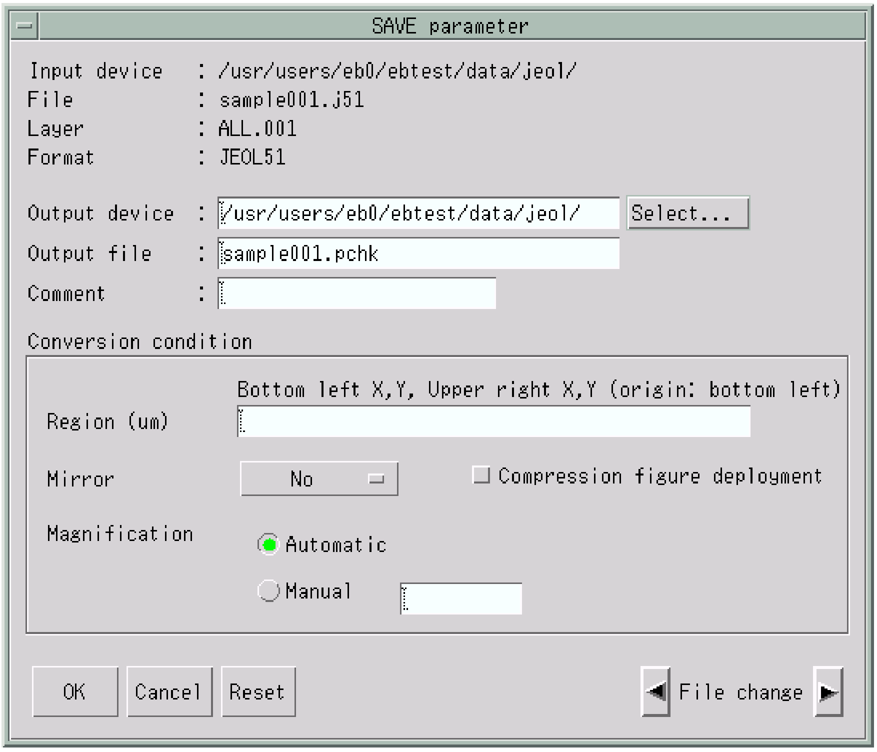


Fig. 71 SAVE parameter window

Output device	Lets you specify the device and directory in which to save the PCHK plotting data file after conversion. By default, the same directory as the input source appears.
Output file	Lets you specify the name of the PCHK plotting data file. You can enter only the characters that can be used in file

	names. If you leave this box blank, the file name will be the input file name with the extension .pchk.
Comment	Lets you specify the comment to be output to the PCHK plotting data file using up to 24 characters. ✍ No comment is output by default.
Conversion condition	The conversion conditions are Region, Mirror, and Magnification.
Region	Lets you set the chip range to be developed in the PCHK plotting data file. Enter the lower left and upper right coordinates, in this order, in μm with the origin at the lower left of the chip. By default, the entire chip is converted.
Mirror	Lets you specify whether or not to invert the patterns in the area to be output. Selecting **X axis** or **Y axis** inverts the patterns around the x or y axis, respectively.
Magnification	Lets you specify the magnification for the display on the view port. The magnification specified here is used if single display is selected for data display. If you select **Automatic**, the program automatically calculates the magnification for displaying the patterns on the entire view port. If you select **Manual**, you can specify the magnification as a real number.

16.2 Command Execution

OK: After you set the parameters, clicking on the **OK** button converts the data.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked on the **File change** button. If you have not clicked on the File change buttons, the default settings are restored.

17 DISPLAYING DATA (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern

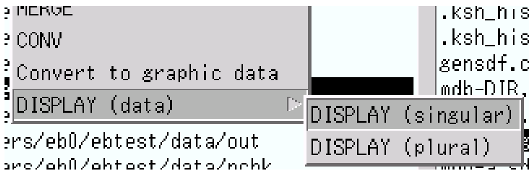
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern

OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The data-display function displays the pattern data contained in a PCHK data file on the display. This chapter explains the method of operating the data-display function.

17.1 Single Display

◆ Select the desired file in the file-selection area, and select **Command(C)–DISPLAY (data)–DISPLAY (singular)**.



17.1.1 Setting parameters

Selecting **DISPLAY (singular)** displays the following window where you can set the parameters.

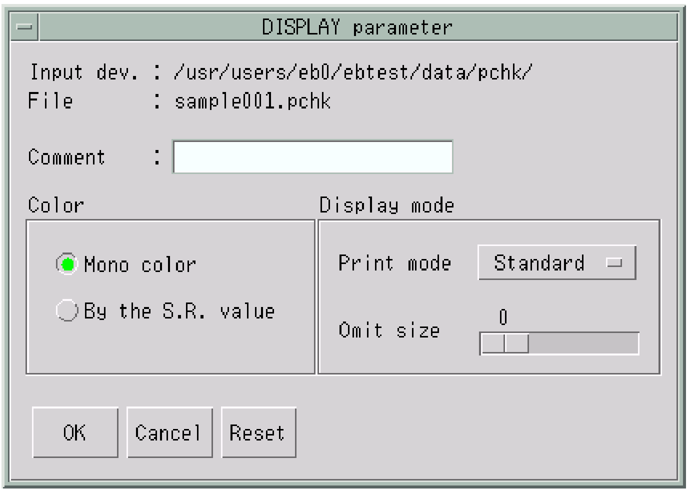


Fig. 72 DISPLAY parameter window

Comment — Lets you specify the comment to be displayed on the Bitmap Display using up to 24 characters. By default, the comment written in the PCHK plotting data file appears.

Color — Lets you specify how to set the colors of the patterns to be displayed on the Bitmap Display. Selecting **Mono color** will display all the patterns in the same color. Selecting **By the S.R. value** will set the display colors in accordance with the shot ranks.

✍ Setting the display colors according to the shot ranks uses the shot rank and color number correspondence file (PCHK1.PRM).

☞ For details, see Appendix 4, "Shot Rank And Color Number Correspondence File."

Print mode — Lets you specify how to write the patterns on the Bitmap Display.

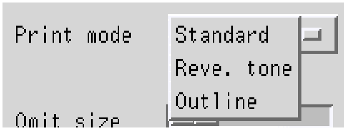


Fig. 73 Print mode option menu

Standard: Paints the inside of the patterns.

Reve. tone: Writes the patterns in the clipping region while performing reverse tone.

Outline: Writes only the outlines of the patterns without painting the inside of the patterns.

Omit size — Lets you specify the omit size by using the scale bar. The program does not display the patterns whose pixels are less than or equal to this setting. If you specify 0, no pattern is omitted.

17.1.2 Command execution

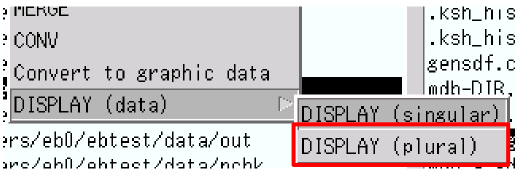
OK: After you set the parameters, clicking on the **OK** button executes the data-display function.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters you changed back to the initial settings.

17.2 Multiple Display



◆ Select desired files in the file-selection area, and select **Command(C)–DISPLAY (data)–DISPLAY (plural)**.

17.3 Setting Parameters

Selecting **DISPLAY (plural)** displays the following window where you can set the parameters.

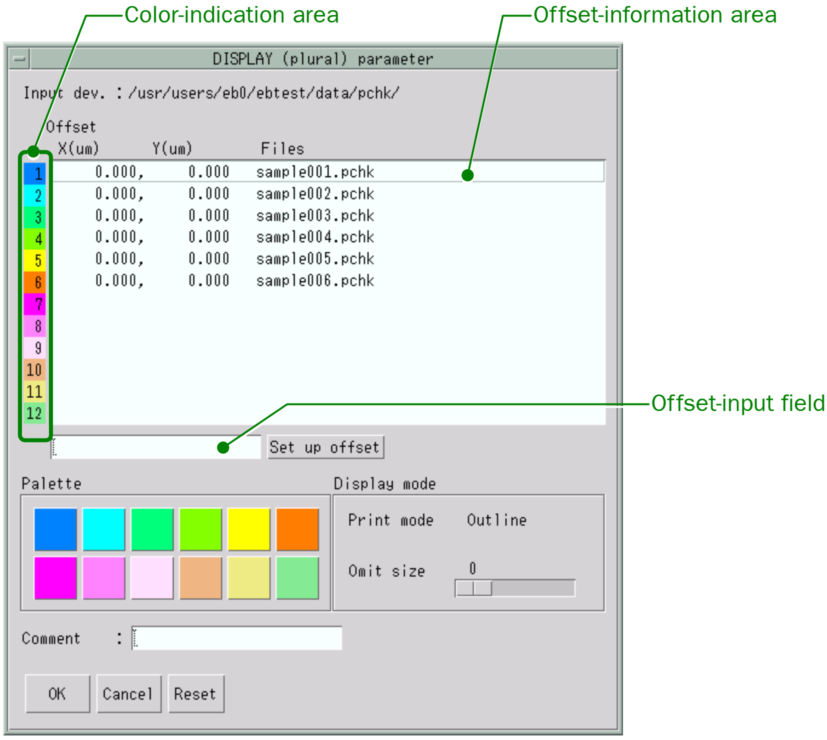


Fig. 74 DISPLAY (plural) parameter window

Offset-information area — Lets you specify the relative distances in the X and Y directions between the data centers, with the first file as the reference. You cannot set the offset for the first file. Clicking on a file in the offset-information area moves the focus to the offset-input field, where you can specify the offset. The offset-input field displays the offset of the selected file.

Offset-input field — Lets you specify the offset (X, Y) in µm.

Set up offset: — Sets the offset for the file selected in the offset-information area. Instead of using this button, you can also press the Return key in the offset-input field.

Palette — Lets you set the colors of the patterns to be displayed on the Bitmap Display for each file. If the offset-information area contains 12 files, you cannot change the colors. Also, you cannot specify the same color for two or more files. Selecting (clicking) a file in the offset-information area and then clicking a color in the palette changes the corresponding color in the color-indication area to the specified color.

☞ You can change the colors in the palette by using the color number and hue correspondence file (PCHK1. PRM). For details, refer to Appendix 5, “Color Number And Hue Correspondence File.”

Omit size	Lets you specify the omit size by using the scale bar. The program does not display the patterns whose pixels are less than or equal to this setting. If you specify 0, no pattern is omitted.
Comment	Lets you specify the comment to be displayed on the Bitmap Display using up to 24 characters. By default, the comment written in the first PCHK plotting data file appears.

17.3.1 Command execution

OK: After you set the parameters, clicking on the **OK** button executes the data-display function.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters you changed back to the initial settings.

18 READING DATA (OPTIONAL FUNCTION)

✍ This function is for the JBX-5000LS/E and JBX-6000FS/E.
✍ Only version 3.804 and earlier versions of JBXFILER can handle this function.

The data-read function converts JEOL51 data files to scanner-format data files. This chapter explains the method of operating the data-read function.

◆ Select desired files in the file-selection area, and select **Command(C)–READ (data)**.

18.1 Setting Parameters

Selecting **READ (data)** displays the following window where you can set the parameters.

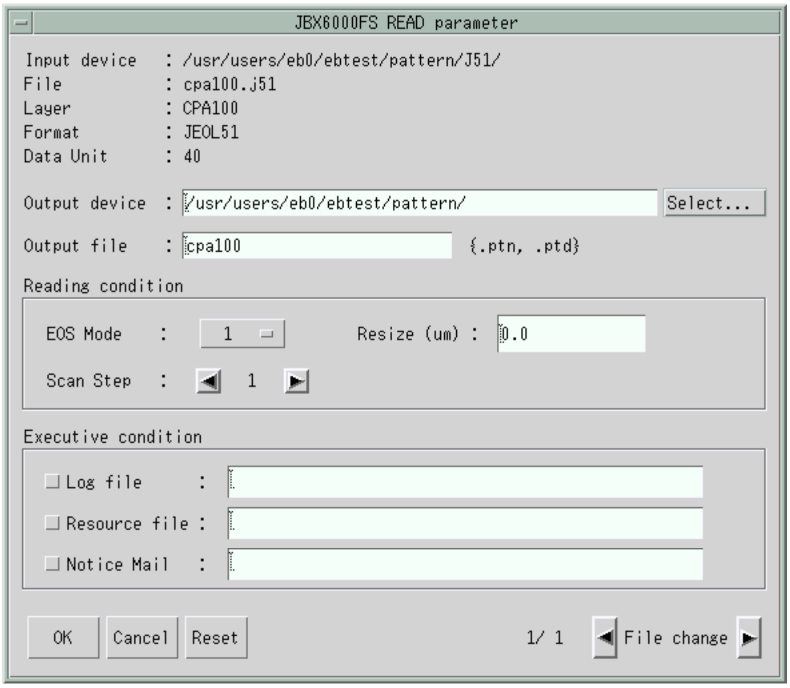


Fig. 75 READ parameter window

Output device	Lets you specify the device and directory in which to save scanner-format data files after conversion. By default, the top line in the JEOL data list (dtab) appears.
Output file	Lets you specify the name of the scanner-format data file. The program automatically appends a file extension (.ptn or .ptd) to the entered name. You can enter only the characters that can be used in file names. The default file name is the input file name with the extension .ptn or .ptd.
Reading condition	Consists of EOS Mode, Scan Step, and Resize.
EOS Mode	Lets you select the EOS mode for writing. The selectable EOS modes depend on the input data unit; only the selectable EOS modes appear on the option menu.

Input data unit	EOS mode
20	1, 2
40	1, 2, 3, 4
80	3, 4
200	5, 6
400	5, 6, 7, 8
800	7, 8

Scan Step

Lets you specify the number of scan steps. The upper limit depends on the EOS mode.

EOS mode	Scan step
1, 2	1 to 8
3, 4	1 to 16
5, 6	1 to 8
7, 8	1 to 16

Resize

Lets you specify the amount of resizing. You can enter a value greater than or equal to 0.0, in μm. The step size depends on the EOS mode. The program rounds the entered value to the nearest multiple of the step size. If you specify 0.0, resizing is not performed.

EOS mode	Step size between amounts of resizing (μm)
1, 2	0.025
3, 4	0.0125
5, 6	0.0025
7, 8	0.00125

Executive condition

Consists of Log file, Resource file, and Notice Mail.

Log file

Lets you specify whether or not to output a log file at each startup. Selecting this button (the button turns green) will output a log file. You can specify any file name. You can also specify a device and directory. The default file name is as follows:

Directory — ${HOME}/eb/log

File name — yyyymmddhhmmss.pread

(yyyy: year, mm: month, dd: day, hh: hour, mm: minute, ss: second)

Resource file

Lets you specify whether or not to read a resource file for command execution. Selecting this button (the button turns green) will read a resource file. You can specify any resource file. By default, the program reads the following resource file:

Directory — ${HOME}/eb/prm

File name — preadrc

Notice Mail — Lets you specify whether or not to send a log by mail when an error is detected. Selecting this button (the button turns green) will send mail. You can specify any destination address. By default, the program sends mail to root.

18.2 Command Execution

OK: After you set the parameters, clicking on the **OK** button executes the data-read function. However, if a file with the same name as the specified output file name already exists, the following confirmation window appears.

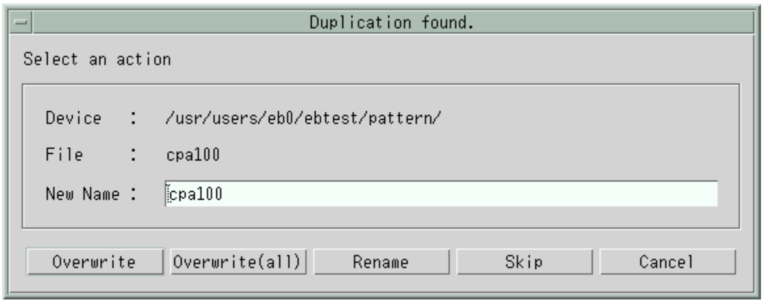


Fig. 76 Duplication found window

New Name — Lets you specify a new file name.

Overwrite: Deletes the existing scanner-format file, and executes the command.

Overwrite (all): Deletes all the duplicated scanner-format files, and executes the command.

Rename: Changes the output file name to the one in the New Name box, and executes the command.

Skip: Cancels the reading of the displayed pattern, and executes the command.

Cancel: Aborts the process and returns you to the READ parameter window.

☞ The results output to the result area are the same as with PREAD. Refer to the pattern-read program manual, “PREAD (Program).”

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked on a File change button. If you have not clicked on the File change buttons, the default settings are restored.

19 SEPARATING PATTERNS (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The pattern-separation function separates the patterns in the specified JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, and JEOL52V3.1 data files. This chapter explains how to use the pattern-separation function.

◆ Select desired files in the file-selection area in the JBXFILER main window, and select **Command(C)–Separate patterns**.

This displays the PSP parameter window.

✍ Without a license to execute this function, the **Separate patterns** command does not appear on the **Command** menu.

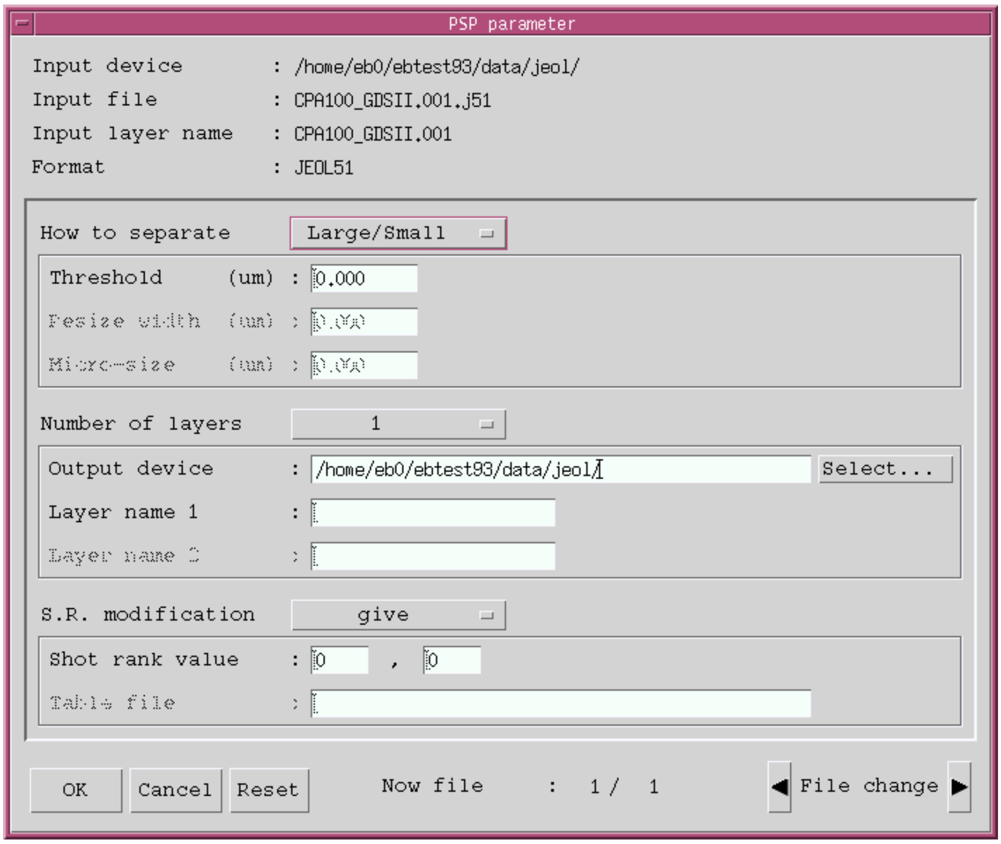


Fig. 77 PSP parameter window

19.1 Setting Parameters

The PSP parameter window lets you set these parameters.

How to separate — Lets you select Large/Small, Outside/Inside, or No separation as the separation method.

Large/Small: Lets you specify the threshold in μm. The threshold must be greater than 0, and the program disregards the fourth and subsequent decimal places.

Outside/Inside: Lets you specify the outline width, resize width, and micro size in μm (see Figure 78). The outline width must be greater than the resize width, and the resize width must be greater than or equal to 0. For each parameter, the program disregards the fourth and subsequent decimal places.

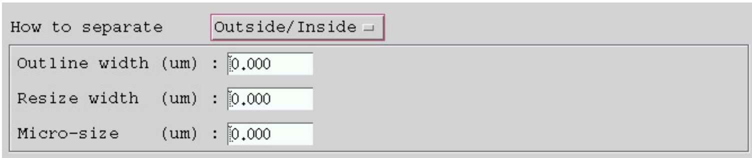


Fig. 78 Separation-parameter area

No separation: Makes the separation-parameter area unavailable. You do not need to set the separation parameters.

Number of layers — Lets you select **1** or **2** as the number of output layers. If you do not want to separate the layers, select **1**; to separate them, select **2**. With **No separation** selected as the separation method, the number of layers is automatically set to **1**.

1: Lets you specify the output device and the output layer name (up to 24 characters). You can enter only the characters acceptable for the output layer name.

2: Changes the relevant parameter area to the one in Figure 79. The method for entering the output device is the same as with **1**. Here is how to specify the output layers:

With **Large/Small** specified as the separation method

As the output layer name (small), specify the layer name for small patterns; as the output layer name (large), specify the layer name for large patterns.

With **Outside/Inside** specified as the separation method

As the output layer name (outside), specify the layer name for outline patterns: as the output layer name (inside), specify the layer name for inside patterns.

Fig. 79 Output-parameter area

S. R. modification — Lets you select give, distribute, or not specify as the shot-rank modification method.

give: In the first box, specify the shot rank for small or outline patterns. In the second box, specify the shot rank for large or inside patterns.

✍ You can enter only numbers.

distribute: Lets you specify the shot-rank table file name (see Figure 80). You can specify a device and directory for the file name. The default device and directory is the current directory.

☞ For details on the shot-rank table file, see Appendix B, “File Specifications” in the manual, “JBX Command User’s Guide.”

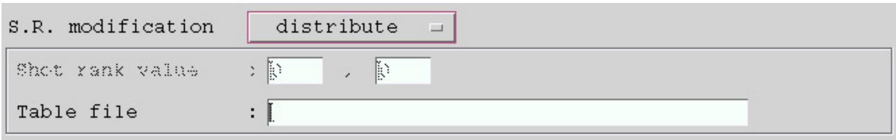


Fig. 80 Shot-rank modification area

Not specify: Makes the shot-rank modification area unavailable. You do not need to set any parameters. In this case, the program uses the same shot ranks for the input data as those for the output data.

19.2 Command Execution

OK: After you set the parameters, clicking the **OK** button displays the PSP confirmation window.

☞ The results output to the result area are the same as with a JBX command. See “JBX Command User’s Guide.”

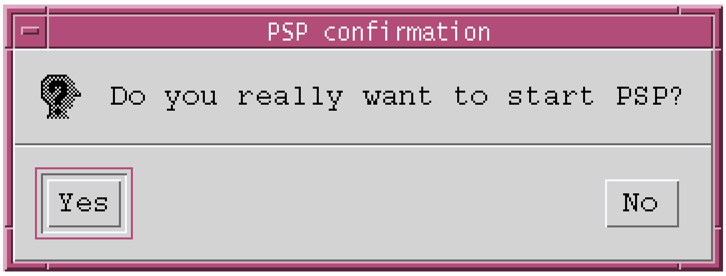


Fig. 81 PSP confirmation window

Yes: Starts the pattern separation.

No: Returns you to the PSP parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked the **File change** button. If you have not clicked the File change buttons, the default settings are restored.

20 CONVERTING THE FORMAT (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The format-conversion function converts the specified JEOL51 format data to JEOL52V3.0 format. This chapter explains how to use the format-conversion function.

◆ Select desired files in the file-selection area in the JBXFILER main window, and select **Command(C)–Convert format**.

This displays the Conversion parameters window.

✍ Without a license to execute this function, the **Convert format** command does not appear on the **Command** menu.

20.1 Setting Parameters (for Version 4.001 and Later Versions of JBXFILER)

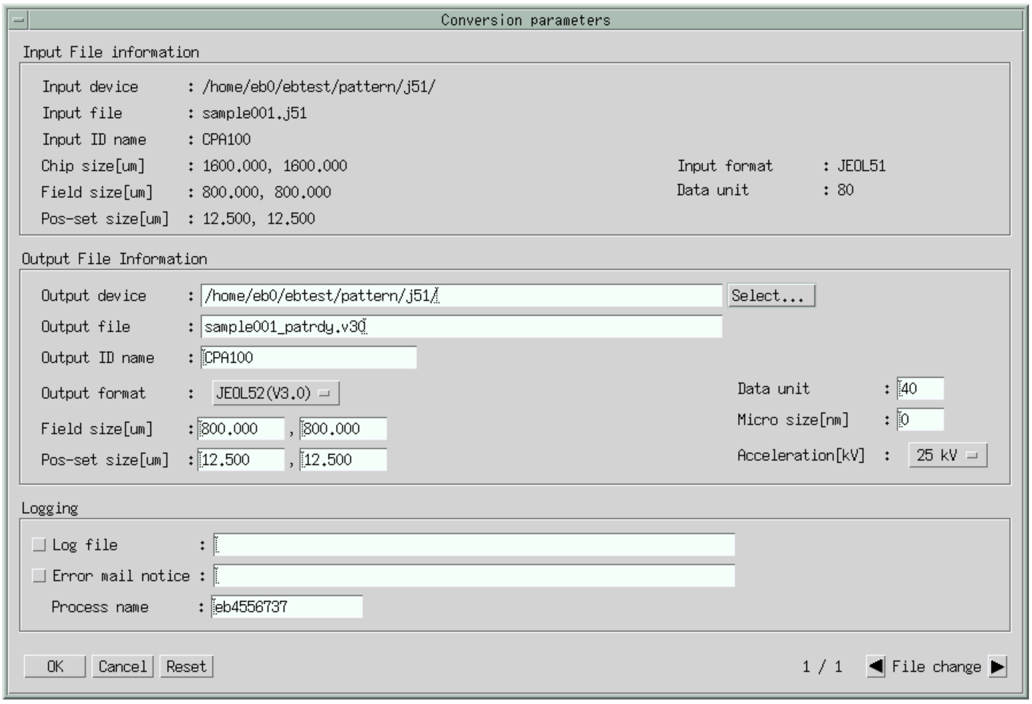


Fig. 82 Conversion parameters window

The Conversion parameters window lets you set these parameters.

■ Output file information

Output device	Lets you specify the destination to which to output a file after conversion. This is a required parameter and you cannot omit setting it.
Output file	Lets you specify the output data file name. You can enter only the characters acceptable in a file name. This is a required parameter and you cannot omit setting it.
Output ID name	Lets you specify the ID name of the output data up to 24 characters. This is a required parameter and you cannot omit setting it.
Field size [μm]	Lets you specify the field size for conversion in μm. The limits depend on the accelerating voltage or the EOS mode.
Pos-set size [μm]	Lets you specify the field size for conversion in μm. The limits depend on the accelerating voltage or the EOS mode.
Data unit	Lets you specify the number of steps per micrometer (μm). You can specify a divisor of 1000, or 2000, 4000, or 8000.

✍ 2000, 4000, or 8000 are available only in the JBX-6300 series.

Micro size [nm]	Lets you specify the micro size for the pattern data, in nm.
Acceleration[kV]	Depends on the system. Changing this parameter changes the limits on the field size and position set size. You can select **25**, **50**, or **100 kV**.

✍ This parameter does not appear in the JBX-6300 series.

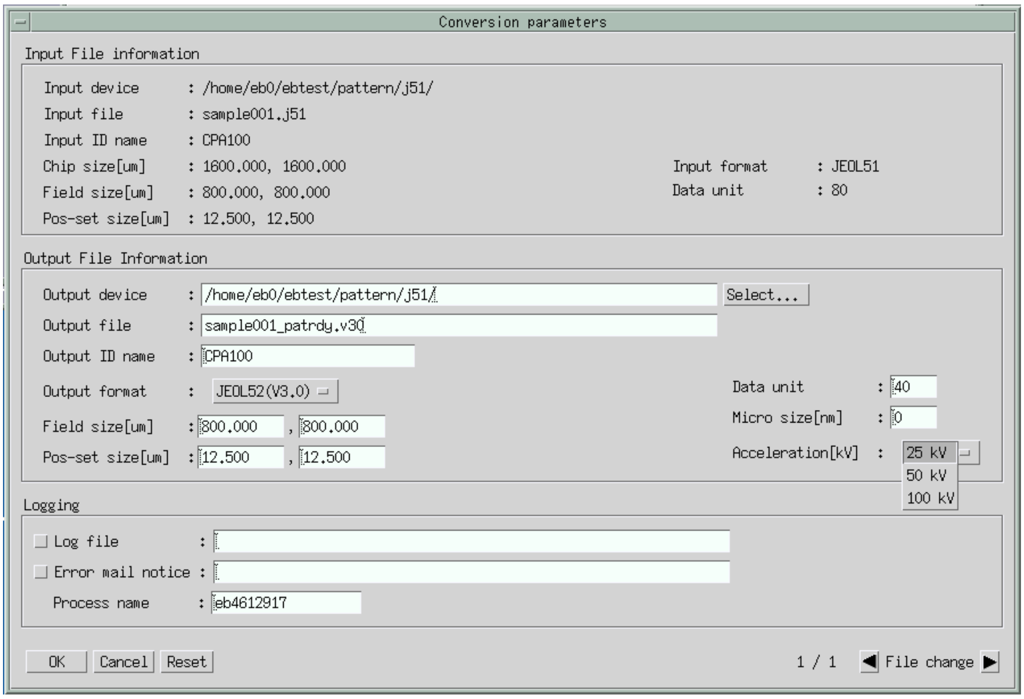


Fig. 83 Conversion parameters window (in systems other than the JBX-6300 series)

EOS mode	Depends on the system. Changing this parameter changes the limits for the field size and position set size. You can select **1**, **2**, **3**, **4**, **5**, or **6**.

✍ This parameter does not appear in systems other than the JBX-6300 series.

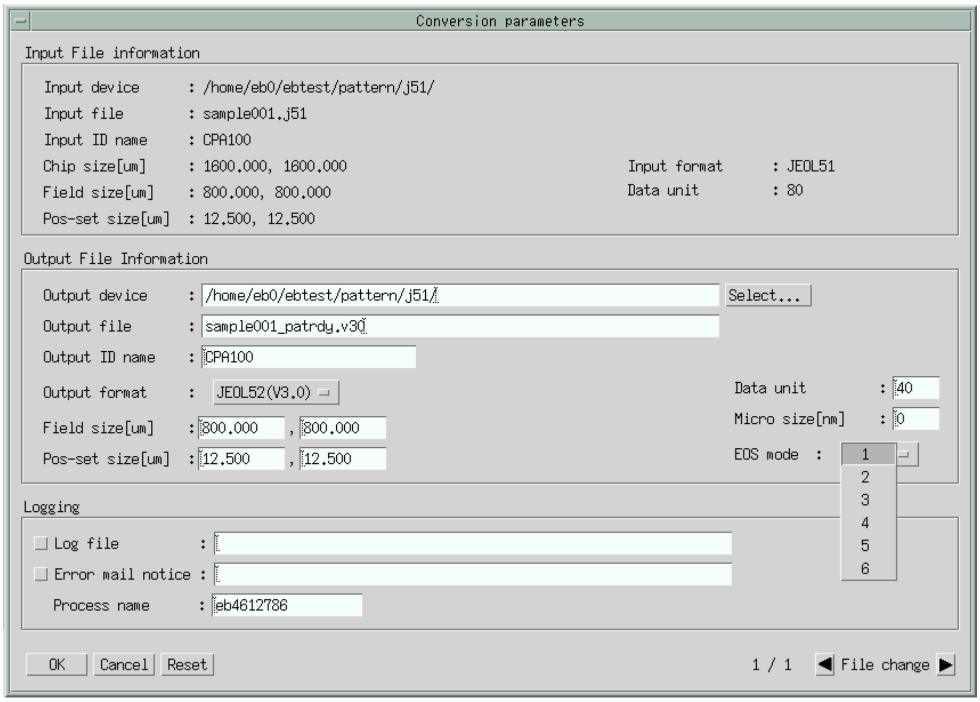


Fig. 84 Conversion parameters window (JBX-6300 series)

■ Logging

Log file — Lets you specify whether or not to output a log file for each startup to a directory other than the system common directory. Selecting the button (the button turns green) will output a log file for each startup. You can specify a device and directory for the file name. The default settings depend on the settings in the initialization file (patrdyrc). If the initialization file does not contain the settings, the program uses these settings:

Directory:	~/eb/log/
File name:	patrdy process ID
File type:	.log

Error mail notice — Lets you specify whether or not to send an error-detection notification mail. Selecting the button (the button turns green) will send an error-detection notification mail. The default mail address depends on the setting in the initialization file.

- no_mail — Refrains from sending mail.
- mail_address — Sends mail to the recorded mail address.

File change — Lets you select a file using the left and right arrow buttons, if multiple input files are selected.

20.2 Command Execution

OK: Converts the format.

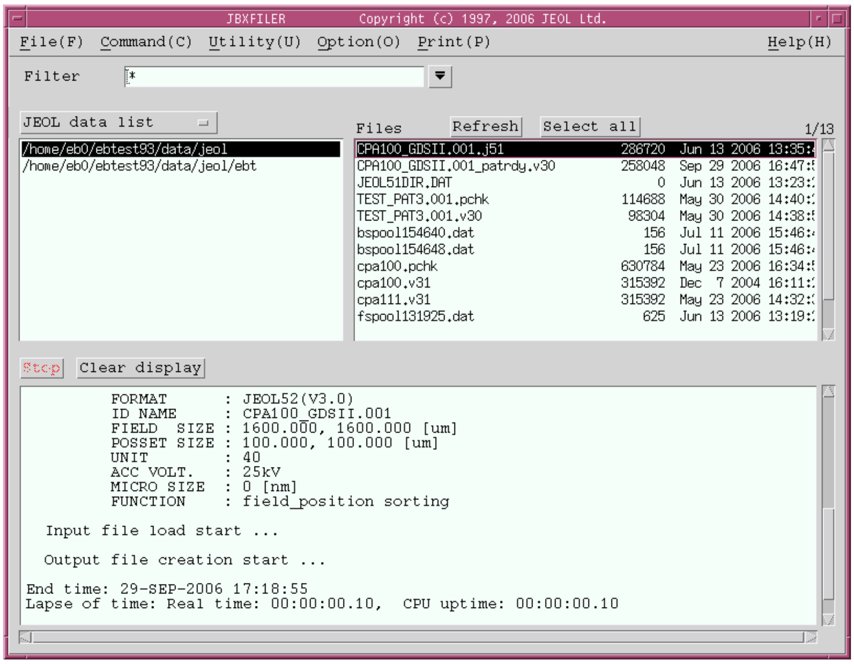


Fig. 85 Example of format conversion

At the same time of the format conversion, the program outputs a history file as follows.

```
------------------------------------------------------------------------------
patrdy Ver 4.302 (2006 07/19)                Copyright (c) 2004, 2006 JEOL Ltd.
Start time: 29-SEP-2006 17:18:55

  INPUT   FILE        : /home/eb0/ebtest93/data/jeol/CPA100_GDSII.001.j51
          FORMAT       : JEOL51(5D)
          ID NAME      : CPA100_GDSII.001
          CHIP    SIZE : 1600.000, 1600.000 [um]
          FIELD   SIZE : 1600.000, 1600.000 [um]
          POSSET SIZE : 100.000, 100.000 [um]
          UNIT         : chip=100, posset=100, figure=40
  OUTPUT FILE         : /home/eb0/ebtest93/data/jeol/CPA100_GDSII.V30
          FORMAT       : JEOL52(V3.0)
          ID NAME      : CPA100_GDSII.001
          FIELD   SIZE : 1600.000, 1600.000 [um]
          POSSET SIZE : 100.000, 100.000 [um]
          UNIT         : 40
          ACC VOLT.    : 25kV
          MICRO SIZE   : 0 [nm]
          FUNCTION     : field_position sorting

  Input file load start ...

  Output file creation start ...

End time: 29-SEP-2006 17:18:55
Lapse of time: Real time: 00:00:00.10,   CPU uptime: 00:00:00.10
```

Fig. 86 Format-conversion history file

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked the **File change** button. If you have not clicked the File change buttons, the default settings are restored.

20.3 Setting Parameters (for Version 3.804 and Earlier Versions of JBXFILER)

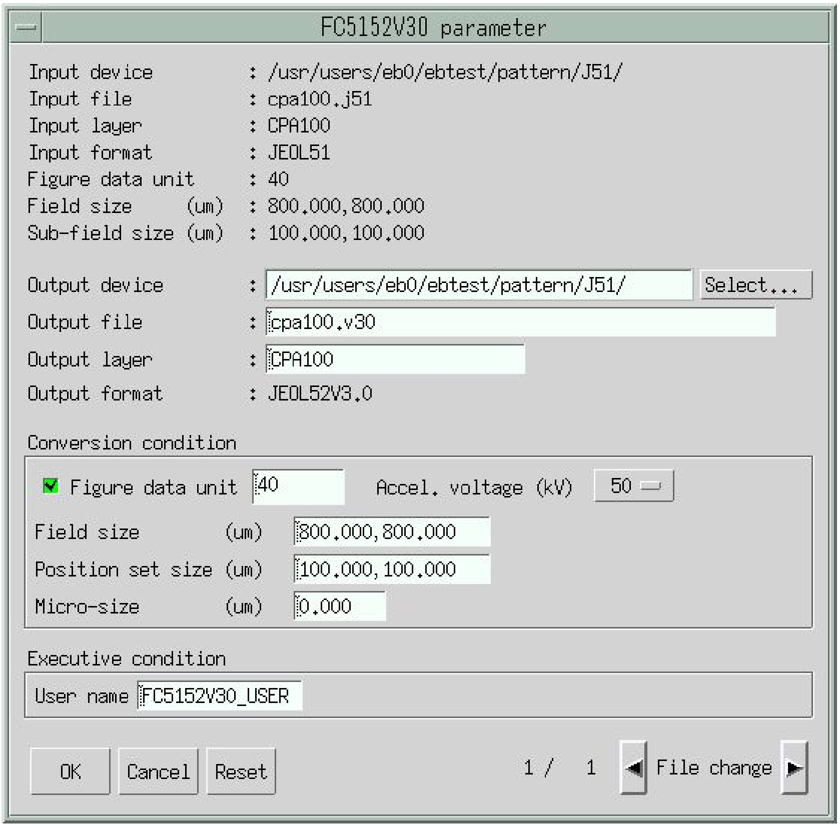


Fig. 87 Conversion parameters window

The conversion parameters window lets you set these parameters.

Output device: Lets you specify the device and directory in which to save a JEOL52V3.0 data file after conversion. By default, the program displays the input device and directory.

Output file: Lets you specify the name of the JEOL52V3.0 data file. You can enter only the characters acceptable for the file name. The default name is the input file name with .v30.

Output layer: Lets you specify the layer name of the JEOL52V3.0 data file up to 24 characters. The default layer name is the same as with the input file.

Conversion conditions

Figure data unit — Lets you specify the pattern data unit for the JEOL52V3.0 data file. If the button is on, the program outputs data in the unit system specified for JEOL52V3.0 by using the value in the box as the pattern data unit. If the button is off, the program disregards the value in the box and outputs data in the unit system specified for JEOL52V1.1. The default setting depends on the pattern data unit for the input file, as follows.

Input data unit	Output data unit
20	20
40	40
80	100
200	200
400	500
800	500

Accel. voltage (kV)

Lets you select the accelerating voltage of the lithography system. The limits on the pattern data unit, field size, and position set size depend on the accelerating voltage.

Field size (μm)

Lets you specify the field size after conversion in μm. All of these requirements must be met:

- Within the field size allowed for JEOL52V3.0 format
- Greater than or equal to the position set size after conversion
- The number of fields is less than or equal to 1534 in both the x and y directions.

✍ Number of fields = Chip size/(New field size – New position set size)
If the new field size is the same as the new position set size, Number of fields = Chip size/New field size.

The default field size is the same as with the input file. However, if the field size is outside the limits allowed for JEOL52V3.0, the program uses the maximum field size.

Position set size (μm)

Lets you specify the position set size after conversion in μm. All of these requirements must be met:

- Within the field size allowed for JEOL52V3.0 format
- Less than or equal to the new or old field size
- The number of position sets is less than or equal to 1534 in both the x and y directions.

✍ Number of position sets = Old field size/New position set size

By default, the subfield size of the input file is used as the position set size. However, if the subfield size is outside the limits on the position set size allowed for JEOL52V3.0, the program uses the maximum position set size.

Micro-size (μm) — Lets you specify the micro size for the pattern data, in the range 0 to 4.000. The default setting is 0.000.

User name — Lets you specify the user name up to 50 characters. The default setting is FC5152V30_USER.

File change — Lets you select a file using the left and right arrow buttons, if multiple input files are selected.

20.4 Command Execution

OK: After you set the parameters, clicking the **OK** button displays the Format conversion confirmation window.

☞ The results output to the result area are the same as with a JBX command. See “JBX Command User’s Guide.”

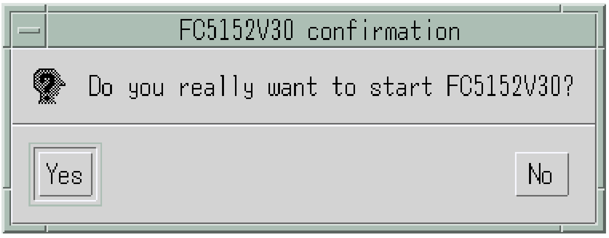


Fig. 88 Format conversion confirmation window

Yes: Starts the format conversion.

No: Returns you to the Conversion parameters window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked the **File change** button. If you have not clicked the File change buttons, the default settings are restored.

21 BLOCKING (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The block function combines the specified 2 to 9 JEOL data files. This chapter explains how to use the block function.

◆ Select desired files in the file-selection area in the JBXFILER main window, and select **Command(C)–Block**.

This displays the Block parameter window.

✍ Actually, you can block up to nine files, although you can select as many files as you like in the file-selection area.

✍ Without a license to execute this function, the **Block** command does not appear on the **Command** menu.

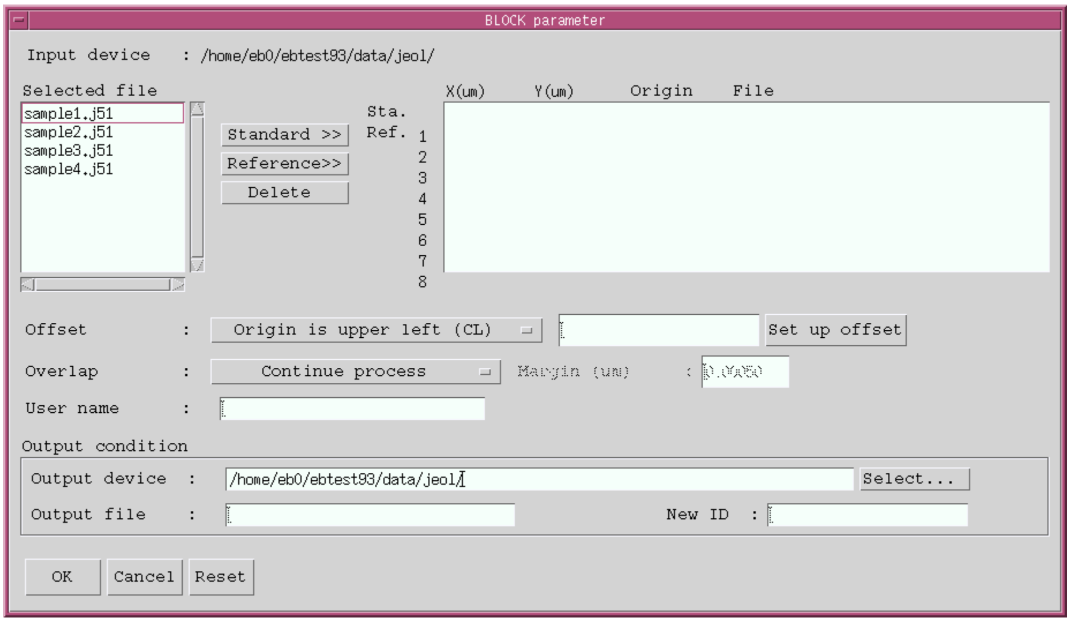


Fig. 89 BLOCK parameter window

21.1 Setting Parameters

The Block parameter window lets you set these parameters.

■ Standard and reference files

You can select the files to combine (the standard and reference files).

Standard file — In the Selected file area, select (click) the file that you want to assign as the standard file, and click the **Standard>>** button. You can specify only one standard file.

Reference files — In the Selected file area, select (click) files that you want to assign as reference files, and click the **Reference>>** button. You can select 1 to 8 reference files. Also, you can assign the same file as many times as you like. After you assign files, the block file information area looks like that shown in the following figure. This area lists the file names, format abbreviations, x, y offset values (in μm), and the specified origin positions. If you want to delete a file from the block file information area, select (click) that file, and click the **Delete** button.

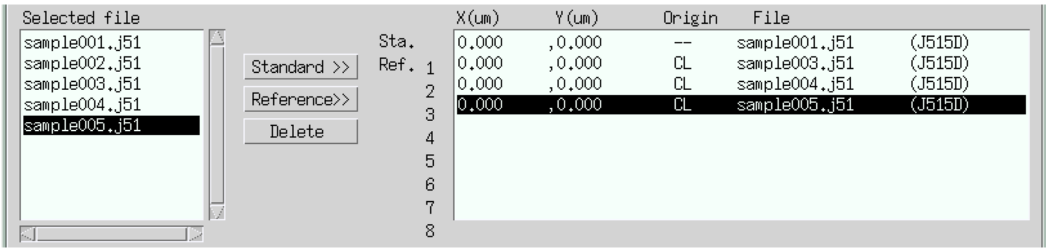


Fig. 90 Block file information area

Format abbreviations

Indication	Format name
J516A	JEOL51 (6A)
J515D	JEOL51
J52	JEOL52
V1170	JEOL52V1.1 (7000)
V21	JEOL52V2.1
V21MP	JEOL52V2.1M
V3090	JEOL52V3.0
V31	JEOL52V3.1

Specified origin position

Indication	Position
CL	Upper left corner of chip
CC	Chip center

Offset — Lets you specify the X, Y spacing (shift amount) between the standard file pattern and the reference file patterns. Here is the procedure. You cannot set the offset for the standard file.

1. Select a file in the block file information area.

   In the block file information area, select a file for which you want to set the offset.

2. Select the origin position.

   Select the origin position for setting the offset.

   •**Origin is upper left (CL)**

   The offset that you enter in step 3 is treated as the relative distance between the chip's upper left corners of the standard pattern and the reference pattern.

   •**Origin is chip center (CC)**

   The offset that you enter in step 3 is treated as the relative distance between the chip centers of the standard pattern and the reference pattern.

3. Enter the offset.

   In the Offset box, enter the x and y offset values in μm.

4. Click **Set up offset**.

   This puts the entered offset into effect. If you select **Origin is upper left (CL)** in step 2, you cannot specify a negative offset value. Setting the offset correctly updates the offset and origin in the block file information area.

Overlap — Lets you specify what to do if the specified chips are overlapping.

- **Continue process** — Outputs the chips even if they are overlapping.
- **Stop process** — Stops the block process if the chips are overlapping.
- **Remove fields (subfields)** — Removes fields or subfields from the standard pattern and then outputs data, if the chips are overlapping. If the reference patterns are overlapping, the reference pattern that was specified first is removed first.

Margin (μm) — Lets you specify the margin in μm in the range 0 to 34.000. You need to set this parameter if you specify **Stop process** or **Remove fields (subfields)** for the overlap specification.

User name — Lets you specify the user name up to 50 characters. The default setting is EBBLOCK_USER.

Output device — Lets you specify the device and directory in which to save the output file after blocking. By default, the input device and directory appears.

Output file — Lets you specify the output file name. This is a required parameter and you must enter it.

New ID — Lets you specify the ID name of the output file. The default setting is the same ID name as with the standard file.

21.2 Command Execution

OK: After you set the parameters, clicking the **OK** button displays the BLOCK confirmation window.

☞ The results output to the result area are the same as with a JBX command. See “JBX Command User’s Guide.”

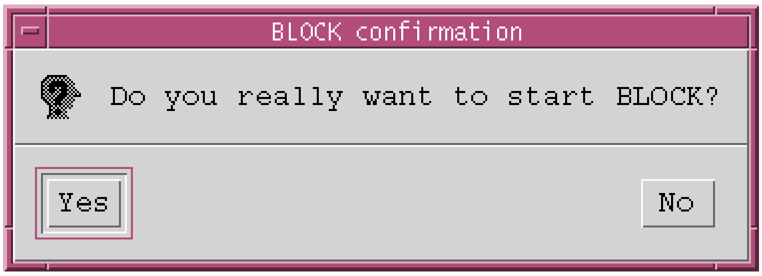


Fig. 91 BLOCK confirmation window

Yes: Starts the block process.

No: Returns you to the BLOCK parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters you changed back to the initial settings.

22 CONVERTING JEOL02 FORMAT (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The JEOL02 format conversion function converts JEOL02 format data to JEOL51 or JEOL52V3.0 format.

22.1 Procedure (for Outputting JEOL51 Format Data)

Here is how to convert JEOL02 format to JEOL51 format.

◆ Select desired files in the file-selection area in the JBXFILER main window, and select **Command(C)–Convert JEOL02 format–JEOL51**.

This displays the FC0251 parameter window.

✍ Without a license to execute this function, **Convert JEOL02 format** and **JEOL51** do not appear on the **Command** menu.

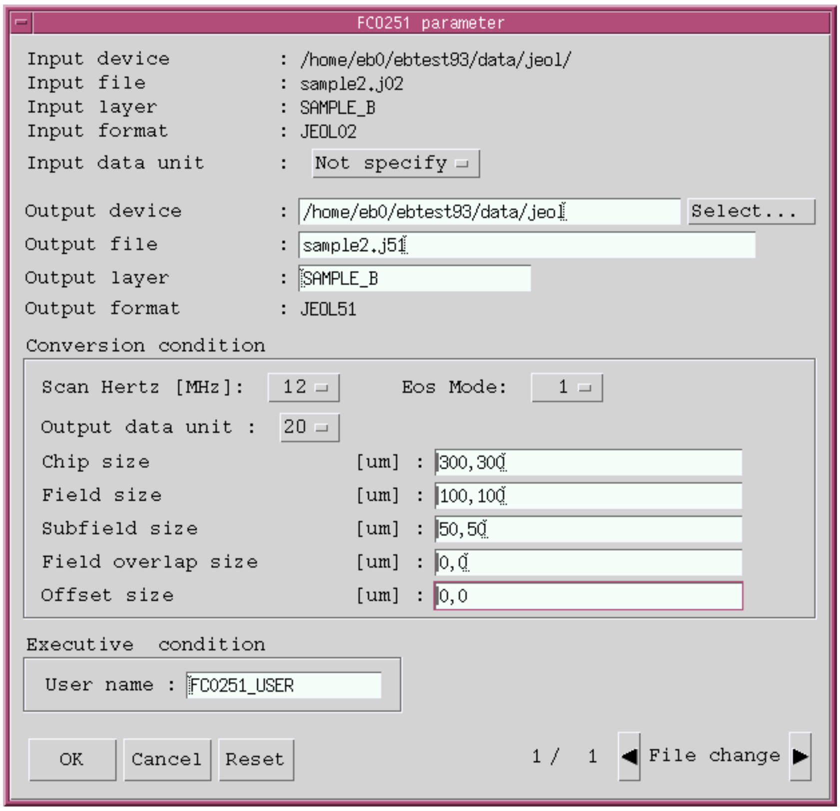


Fig. 92 FC0251 parameter window (for outputting in JEOL51 format)

22.1.1 Setting parameters

The FC0251 parameter window lets you set these parameters.

Input data unit	Lets you select the data unit of the input file.
Output device	Lets you specify the device and directory in which to save a JEOL51 data file after conversion. By default, the top line in the JEOL data list (disklist) appears.
Output file	Lets you specify the file name of the JEOL51 data file. You can enter only the characters acceptable in a file name. The default name is the input file name with .j51.
Output layer	Lets you specify the layer name of the JEOL51 data file with up to 24 characters. The default layer name is the same as with the input file.

■ Conversion conditions

Scan Hertz [MHz]	Lets you specify the scanning frequency of the system.
EOS mode	Lets you specify the EOS mode of the system.
Output data unit	Lets you specify the number of steps per micrometer (μm).
Chip size [μm]	Lets you specify the input data range in μm. Format: chip_x, chip_y This is a required parameter and you must specify it.
Field size [μm]	Lets you specify the field size in μm. Format: field_x, field_y
Subfield size [μm]	Lets you specify the subfield size in μm. Format: subfield_x, subfield_y
Field overlap size [μm]	Lets you specify the field overlap size in μm. Format: over_x, over_y
Offset size [μm]	Lets you specify the JEOL51 data position at which to output the JEOL02 data origin, in μm. Format: offset_x, offset_y

■ Executive condition

User name	Lets you specify the user name up to 50 characters.
File change	Lets you select a file using the left and right arrow buttons, if multiple input files are selected.

22.1.2 Command execution

OK: Displays the FC0251 confirmation window.

☞ The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

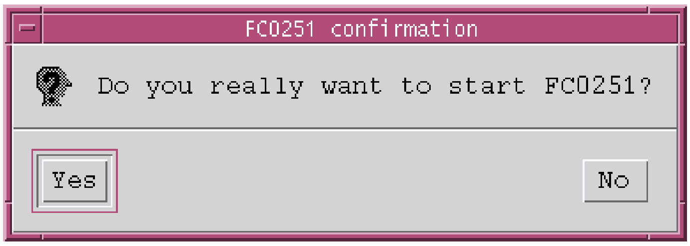


Fig. 93 FC0251 confirmation window

Yes: Starts to convert JEOL02 format.

No: Returns you to the FC0251 parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked a File change button. If you have not clicked the File change buttons, the default settings are restored.

22.2 Procedure (for Outputting JEOL52V3.0 Format Data)

Here is how to convert JEOL02 format to JEOL52V3.0 format.

◆ Select desired files in the file-selection area in the JBXFILER main window, and select **Command(C)–Convert JEOL02 format–JEOL52V3.0**.

This displays the FC0252V30 parameter window.

✍ Without a license to execute this function, **Convert JEOL02 format** and **JEOL52V3.0** do not appear on the **Command** menu.

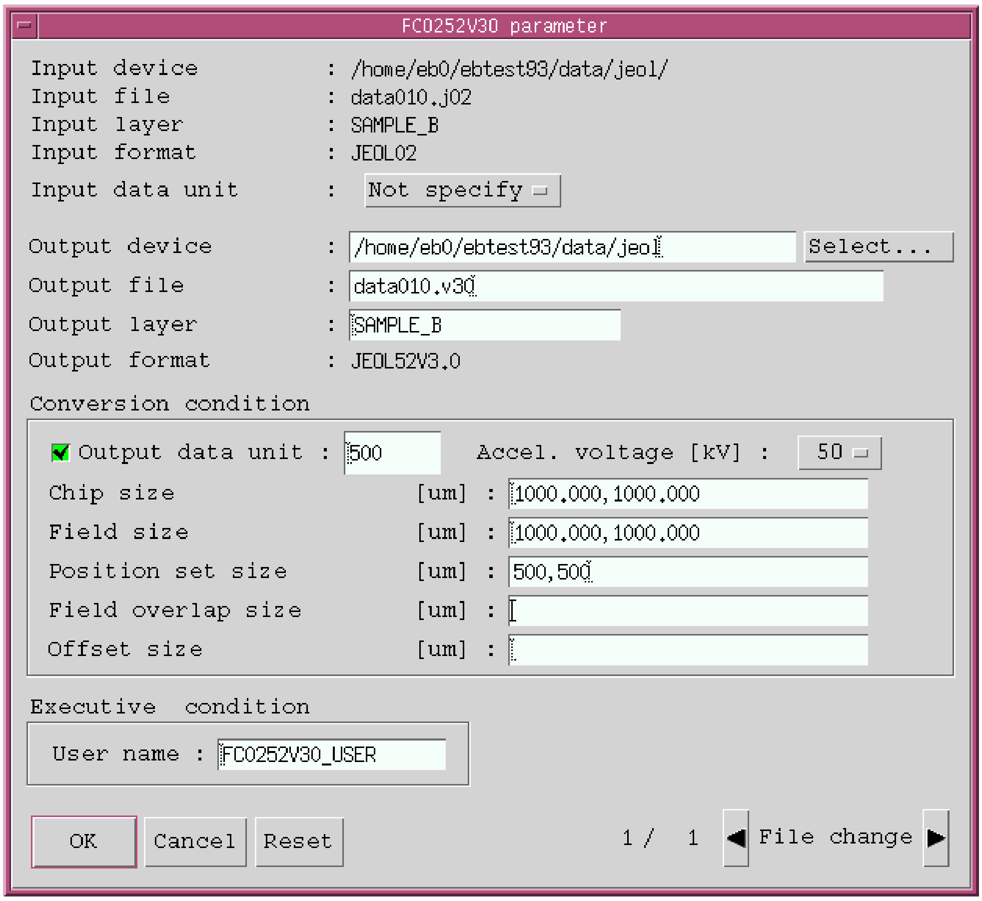


Fig. 94 FC0252V30 parameter window (for outputting in JEOL52V3.0 format)

22.2.1 Setting parameters

The FC0252V30 parameter window lets you set these parameters.

Input data unit	Lets you select the data unit of the input file. If you want to change the data unit of a JEOL02 data file with a data unit of 20, 40, or 200 and then convert that file to JEOL52V3.0 data, select the data unit of the JEOL02 data file. With **Not specify** selected, the program assumes that the JEOL02 data has the data unit specified in the Output data unit box in the Conversion condition section.
Output device	Lets you specify the device and directory in which to save a JEOL52V3.0 data file after conversion. By default, the top line in the JEOL data list (disklist) appears.
Output file	Lets you specify the file name of the JEOL52V3.0 data file. You can enter only the characters acceptable in a file name. The default name is the input file name with .v30.
Output layer	Lets you specify the layer name of the JEOL52V3.0 data file with up to 24 characters. The default layer name is the same as with the input file.

■ Conversion conditions

Output data unit: Lets you specify the pattern data unit of the JEOL52V3.0 data file. All of these requirements must be met:

- Within the pattern data unit range specified for each accelerating voltage

	Accelerating voltage [kV]		
	25	50	100
Maximum	1	1	1
Minium	500	500	1000

- Divisor of 1000
- Integer multiple of the input data unit, if that unit is specified

If the button is on, the program outputs data in the unit system specified for JEOL52V3.0, using the value in the box as the pattern data unit. If the button is off, the program disregards the value in the box and outputs data in the unit system specified for JEOL52V1.1.

Accel. voltage [kV]

Lets you select the accelerating voltage of the lithography system. The limits for the pattern data unit, field size, and position set size depend on the accelerating voltage.

Chip size [µm]: Lets you specify the input data range in µm.
Format: chip_x, chip_y
This is a required parameter and you must specify it.

Field size [µm]: Lets you specify the field size in µm.
Format: field_x, field_y
All of these requirements must be met:

- Within the field size range specified for each accelerating voltage

	Accelerating voltage [kV]		
	25	50	100
Maximum	50.000	50.000	50.000
Minimum	2000.000	1000.000	500.000

- Less than or equal to the chip size

The default size is the same as the chip size. However, if the chip size is outside the upper limit of the field size, the maximum field size is used.

Position set size [μm]

Lets you specify the position set size in μm.
Format: posset_x, posset_y
All of these requirements must be met:

- Within the position set size range specified for each accelerating voltage

	Accelerating voltage [kV]		
	25	50	100
Maximum	0.100	0.100	0.100
Minimum	2000.000	1000.000	500.000

- Less than or equal to the field size (or less than or equal to half the field size, if the chip consists of multiple fields)

The default size is the same as the field size. However, it is half the field size if the data consists of multiple fields.

Field overlap size [μm]

Lets you specify the field overlap size in μm.
Format: over_x, over_y
This parameter is in effect only if the chip consists of multiple fields. The default size is the same as the position set size (if the chip consists of one field, the default size is (0.000, 0.000)).

Offset size [μm]
Lets you specify the JEOL52V3.0 data position at which to output the JEOL02 data origin, in μm.
Format: offset_x, offset_y
The default size is (0.000, 0.000).

■ Executive condition

User name
Lets you specify the user name up to 50 characters. The default name is FC0252V30_USER.

File change
Lets you select a file using the left and right arrow buttons, if multiple input files are selected.

22.2.2 Command execution

OK: Displays the FC0252V30 confirmation window.

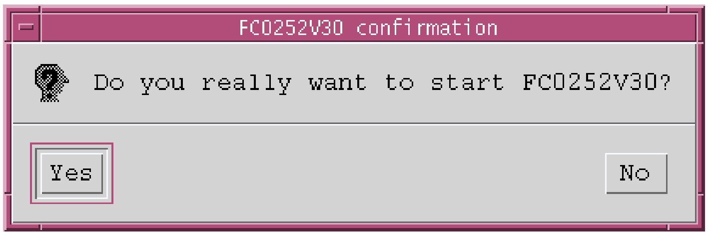


Fig. 95 FC0252V30 confirmation window

Yes: Starts to convert JEOL02 format.

The results output to the result area are the same as with a JBX command. See "JBX Command User's Guide."

No: Returns you to the FC0252V30 parameter window.

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked the **File change** button. If you have not clicked the File change buttons, the default settings are restored.

23 PREPARING PATTERNS (OPTIONAL FUNCTION)

✍ In this chapter, windows in some figures show the directory configuration, which depends on the model as follows:

Tru64 UNIX: /usr/users/eb0/•••
Example: /usr/users/eb0/ebtest/pattern
Solaris: /home/eb0/•••
Example: /home/eb0/ebtest/pattern
OpenVMS: DKA100:[SAMPLE.PATTERN•••]
Example: DKA100:[SAMPLE.PATTERN.JEOL01]

The pattern-preparation function (PATRDY) converts data in various JEOL formats other than JEOL51 to JEOL52 (V1.1/V3.0/V3.1) format data that do not contain a map library, and converts data that cannot be written due to system limitations to data that fall within the limitations. Also, the function divides the various JEOL format data into fields, and converts them to JEOL52 (V1.1/V3.0/V3.1) format data.

Here is how to use the pattern-preparation function:

◆ Select desired files in the file-selection area, and select **Command(C)–PATRDY**.

This displays the Conversion parameters window.

✍ Without a license to execute this function, **PATRDY** does not appear on the **Command** menu.

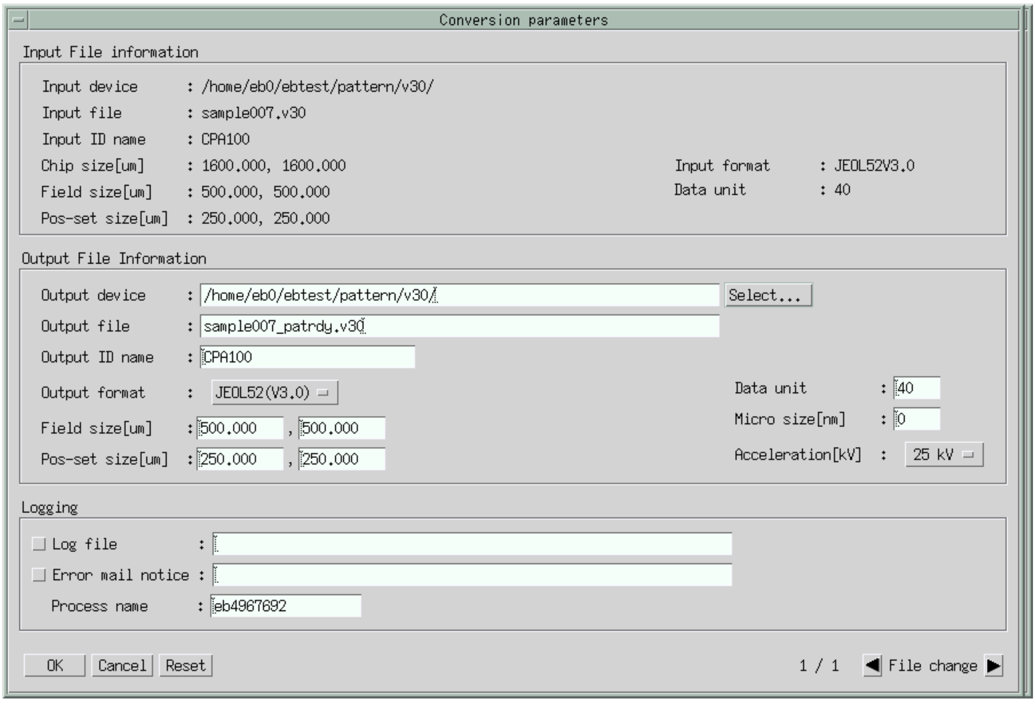


Fig. 96 Conversion parameters window

23.1 Setting Parameters

The Conversion parameters window lets you set these parameters.

■ Output file information

- Output device

  This box lets you specify the destination to which to output a file after conversion.

  This is a required parameter and you must set it.
- Output file

  This box lets you specify the output data file name. You can enter only the characters acceptable in a file name.

  This is a required parameter and you must set it.
- Output ID name

  This box lets you specify the ID name of the output data up to 24 characters.

  This is a required parameter and you must set it.
- Output format

  This option menu lets you select the output data format.

  This is a required parameter and you must set it.
- Field size [µm]

  These boxes let you specify the field size for conversion in µm. The limits depend on the accelerating voltage or the EOS mode.
- Pos-set size [µm]

  These boxes Let you specify the field size for conversion in µm. The limits depend on the accelerating voltage or the EOS mode.
- Data unit

  This box lets you specify the number of steps per micrometer (µm). You can specify a divisor of 1000, or 2000, 4000, or 8000.

  You can specify 2000, 4000, or 8000 only in the JBX-6300 series.
- Micro size [nm]

  This box lets you specify the micro size for the pattern data, in nm.
- Acceleration [kV]

  Depends on the system. Changing this parameter changes the limits for the field size and position set size. You can select 25, 50, or 100 kV.

  This parameter does not appear in the JBX-6300 series.

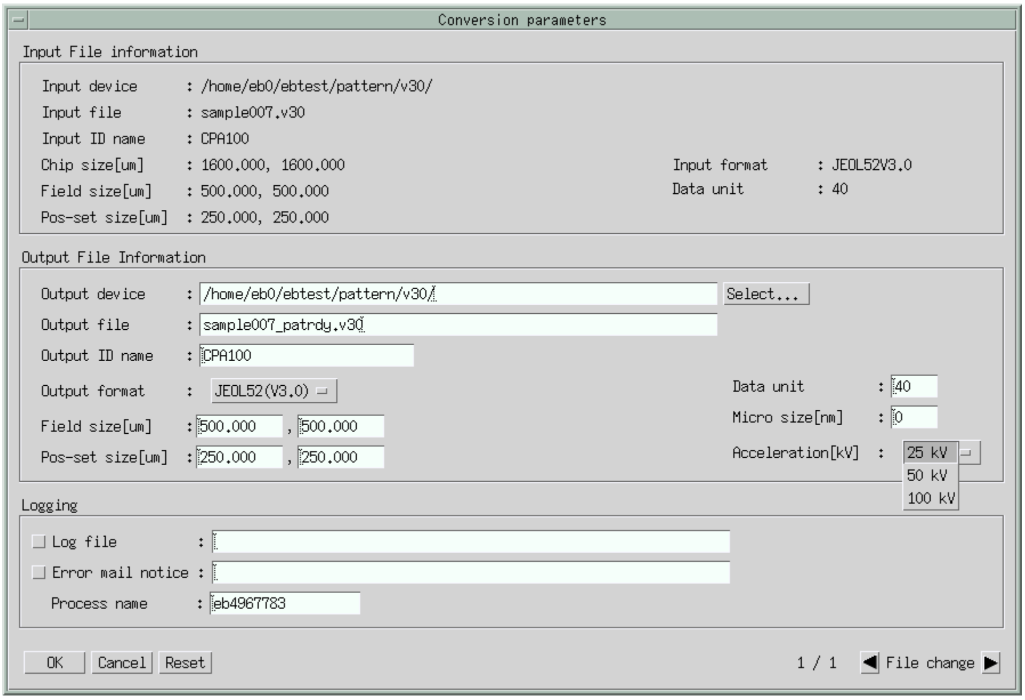


Fig. 97 Conversion parameters window (in systems other than the JBX-6300 series)

- EOS mode

  This parameter depends on the system. Changing this parameter changes the limits for the field size and position set size. You can select **1**, **2**, **3**, **4**, **5**, or **6**.

  ✍ This parameter does not appear in systems other than the JBX-6300 series.

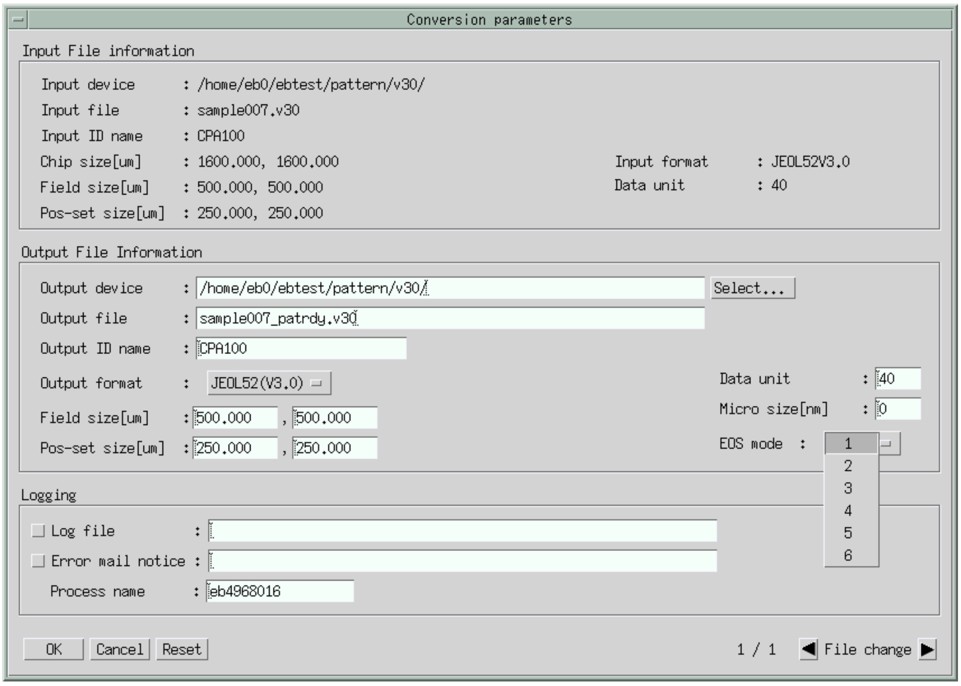


Fig. 98 Conversion parameters window (JBX-6300 series)

Logging

- Log file

  This box lets you specify whether or not to output a log file for each startup to a directory other than the system common directory. Selecting this button (the button turns green) will output a log file for each startup. You can specify a device and directory for the file name. The default settings depend on the settings in the initialization file (patrdyrc). If the initialization file does not contain the settings, the program uses these settings:

Directory:	~/eb/log/
File name:	patrdy process ID
File type:	.log

- Error mail notice

  This box lets you specify whether or not to send an error-detection notification mail. Selecting this button (the button turns green) will send an error-detection notification mail. The default mail address depends on the setting in the initialization file (patrdyrc).

  no_mail: Refrains from sending mail.
  mail_address: Sends mail to the recorded mail address.

- File change

  These left and right arrow buttons let you select a file, if multiple input files are selected.

23.2 Command Execution

OK: Executes the pattern-preparation function.

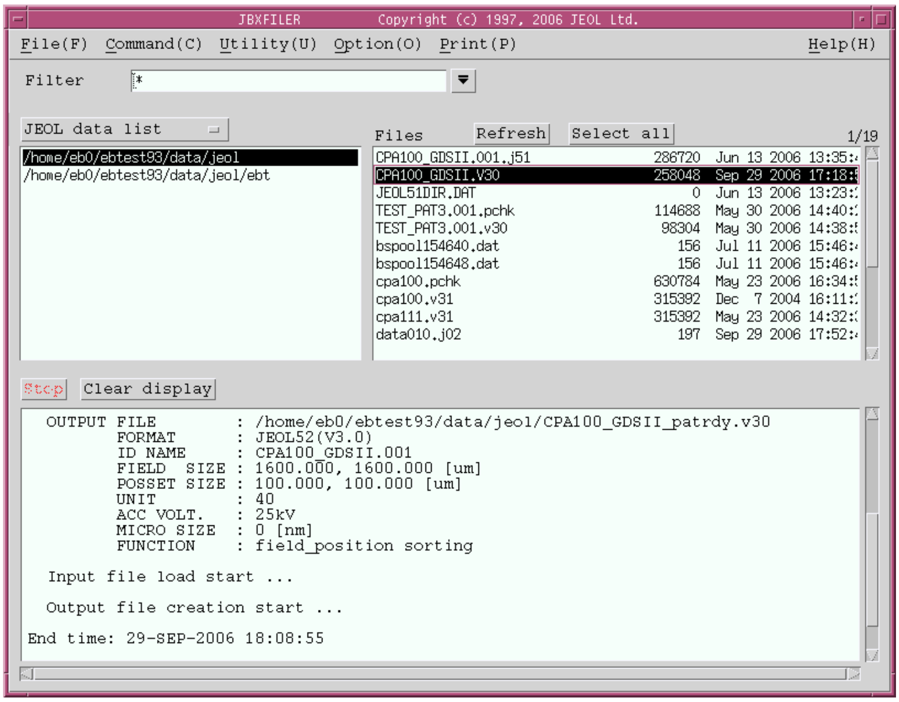


Fig. 99 Example of pattern preparation

At the same time of the pattern preparation, the program outputs a history file as follows.

```
---------------------------------------------------------------------------
patrdy Ver 4.302 (2006 07/19)                    Copyright (c) 2004, 2006 JEOL Ltd.
Start time: 29-SEP-2006 18:09:47

  INPUT   FILE          : /home/eb0/ebtest93/data/jeol/CPA100_GDSII.V30
          FORMAT          : JEOL52(V3.0)
          ID NAME         : CPA100_GDSII.001
          CHIP     SIZE : 1600.000, 1600.000 [um]
          FIELD    SIZE : 1600.000, 1600.000 [um]
          POSSET SIZE : 100.000, 100.000 [um]
          UNIT          : 40
  OUTPUT FILE           : /home/eb0/ebtest93/data/jeol/CPA100_GDSII_patrdy.v30
          FORMAT          : JEOL52(V3.0)
          ID NAME         : CPA100_GDSII.001
          FIELD    SIZE : 1600.000, 1600.000 [um]
          POSSET SIZE : 100.000, 100.000 [um]
          UNIT          : 40
          ACC VOLT.     : 25kV
          MICRO SIZE    : 0 [nm]
          FUNCTION        : field_position sorting

  Input file load start ...

  Output file creation start ...

End time: 29-SEP-2006 18:09:47
Lapse of time: Real time: 00:00:00.10,    CPU uptime: 00:00:00.10
```

Fig. 100 Pattern-preparation history file

Cancel: Cancels the process and returns you to the main window of JBXFILER.

Reset: Resets the parameters to the settings that were current when you clicked the **File change** button. If you have not clicked the File change buttons, the default settings are restored.

24 APPENDICES

24.1 List of Messages

Here is a list of messages used in JBXFILER.

The file doesn't exist.

Meaning: 1. No file matches the search name.
2. The selected file does not exist.

Action: 1. Set the search name according to the existing file, or select the device and directory that contains the file.
2. Click the Refresh button, and then select a file again.

The error occurred in the search finishing step of the file.
It failed in the management after the file search.

Meaning: An error occurred in runtime processing (file-search end process).

Action: Contact your JEOL service office.

The input form of the date is illegal.

Meaning: The date entry format is incorrect.

Action: Enter the date in the "mmm dd yyyy hh:mm:ss" format.

mmm:	month (3 letters)
dd:	day
yyyy:	year
hh:	hour (in 24-hour notation)
mm:	minute
ss:	second

The relations of the size are illegal.

Meaning: The specified size is incorrect (the lower- and upper-limit settings conflict with each other).

Action: Enter correct conditions.

The relations of the date are illegal.

Meaning: The specified date is incorrect (the before and after settings conflict with each other).

Action: Enter correct conditions.

The file isn't selected.

Meaning: No file is selected.

Action: Select one or more files before execution.

The file to agrees in the selection condition doesn't exist.

Meaning: No file matches the search condition.

Action: Change the search condition.

The layer isn't found.

Meaning: No layer matches the search condition.

Action: Change the layer search condition.

Select a layer name.
Meaning: No layer name is selected.
Action: Select a layer name.

New ID name isn't specified.
Meaning: No new ID name is specified.
Action: Specify all the new ID names.

It is ignored except for the first file.
Meaning: Two or more files are selected.
Action: Select only one file.

Even the 12th adopted from the beginning.
Meaning: 13 or more files are selected.
Action: Select 12 or fewer files.

Direction file failed to create.
Meaning: Creation of the direction file failed.
Action: Check the ~/eb/command directory or the directory specified by the environment variable CNVSCRPATH to see if a file can be created there.

The same color as other files can't be specified.
Meaning: The color you are trying to specify is the same as the color of another file.
Action: Specify a color that is different from the colors of the other files.

Offset can't be specified to be a standard file.
Meaning: You are trying to set the offset for the standard file.
Action: You cannot set the offset for the standard file.

Failed to create information table.
Meaning: Failed to allocate the internal table.
Action: Check the settings related to the memory of the system. Select fewer files.

Enter the new ID name.
Meaning: No new ID name is specified.
Action: Specify a new ID name.

There is a file which isn't given a New ID name.
Meaning: Not all the new ID names are set.
Action: Set all the new ID names.

The format with no executive permission was found.
The format which isn't available (isn't defined) was found.
Meaning: The format of the selected file is not executable.
Action: Contact your JEOL service office.

Failed to investigate the format name.
The format can't be distinguished.

Meaning: Some of the selected files are in formats that JBXFILER does not know.
Action: Select files in formats that JBXFILER supports.

The format which is not corresponded to was found.

Meaning: The function to be performed cannot handle the selected files.
Action: Deselect the unsupported files from the selected files.

It exceeds the number (99) which can be done backup.

Meaning: The number of files you specified for backup exceeds 99.
Action: Specify 99 or fewer files.

All the formats of files to be saved support aren't the same.

Meaning: Not all the selected files are in the same format.
Action: Select files that are all in the same format.

The output device isn't specified.
Output device is not specified.

Meaning: The output device is not specified.
Action: Specify the output device.

This function can't be used at present.

Meaning: The selected function is unavailable.
Action: Contact your JEOL service office.

A files which is not in the JEOL01 format is specified.

Meaning: Some of the selected files are not in JEOL01 format.
Action: Select only JEOL01 format files.

Even the ninth is adopted from the beginning.

Meaning: 10 or more files are selected.
Action: Select 9 or fewer files.

Dump start record designation is illegal.

Meaning: The start record is incorrect.
Action: The start record must be smaller than the end record.

Dump end record designation is illegal.

Meaning: The end record is incorrect.
Action: The end record must be greater than the start record. It must be less than or equal to the record size.

The designation of the upper left coordinate is illegal.

Meaning: The specified coordinates are incorrect.
Action: Set the coordinates correctly. Check the coordinates together with the lower right coordinates.

The designation of the lower right coordinate is illegal.
Meaning: The specified coordinates are incorrect.
Action: Set the coordinates correctly. Check the coordinates together with the upper left coordinates.

The parameter designation of the screen is illegal.
Meaning: The window parameter settings are incorrect.
Action: Set the parameters correctly.

The designation of the magnification is illegal.
Meaning: The specified magnification is incorrect.
Action: Set the parameter correctly.

Failed to re-read a system parameter file.
Failed to read a system parameter file.
Meaning: Reading of the system parameter file (ebsys.prm) failed.
Action: Check the read privilege on the system parameter file. The ~ebsys/config directory contains the system parameter file.

Resizing can't be specified. Resizing will be cleared.
Meaning: The conditions do not allow resizing to be specified, so resizing is forcibly canceled.
Action: Check the relationship between the EOS mode and the scanning step.

Failed to investigation a format of the magnetic tape.
Meaning: The investigation of the format of the magnetic tape failed. The loaded magnetic tape is in an unsupported format.
Action: Load a magnetic tape in a supported format.

The format of the magnetic tape can't be used at present.
Meaning: The tape in use is in an unusable format.
Action: Contact your JEOL service office.

A device directory to restore isn't specified.
Meaning: The device and directory to be restored are not specified.
Action: Specify a device and directory.

The object of the restoring isn't selected.
Meaning: Nothing is selected to be restored.
Action: Select what is to be restored.

Failed to create the direction file of restoring.
Meaning: Creation of the direction file for restoring failed.
Action: Check the ~/eb/command directory and the directory specified by the environment variable CNVSCRPATH to see if a file can be created there.

The designation of field position is illegal.
Meaning: The field position is outside the chip.
Action: Specify coordinates inside the chip.

Failed to create the parameter file for the printing.
Meaning: Creation of the parameter file failed.
Action: Check the device and directory where you tried to create the file to see if a file can be created there.

Failed to open the parameter file for the printing.
Meaning: Opening of the parameter file failed.
Action: Check to see if the parameter file exists, and if you have read privilege.

The format which can't be used for the input file of CONV was found.
Meaning: Some CONV input files are in formats that are not specifiable.
Action: Specify only JEOL52, JEOL52V1.1, JEOL01, STREAM, JEOL52V3.0, or JEOL52V3.1 format files.

The designation of the micro size is illegal.
The designation of resizing is illegal.
The designation of the amount of frame is illegal.
The designation of scaling value is illegal.
Meaning: The specified value is incorrect.
Action: Set the parameter correctly.

The designation of the chip size is illegal.
The designation of the field size is illegal.
The designation of the sub-field size is illegal.
The designation of the position set is illegal.
Meaning: The specified value is incorrect.
Action: Set the parameter correctly.

The designation of the maximum beam size is illegal.
The designation of the maximum beam area is illegal.
The designation of the trapezoid beam size is illegal.
The designation of the arc length is illegal.
Meaning: The specified value is incorrect.
Action: Set the parameter correctly.

The contents of the conversion parameter were initialized.
Meaning: The parameters related to the displayed file were initialized.
Action: None.

The coordinate designation of the screen is illegal.
Meaning: The window specification is incorrect.
Action: Specify the coordinate position in the chip with the origin at the lower left.

The number of the sub-field division exceeds limitation value.
Meaning: The number of field divisions exceeds the limit.
Action: Check the chip size, field size, and subfield size.

Failed to create the parameter file for CONV.
Meaning: Creation of the parameter file failed.
Action: Check the device and directory where you tried to create the file to see if a file can be created there.

Failed to open the parameter file for CONV.
Meaning: Opening of the parameter file failed.
Action: Check to see if the parameter file exists, and if you have read privilege.

The conversion range of STREAM is illegal.
Meaning: The chip area specification in the STREAM format is incorrect.
Action: Specify correct values.

The parameter file is illegal.
Meaning: Reading of the parameter file failed.
Action: Check the read privilege of the parameter file.

It exceeded number (32) which a layer number could be registered.
Meaning: There are more than 32 layer number settings.
Action: Set 32 or fewer layer numbers.

The top structure name isn't specified.
Meaning: The top-structure name is not set.
Action: Set the top-structure name.

The layer number isn't set.
Meaning: The layer number is not set.
Action: Set the layer number.

There is a file which the STREAM parameter isn't set up to.
Meaning: The top-structure name, chip area, and layer number are not specified.
Action: Set all the parameters.

The layer number isn't specified.
Meaning: The layer number and data type are not specified.
Action: Set the parameters correctly.

The default file is illegal.
Meaning: The contents of the default file (default.prm) are incorrect.
Action: Check the contents of the default file.

EBSYSIDTALL.EXE is illegal.
Meaning: The contents of ebsysidtall.exe are incorrect.
Action: Contact your JEOL service office.

The designation of the layer number is illegal.
Meaning: The layer number and data type are specified incorrectly.
Action: Set the parameters correctly.

Failed to create the file for the printer output.
Meaning: Creation of the temporary file for printing failed.
Action: Check the ~/eb/command directory and the directory specified by the environment variable CNVSCRPATH to see if a file can be created there.

There is no thing selected in the result indication area.
Meaning: Nothing is selected in the result area.
Action: Select something in the area.

There are no contents of the indication in the result indication area.
Meaning: The result display area is empty.
Action: Execute the command when something is displayed.

Failed to get the information of the JBX command image files.
Meaning: Acquisition of the image information on the JBX command failed.
Action: Contact your JEOL service office.

It can't be specified in the magnification by 0.
Meaning: You cannot specify zero for the magnification.
Action: Enter a magnification other than 0.

The model which can be outputted isn't defined.
Meaning: The models to which data can be output are not specified.
Action: Contact your JEOL service office.

The ID name has already existed.
Meaning: The specified ID name already exists.
Action: Change the ID name.

The specified ID name is used with others.
Meaning: The specified ID name is already used in other file settings.
Action: Change the ID name to an unused one.

The default file is illegal.
Meaning: The default file (default.prm) is incorrect.
Action: Check the contents of the default file.

The relations of the parameters in DEFAULT.PRM are illegal.
Meaning: Although the parameter settings in the default file (default.prm) are correct, the combination of the outline, resize, reverse tone, and shot modulation parameters is incorrect.
Action: Check the contents of the default file.

Reverse tone can't be specified. (DEFAULT.PRM)

Meaning: You cannot specify reverse tone in the default file (default.prm).

Action: Check the contents of the default file. You cannot specify it if outlining is not specified.

Resizing can't be specified. (DEFAULT.PRM)

Meaning: You cannot specify resizing in the default file (default.prm).

Action: Check the contents of the default file. You cannot specify it if outlining is not specified.

There is a wrong point in the parameter file.

Meaning: The contents of the parameter file are incorrect.

Action: Check the contents of the parameter file.

The relations of the parameter of the parameter file are illegal.

Meaning: Although the parameter settings in the parameter file are correct, the combination of the outline, resize, reverse tone, and shot modulation parameters is incorrect.

Action: Check the contents of the parameter file.

The reverse tone can't be specified. (parameter file)

Meaning: You cannot specify reverse tone in the parameter file.

Action: Specify it together with outlining.

Resize can't be specified. (parameter file)

Meaning: You cannot specify resizing in the parameter file.

Action: Specify it together with outlining.

It failed in the creation of a sub-process.

Meaning: An error occurred during sub-process creation from the runtime routine.

Action: Contact your JEOL service office.

The result of the command execution is illegal.

Meaning: The result of the execution of the JBX command is incorrect.

Action: Check the message displayed as the result, referring to the list of the JBX command messages.

The information of file failed to get.

Meaning: Acquisition of the file information failed.

Action: Check the device and directory selected in the device-selection area to see if it contains files without read privilege.

The device-list failed to read.

Meaning: Reading of the device list failed.

Action: Check the read privilege on the device list.

The designation of the offset is illegal.

Meaning: The offset specification is incorrect.
Action: Set the offset correctly.

The device-list failed to write.

Meaning: Writing of the device list failed.
Action: Check the write privilege of the device list.

The contents of the device-list are empty.

Meaning: The device list is empty.
Action: Specify devices in the device list.

The file which can’t be overlapped is ignored.

Meaning: There are files whose field sizes are too large for overlapping.
Action: Check the field sizes, and select files that can be used for overlapping.

As for all the selected files as well, it can’t be overlapped.

Meaning: The field sizes of all the files are too large for overlapping.
Action: Check the field sizes, and select files that can be used for overlapping.

The parameter file doesn’t exist.

Meaning: The parameter file does not exist.
Action: Check to see if the specified file exists.

The data file for the PCHK-plot is illegal.

Meaning: The specified file is not a PCHK plotting data file.
Action: Specify a PCHK plotting data file.

The chip size was rounded in the appropriate unit.
The field size was rounded in the appropriate unit.
The sub-field size was rounded in the appropriate unit.
The position set was rounded in the appropriate unit.
The micro size was rounded in the appropriate unit.
The modified chip size was rounded in the appropriate unit.

Meaning: The specified size was rounded to ensure consistency with the output data unit.
Action: If the rounded size is improper, specify the size again.

The chip size, the field size and the sub-field size were rounded in the appropriate unit.
The chip size, the field size and the position set size were rounded in the appropriate unit.
The chip size and the field size were rounded in the appropriate unit.
The chip size and the sub-field size were rounded in the appropriate unit.
The chip size and the position set size were rounded in the appropriate unit.
The field size and the sub-field size were rounded in the appropriate unit.
The field size and the position set size were rounded in the appropriate unit.
The field size and the micro-size were rounded in the appropriate unit.

The position set size and the micro size were rounded in the appropriate unit.
The field size, the position set size and the micro size were rounded in the appropriate unit.

Meaning: The specified size was rounded to ensure consistency with the output data unit.
Action: If the rounded size is improper, specify the size again.

There is a mistake in the designation of amount of shift.

Meaning: Either the x or y shift amount, but not both, is specified.
Action: Specify both the x and y shift amounts.

A specified output data unit is outside the limitation range.
A specified shift amount is outside the limitation range.

Meaning: The specified output unit (shift amount) is outside the limits.
Action: Specify a value within the limits.

The format which can't be used for the input file of PSP was found.

Meaning: Some specified files are in formats that the pattern-separation function cannot handle.
Action: Select JEOL51, JEOL52, JEOL52V1.1, JEOL52V2.1, JEOL52V3.0, or JEOL52V3.1 format files.

The threshold value isn't specified.

Meaning: The threshold is not specified.
Action: Enter the threshold.

The threshold value is illegal. boundary value > 0

Meaning: The specified threshold is zero or less, or has four or more decimal places.
Action: Specify the threshold within the limits.

The outline width isn't specified.

Meaning: The outline width is not specified.
Action: Enter the outline width.

The outline width is illegal. outline width > 0

Meaning: The specified outline width is zero or less, or has four or more decimal places.
Action: Enter the outline width within the limits.

The amount of resizing isn't specified.

Meaning: The resize amount is not specified.
Action: Enter the resize amount.

The resize amount is illegal. Amount of resizing > 0

Meaning: The specified resize amount is zero or less, or has four or more decimal places.
Action: Enter the resize amount within the limits.

The input range is outline width > amount of resizing > 0.

Meaning: The relationship between the outline width and resize amount is incorrect.

Action: Check the relationship between the outline width and resize amount, and specify them again.

The micro size isn't specified.

Meaning: The micro size is not specified.

Action: Enter the micro size.

The layer name of the small figures or the outline figures isn't specified.

Meaning: The layer name for small or outline patterns is not specified.

Action: Specify the layer name for small or outline patterns.

The layer name of the big figures or the inside figures isn't specified.

Meaning: The layer name for large or inside patterns is not specified.

Action: Specify the layer name for large or inside patterns.

The layer name overlaps.

Meaning: The layer names for small and large patterns or for outline and inside patterns are the same.

Action: Specify different layer names.

The shot rank value isn't specified.

Meaning: The shot rank is not specified (in two places).

Action: Specify the shot rank in two places.

The shot rank value is illegal.

Meaning: The shot rank is outside the limits (in two places).

Action: For data formats other than JEOL52V3.0, specify the shot rank in the range $0 \leq$ shot rank ≤ 63.
For the JEOLV3.0 (JBX-6300 series) data format, specify the shot rank in the range $0 \leq$ shot rank ≤ 255.
For the JEOL52V3.0 (JBX-9300FS) data format, specify the shot rank in the range $0 \leq$ shot rank ≤ 1023.

The table file name isn't specified.

Meaning: The shot-rank table file name is not specified.

Action: Specify the shot-rank table file name.

The format which can't be used for the input file of FC5152V30 is found.

Meaning: Some specified files are in formats that the format-conversion function cannot handle.

Action: Select JEOL51 format data.

The output file name is illegal.

Meaning: The output file name is incorrect.

Action: Specify a correct output file name.

The designation value of the field size is outside the limitation range.
The designation value of the position set size is outside the limitation range.

Meaning: The specified field size (position set size) is outside the limits.
Action: Specify a value within the limits.

Field size is too small to process convert in these conditions.
Position set size is too small to process convert in these condition.

Meaning: Because the specified field size (position set size) is too small, the number of fields (position sets) in the x or y direction exceeds the limit.
Action: Increase the field size (position set size).

The field size for output must be equal to or smaller than that of the input data.
The position set size for output must be equal to or smaller than that of the input data.

Meaning: The specified field size (position set size) is larger than the field size of the input data.
Action: Specify a value that is less than or equal to the field size of the input data.

The relations between the field size and the position set size are illegal.

Meaning: The specified position set size is larger than the field size.
Action: Specify a field size that is larger than the position set size.

The format which can't be used for input file of BLOCK was found.

Meaning: Some input files are in formats that the block function cannot combine.
Action: Specify JEOL51, JEOL52, JEOL52V1.1 (7000), JEOL52V2.1, JEOL52V2.1M, JEOL52V3.0, or JEOL52V3.1 format files.

The standard file has already been specified.

Meaning: A standard file is already specified, yet you are trying to specify another.
Action: You can specify only one standard file. If you want to change the standard file, first deselect it and then specify a new one.

The reference file can't be specified any further.

Meaning: The reference files are all specified, yet you are trying to specify another.
Action: You can specify up to eight reference files. If you want to change a reference file, first deselect an unnecessary file and then specify a new one.

The standard file isn't specified.

Meaning: The standard file is not specified.
Action: Specify the standard file.

The reference file isn't specified.

Meaning: No reference file is specified.
Action: Specify one to eight reference files.

The format doesn't correspond.

Meaning: The formats of the files to combine are not the same.
Action: To combine files, select files in the same format.

The margin is the outside the limitation range.
Meaning: The specified margin is incorrect.
Action: Set the margin correctly.

Some files which are not JEOL02 format is specified.
Meaning: Some selected files are not in JEOL02 format, so the JEOL02 format-conversion function cannot handle them.
Action: Select only JEOL02 format files.

The designated value of the chip size is outside the limitation range.
The designated value of the overlap size is outside the limitation range.
The designated value of the offset size is outside the limitation range.
Meaning: The specified value is outside the limits.
Action: Specify a value within the limits.

The designation of the overlap size is illegal.
The designation of the offset size is illegal.
Meaning: The specified field-overlap size or offset size is incorrect.
Action: Set the size correctly.

The chip size isn't specified.
Meaning: The chip size is not specified.
Action: Set the chip size.

A result of a calculation of the modified chip size exceeds limitation value.
Meaning: The modified chip size calculated from the specified conversion parameters is outside the limits.
Action: Set the conversion parameters so that the modified chip size falls within the limits.

The specified output unit parameter isn't recognized in the format.
Meaning: The specified value is not allowed for the format.
Action: Specify a divisor of 1000 in the range 1 to 1000. If the output format is JEOL52V3.0 (6300), you can also specify 2000, 4000, and 8000.

The process information table is full. Please carry out after the process end under present execution.
Meaning: The process cannot be executed because the number of registered processes exceeded the limit.
Action: After the running process completes, execute the desired process again.

Unmatched ".
Unmatched '.
Meaning: The number of double quotes (") or single quotes (') in the command line specified in the command-input process is incorrect.
Action: Be sure to use quotation marks in pairs.

The specified chip size is over convertible size (2000000um).
Meaning: The specified chip size exceeds the limit for data conversion.
Action: Specify a chip size less than or equal to 2000000 μm.

A file which is not in the JEOL01, PG3000, or PG3600 format is specified.
Meaning: Some specified files are not in JEOL01, PG3000, or PG3600 format.
Action: Specify only JEOL01, PG3000, or PG3600 format files.

Please choose the file of the same format.
Meaning: The selected files are in multiple formats.
Action: Select only files with the same format.

The specified offset is outside the limitation range. (0.0 – 9999999.9)
The specified offset is outside the limitation range. (–9999999.9 – 9999999.9)
Meaning: The specified offset is outside the limits.
Action: Set the offset within the limits.

Failed to create the file for batch starting.
Meaning: Creation of the batch-startup file for data conversion failed.
Action: Check the ~/eb/command directory or the directory specified by the environment variable CNVSCRPATH to see if a file can be created there.

The output file name is not specified.
Meaning: The output file name is not specified.
Action: Specify the output file name.

A space cannot be used for the head of an output layer name. The space was deleted.
A space cannot be used for the head of ID name of JEOL format except JEOL51. The space was deleted.
Meaning: Space characters at the beginning of the layer name (ID name) were deleted.
Action: Do not use space characters at the beginning.

The name is already used to another file.
Meaning: The specified file name is used for another file specification.
Action: Specify a different name.

The same name already exists.
Meaning: A file with the same name exists.
Action: Specify a different name.

The name has invalid character(s).
Meaning: The file name contains unusable characters.
Action: Remove the invalid characters, and specify a name again.

24.2 JBXFILER Resource File

The filer resource file defines the resources of JBXFILER.

The public resources are screen colors, filter templates, and ones specific to JBXFILER. Operation is not guaranteed if you modify other resources; do it at your own risk.

File name	Jbxdtp
File attribute	Text file
Location	~ebsys/config/X11/ja_JP.eucJP ~ebsys/config/X11/en_US.ISO8859-1
Contents	Template settings, screen colors, and resources specific to JBXFILER
Format	Same as that of the X-Window resource file. The resources specific to JBXFILER are as follows.

printQueName	Specifies the name of the printer to use. When omitted, the default printer is used.
keepLine	Specifies the maximum number of lines that the result area can save. The default setting is 5000.
language	Specifies the display language. Specify 0 for Japanese or 1 for English. The default setting is 0 (Japanese). ✍ This item takes effect only in OpenVMS version of JBXFILER.
netTime	Specifies the time interval for network monitoring, in seconds. The default setting is 3600 (s).

Example

```
!
!    Templates (up to seven settings)
!
Jbxdtp*pb_filter1.labelString : *                ! Template 1
Jbxdtp*pb_filter2.labelString : *.J51            ! Template 2
Jbxdtp*pb_filter3.labelString : *.J52            ! Template 3
Jbxdtp*pb_filter4.labelString : *.v11            ! Template 4
Jbxdtp*pb_filter5.labelString : *.v30            ! Template 5
Jbxdtp*pb_filter6.labelString : *.v31            ! Template 6
Jbxdtp*pb_filter7.labelString : *.pchk           ! Template 7
!
!    Color definitions
!
Jbxdtp*traversalHighlight          :  DimGray
Jbxdtp*topShadowColor              :  GhostWhite

Jbxdtp*background                  :  LightGray
Jbxdtp*bottomShadowColor           :  DimGray
Jbxdtp*shadowThichness             :  4
Jbxdtp*XmScrollBar*background      :  LightGray
Jbxdtp*XmScrollBar*troughColor     :  Gray
Jbxdtp*XmList.background           :  Mintcream
Jbxdtp*XmTextField.background      :  Mintcream
Jbxdtp*XmText.background           :  Mintcream
!
!    Application-specific resources
!
Jbxdtp*printQueName         : XXXXX      ! Printer name
Jbxdtp*keepLine             : 2000       ! Number of lines to save
Jbxdtp*language             : 0          !   Display   language   [0:
Japanese]
```

24.3 Data Conversion Parameter File

The data conversion parameter file contains the parameter settings for data conversion. This is a file for CONV; JBXFILER also uses this file.

File name	Any file name	
File attribute	Text file	
Location	Any directory on UNIX	
Contents	Parameter settings for data conversion	
Format	LAYER_NAME	Layer name
	MICRO_SIZE	Micro size (μm)
	MAGNIFICATION	Magnification (integer)
	OUTPUT_STEP	Output data unit
	CHIP_SIZE	Chip size x, y (μm)
	FIELD_SIZE	Field size x, y (μm)
	SUBFIELD_SIZE	Subfield size x, y (μm)
	SCALING	Scaling value
	RESIZE	Resize amount (μm)
	MIRROR	Mirror (NO/X/Y)
	ROTATION	Rotation (NO/90/180/270)
	OUTLINE	YES/NO
	SHOT_MODU	Y/N Y area shot rank, line shot rank Y area shot rank
	REVERSE_TONE	NO/frame size x, y (μm)
	WINDOW	NO POS lower left x, y, upper right x, y (μm) NEG lower left x, y, upper right x, y (μm)
	ARCH_LENGTH	Arch length (μm)
	DATA_UNIT	MICRON/INCH
	MAX_BEAM_AREA	Maximum beam area (μm)
	MAX_BEAM_SIZE	Maximum beam size (μm)
	TRAP_BEAM_SIZE	Trapezoid beam size (μm)

● Example

```
! ebconv parameter file created by JBXdtp-filer
LAYER_NAME          TEST-ID_NAME
MICRO_SIZE          1, 2
MAGNIFICATION       2
OUTPUT_STEP         40
CHIP_SIZE           2000.000, 2000.000
FIELD_SIZE          1000.000, 1000.000
SUB_FIELD_SIZE      100.000, 100.000
SCALING             3.000
RESIZE              20.0
MIRROR              X
ROTATION            90
OUTLINE             YES
REVERSE_TONE        NO
WINDOW              NO
```

24.4 Shot Rank And Color Number Correspondence File

The shot rank and color number correspondence file is for the data-display function. JBXFILER also uses this file; however, the contents and format of the file are modified for the support of the 1024-step shot ranking and increase in the number of display colors.

File name	pchk1.prm	
File attribute	Text file	
Location	Directory specified by the environment variable CNVUSRPATH (default: ~/eb/prm)	
Contents	Identifier	#RANK16, #RANK64, #RANK1024, #PATTERN
	Shot rank	0 to 15, 0 to 63, 0 to 1023
	File number	1 to 12
	Color number	1 to 1024

● Format

a. Identifier

Start rank, End rank, Color number
Start rank, End rank, Color number A-Color number B

- You can specify a color number for displaying patterns whose ranks are in between the start and end ranks.
- The identifier indicates the number of shot rank steps. If you want to specify color numbers in the 16-step shot ranking, write #RANK16.
- For consecutive shot ranks, if you want to specify color numbers that increase or decrease at regular intervals, connect the first and last color numbers with a hyphen (-) (the interval between two color numbers is calculated by dividing the difference between color numbers A and B by the difference between the start and end ranks).

b. Identifier

File number, Color number

- You can specify a color number for displaying the patterns of each file in the multiple-display process.
- The identifier indicates that a color is specified for each file (#PATTERN is the only identifier available for this purpose). The file number indicates the display sequence.

✍ If no identifier is specified, the program assumes that the 64-step shot ranking is specified.

● Example

#RANK1024	: Color number settings for 1024-step shot ranking	
0, 0, 1	: Shot rank = 0	Color number = 1
1, 63, 2	: Shot ranks = 1 to 63	Color number = 2
64, 100, 3-39	: Shot ranks = 64 to 100	Color number = 3 to 39
101, 500, 103	: Shot ranks = 101 to 500	Color number = 103
501, 999, 500	: Shot ranks = 501 to 999	Color number = 500
1000, 1023, 1000	: Shot ranks = 1000 to 1023	Color number = 1000

(1024-step rank/color correspondence)

#RANK16	: Color number settings for 16-step shot ranking	
0, 3, 1	: Shot ranks = 0 to 3	Color number = 1
4, 15, 325-380	: Shot ranks = 4 to 15	Color numbers = 325 to 380

(16-step rank/color correspondence)

#RANK64	: Color number settings for 64-step shot ranking	
0, 19, 10-29	: Shot ranks = 0 to 19	Color numbers = 10 to 29
20, 39, 220-239	: Shot ranks = 20 to 39	Color numbers = 220 to 239
40, 59, 555-574	: Shot ranks = 40 to 59	Color numbers = 555 to 574
60, 63, 990-996	: Shot ranks = 60 to 63	Color numbers = 990 to 996

(64-step rank/color correspondence)

#PATTERN	: File/color number settings for multiple-data display	
1, 1	: File number = 1	Color number = 1
2, 100	: File number = 2	Color number = 100
3, 200	: File number = 3	Color number = 200
10, 300	: File number = 10	Color number = 300
12, 999	: File number = 12	Color number = 999

(File/color correspondence for multiple-data display)

24.5 Color Number And Hue Correspondence File

The color number and hue correspondence file is for the data-display function. JBXFILER also uses this file; however, the writable color numbers are modified in accordance with the increase in the number of display colors.

Filename	pchk1.col
File attribute	Text file
Location	Directory specified by the environment variable CNVUSRPATH (default: ~/eb/prm)
Contents	Color number and brightness Color number: 0 to 1028 Brightness: 0 to 255 The program uses color numbers 0 and 1025 to 1028. 0: Background color 1: Palette 1 2: Palette 2 . . . 1024: Palette 1024 1025: Color for displaying patterns whose ranks are not displayed 1026: Message color 1027: Window frame/menu color 1028: Dashed lines for indicating sections
Format	① color, red, blue, green ② color1-color2, red1-red2, blue1-blue2, green1-green2

color, color1, color2	: Color numbers
red, red1, red2	: Red brightness
blue, blue1, blue2	: Blue brightness
green, green1, green2	: Green brightness

Method ① sets the hue for one color number.
Method ② sets the hues for a range of color numbers at once, so that the hues gradually change one by one.

● Example

```
0,    0,    0,    0    ...①
1,   79,   79,  255
2,  175,   79,  255
3,  127,  127,    0
    10-19,100-190,255-255,0-0    ...②
.
.
.
```

① Hue setting for one color number

For color number 0, the red brightness is set to 0, the blue brightness is set to 0, and the green brightness is set to 0.

② Hue setting for a range of color numbers (gradation setting)

The hues of color numbers 10 to 19 are set at once, as follows:

Red brightness:	100, 110, 120... (in steps of 10) are assigned to color number 10 and subsequent numbers, respectively.
Blue brightness:	255 is assigned to all the color numbers 10 to 19.
Green brightness:	0 is assigned to all the color numbers 10 to 19.

- If you want to specify the same brightness for a range of color numbers, specify the same value on both sides of a hyphen (-).
- For a range of color numbers, if you want to specify brightness values that increase or decrease at regular intervals, connect the first and last brightness values with a hyphen (-) (the interval between two brightness values is calculated by dividing the difference between the first and last brightness values by the difference between the first and last color numbers.)

24.6 Network Setup File (OpenVMS version)

The network setup file is for the network function. Systems that support networking use this file.

File name	NETCONNECT.PRM
File attribute	Text file
Location	Directory specified by the environment variable CONV$DISK0: (default: ~/eb/prm)
Contents	Name of the host to be NFS mounted, and information required for NFS mounting
Format	DNFSn: Hostname Pathname n: Integer greater than or equal to 1

● Example

```
dnfs2:   jst01   /mnt/users/dir
```

24.7 Tables of the Limits on the Data-Conversion Parameters

The following tables show the parameter limits and default values for setting the parameters of the data-conversion function.

■ Table of the parameter limits and default values for the JBX-5000LS/E, JBX-6000FS, and JBX-6000FS/E

Parameter	Limits	Default value
Output file	Up to 84 characters (including file type)	Output layer name + .J51
Output layer	Up to 24 characters	Input layer name
Chip size	0.001 to 130000.0 μm	Depends on the input format. ✍ Must be an integer multiple of the output unit. Maximum number of subfields: 12000×12000
Field size	10.0 to 1600.0 μm	1600.0/800.0/160.0/80.0 μm ✍ Depends on the output unit.
Subfield size	1.25 to 100.0 μm	100.0/50.0/10.0/5.0 μm ✍ Depends on the output unit.
Magnification	1/1000 to 10	1
Mirror	None/X axis/Y axis	None
Output unit (step)	20/40/80/200/400/800	40
Scaling	0.1 to 20.0	1.0
Rotation	None/90°/180°/270°	None
Special conversion	No conversion/Outline/Shot modulation 1/Shot modulation 2	Outline
Special conversion/shot modulation 2	Area pattern rank: 0 to 63 Single-line rank: 0 to 15	0 for both area patterns and single lines
Resizing (μm)	–32.0 to 32.0 μm	0.0 (no resizing)
Reverse tone (μm)	None/0.0 to 2000.0 μm	None
Clipping	None/Region in the chip size	None
2 time drawing (μm)	Cannot be specified.	Cannot be specified.
Maximum beam area	Cannot be specified.	Cannot be specified.
Maximum beam size	Cannot be specified.	Cannot be specified.
Trapezoid beam size	Cannot be specified.	Cannot be specified.
Arch length (JEOL01)	0.05 to 1000.0 μm	1.0 μm
Data unit (PG3000/PG3600)	micron/inch	micron

✍ These parameters are further limited depending on the conditions for writing.
- Field size
- Subfield size

● Output unit (step)

- JBX-5000LS/E

EOS mode	Maximum field size	Subfield size	Accelerating voltage	Deflector	Output (µm)
1	1600 µm	25 to 100 µm	25 kV	2nd	0.05, 0.025
2	1600 µm	25 to 100 µm	25 kV	2nd	0.05, 0.025
3	800 µm	25 to 100 µm	50 kV	2nd	0.05, 0.025
4	800 µm	25 to 100 µm	50 kV	2nd	0.05, 0.025
5	160 µm	2.5 to 10 µm	25 kV	1st	0.005, 0.0025
6	160 µm	2.5 to 10 µm	25 kV	1st	0.005, 0.0025
7	80 µm	2.5 to 10 µm	50 kV	1st	0.005, 0.0025
8	80 µm	2.5 to 10 µm	50 kV	1st	0.005, 0.0025

- JBX-6000FS, JBX-6000FS/E

EOS mode	Maximum field size	Subfield size	Accelerating voltage	Deflector	Output (µm)
1	1600 µm	25 to 100 µm	25 kV	2nd	0.05, 0.025
2	1600 µm	25 to 100 µm	25 kV	2nd	0.05, 0.025
3	800 µm	12.5 to 50 µm	50 kV	2nd	0.025, 0.0125
4	800 µm	12.5 to 50 µm	50 kV	2nd	0.025, 0.0125
5	160 µm	2.5 to 10 µm	25 kV	1st	0.005, 0.0025
6	160 µm	2.5 to 10 µm	25 kV	1st	0.005, 0.0025
7	80 µm	1.25 to 5 µm	50 kV	1st	0.0025, 0.00125
8	80 µm	1.25 to 5 µm	50 kV	1st	0.0025, 0.00125

Table of the parameter limits and default values for the JBX-9000MV

Parameter	Limits	Default value
Output file	Up to 84 characters (including file type)	Output layer name + .v30
Output layer	Up to 24 characters	Input layer name
Chip size	0.001 to 230000.0 µm	Depends on the input format. Must be an integer multiple of the output unit.
Field size	50.0 to 2000.0 µm	1500.0 µm
Position set size	0.1 to 2000.0 µm	100.0 µm
Magnification	1/1000 to 20	1
Mirror	None/X axis/Y axis	None
Output unit (step)	1 to 1000 (Only the divisors of 1000 can be specified.)	Combination of 1000, 100, and 40*
Scaling	0.1 to 20.0	1.0
Rotation	None/90°/180°/270°	None
Micro size (µm)	0.0 to 2.0 µm	0.0 µm
Special conversion	No conversion/Outline/Shot modulation 1	Outline
Resizing (µm)	–32.0 to 32.0 µm	0.0 (no resizing)
Reverse tone (µm)	None/0.0 to 2000.0 µm	None
Clipping	None/Region in the chip size	None
2 time drawing (µm)	None/0.0 to 4.0 µm	None
Maximum beam area	Cannot be specified.	Cannot be specified.
Maximum beam size	Cannot be specified.	Cannot be specified.
Trapezoid beam size	Cannot be specified.	Cannot be specified.
Arch length (JEOL01)	Cannot be specified.	Cannot be specified.
Data unit (PG3000/PG3600)	micron/inch	micron

* The default settings are: unit of chip size = 1000, unit of position set value = 100, and unit of pattern data= 40. If you specify any output unit, all the units of chip size, position set value, and pattern data will be the same.

■ Table of the parameter limits and default values for the JBX-9300FS

Parameter	Limits	Default value
Output file	Up to 84 characters (including file type)	Output layer name + .v30
Output layer	Up to 24 characters	Input layer name
Chip size	0.001 to 230000.0 μm	Depends on the input format. Must be an integer multiple of the output unit.
Field size	50.0 to 2000.0 μm	Version 4.000 or earlier versions of JBXFILER: 1500.0 μm Version 4.000 or later versions of JBXFILER: 500.0 μm
Position set size	0.1 to 2000.0 μm	Version 4.000 or earlier versions of JBXFILER: 100.0 μm Version 4.000 or later versions of JBXFILER: 500.0 μm
Magnification	1/1000 to 20	1
Mirror	None/X axis/Y axis	None
Output unit (step)	1 to 1000 (Only the divisors of 1000 can be specified.)	Version 4.000 and earlier versions of JBXFILER: Combination of 1000, 100, and 40* Version 4.000 and later versions of JBXFILER: Combination of 1000, 1000, and 1000**
Scaling	0.1 to 20.0	1.0
Rotation	None/90°/180°/270°	None
Micro size (μm)	0.0 to 2.0 μm	0.0 μm
Special conversion	No conversion/Outline/Shot modulation 1	Outline
Resizing (μm)	–32.0 to 32.0 μm	0.0 (no resizing)
Reverse tone (μm)	None/0.0 to 2000.0 μm	None
Clipping	None/Region in the chip size	None
2 time drawing (μm)	Cannot be specified.	Cannot be specified.
Maximum beam area	Cannot be specified.	Cannot be specified.
Maximum beam size	Cannot be specified.	Cannot be specified.
Trapezoid beam size	Cannot be specified.	Cannot be specified.
Arch length (JEOL01)	Cannot be specified.	Cannot be specified.
Data unit (PG3000/PG3600)	micron/inch	micron

*The default settings are: unit of chip size = 1000, unit of position set value = 100, and unit of pattern data= 40. If you specify any output unit, all the units of chip size, position set value, and pattern data will be the same.
**The default settings are: unit of chip size = 1000, unit of position set value = 1000, and unit of pattern data= 1000.

Table of the parameter limits and default values for the JBX-6300 series

Parameter	Limits	Default values
Output file	Up to 84 characters (including file type)	Output layer name + .v30
Output layer	Up to 24 characters	Input layer name
Chip size	0.001 to 230000.0 µm	Depends on the input format. Must be an integer multiple of the output unit.
Field size	2.5 to 2000.0 µm	500.0 µm
Position set size	0.1 to 2000.0 µm	500.0 µm
Magnification	1/1000 to 20	1
Mirror	None/X axis /Y axis	None
Output unit (step)	1 to 1000 (Only the divisors of 1000 can be specified.) 2000, 4000, 8000	Combination of 1000, 1000, and 1000*
Scaling	0.1 to 20.0	1.0
Rotation	None/90°/180°/270°	None
Micro size (µm)	0.0 to 2.0 µm	0.0 µm
Special conversion	No conversion/Outline/Shot modulation 1	Outline
Resizing (µm)	-32.0 to 32.0 µm	0.0 (no resizing)
Reverse tone (µm)	None/0.0 to 2000.0 µm	None
Clipping	None/Region in the chip size	None
2 time drawing (µm)	Cannot be specified.	Cannot be specified.
Maximum beam area	Cannot be specified.	Cannot be specified.
Maximum beam size	Cannot be specified.	Cannot be specified.
Trapezoid beam size	Cannot be specified.	Cannot be specified.
Arch length (JEOL01)	Cannot be specified.	Cannot be specified.

* The default settings are: unit of chip size = 1000, unit of position set value = 1000, and unit of pattern data= 1000. If you specify any output unit, all the units of chip size, position set value, and pattern data will be the same.

24.8 JBXFILER Default File

The JBXFILER default file contains the default values of the parameters of each function. At present, you can specify only the output device parameters.

File name	Jbxdtp.default
File attribute	Text file
Location	Directory specified by the environment variable CNVUSRPATH (default: ~/eb/prm)
Summary	This file contains the default values of the parameters of each function. It is read only one time when JBXFILER starts.
Format	***command*keyword:△directory***

command	Specifies the command for setting the default value. You can specify these commands: CONV, SAVE, MERGE, PREAD6FS
keyword	Specifies the parameter for setting the default value. ✍ At present, you can specify only OUTDEV (output device).
directory	Specifies a directory as the default value.
△	Indicates one or more blank characters (space or tab).

● Example

CONV*OUTDEV:/home/eb0/ebtest/pattern

Default output destination for covering data

MERGE*OUTDEV:/home/eb0/ebtest/cad

Default output destination for merging data

SAVE*OUTDEV:/home/eb0/ebtest/pchk

Default output destination for converting display data

PREAD6FS*OUTDEV:/home/eb0/ebtest/pattern

Default output destination for reading data

✍ The directory configuration varies depending on the model.

24.9 Scanner-Format Data Management Table Operation Command (setdtab)

Overview

The setdtab command manages the directories of scanner-format data (pattern data resulting from data reading). It has these four functions:

- Displaying how to use the setdtab command
- Displaying the dtab settings
- Adding a directory to dtab
- Removing a directory from dtab

Operation

Displaying how to use the setdtab command

You can display how to use setdtab, as follows:

Command:	(ebtest)/ setdtab
Display example:	Usage: setdtab show setdtab add directory [dirnum] setdtab remove dirnum

Displaying the dtab settings

You can display the present directory-name and management-number settings of dtab, as follows:

Command:	(ebtest)/ setdtab show
Display example:	0 /usr/users/eb0/ebtest/pattern/ptn 1 /usr/users/eb0/ebtest/jst

Adding a directory to dtab

You can add a directory to dtab, as follows:

Command:	(ebtest)/ setdtab add directory [dirnum]
Display example:	No display

Removing a directory from dtab

You can remove a directory from dtab, as follows:

Command:	(ebtest)/ setdtab remove dirnum
Display example:	No display

24.10 Library Parameter File

File name	Any file name (default file type: .inf)
File attribute	Text file
Location	Directory specified by the environment variable CNVUSRPATH (default: ~/eb/prm)
Contents	Information on library decompression
Format	UNIT X_1, Y_1, X_2, Y_2 ← Specification of unit region AREA X_3, Y_3, X_4, Y_4 ← Specification of decompression area

X_1, Y_1: Lower left coordinates of unit
X_2, Y_2: Upper right coordinates of unit
X_3, Y_3: Lower left coordinates of decompression area
X_4, Y_4: Upper right coordinates of decompression area

Rules

- The unit is the repetition unit, and the decompression area is the repetition region.
- The decompression area must be an integer multiple of the unit.
- You can specify up to 2,000 units (256 units for JEOL52).
- You can specify up to 40,000 areas.
- Specify the coordinates in μm. The program rounds the coordinates to three decimal places.
- The coordinates are in the first quadrant (with the origin at the lower left).
- Express the coordinates of the unit and decompression area as positions on the source data.
- Insert one or more spaces between UNIT or AREA and the coordinates.
- Separate the coordinates with commas (,).
- You can use any number of spaces between data.
- The specification of the unit must precede the specification of the decompression area.
- The pattern to be decompressed in the decompression area is the pattern of the unit specified immediately before the area specification.
- The program regards a line whose first column begins with an asterisk (*) or a blank line as a comment line.
- Nested structure is not allowed.

● Example of file content

```
* U1 is unit of area A, B, F
UNIT x4, y4, x3, y3
AREA x0, y0, x1, y1
AREA x2, y2, x3, y3

* U2 is unit of area C, D, E
UNIT x5, y6, x6, y7
AREA x5, y5, x7, y7
AREA x8, y5, x9, y7
AREA x10, y5, x11, y7
```

● Example of decompression

The following is an example of decompression where data are converted using the library parameter file shown in the example of file content. Here, unit U1 is decompressed to the decompression areas A and B, and unit U2 is decompressed to the decompression areas C, D, and E.